

Week 3

Analysis Techniques

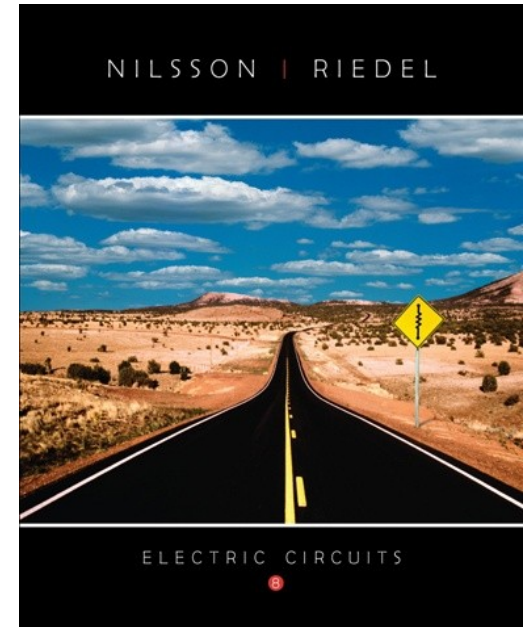
Textbook (for Circuits only)

J. W. Nilsson, S. A. Riedel

Electric Circuits

Pearson Prentice Hall

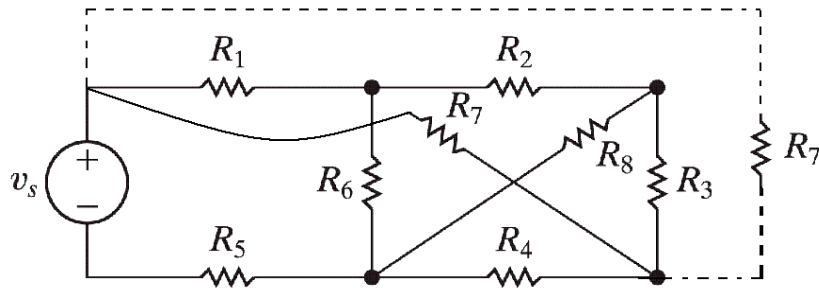
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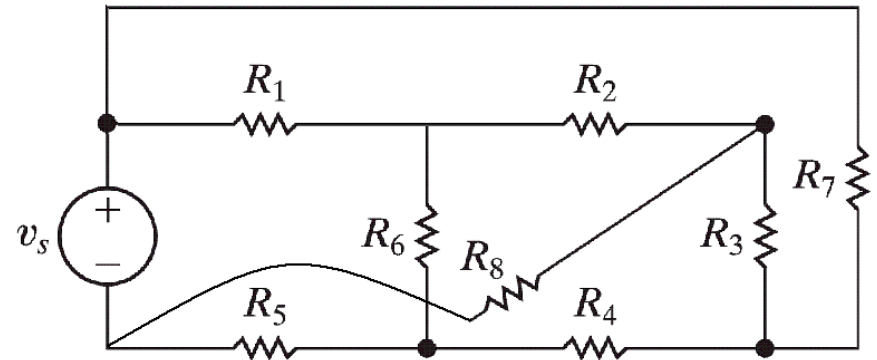
The Heart of the Course

Nodal and Mesh analysis techniques are used to solve circuits

Planar Circuits: Can be drawn on a plane with no crossing branches



Planar
Nodal, Mesh

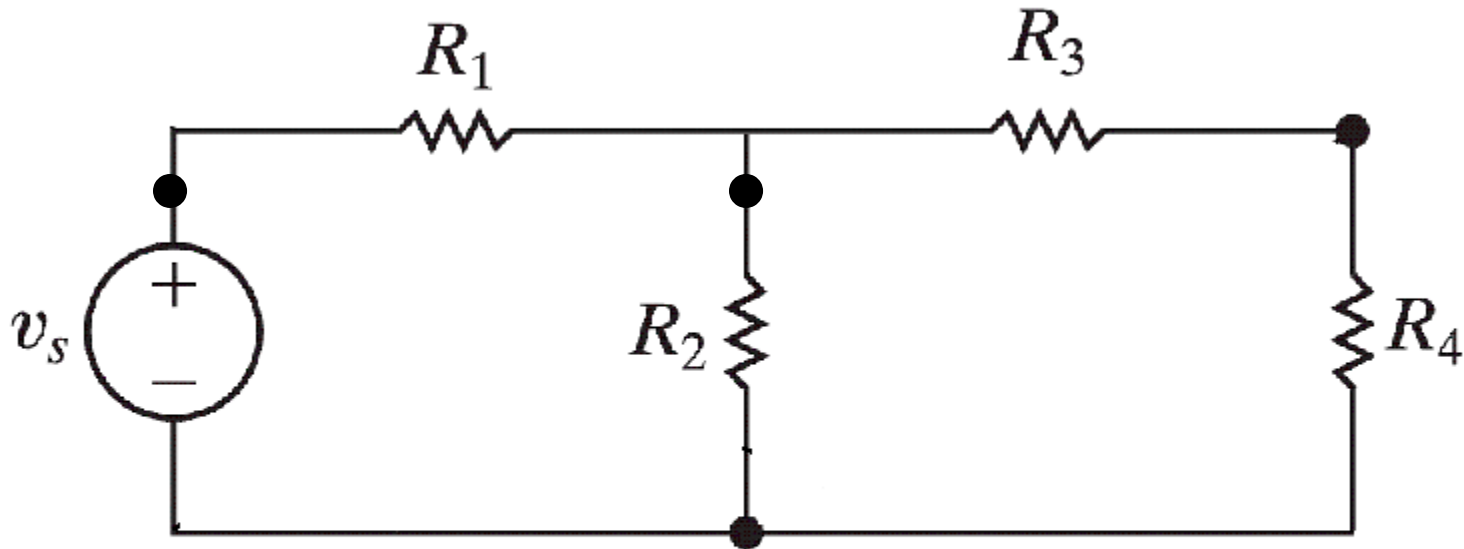


Non Planar
Nodal Only

Most Important Terms

Node	Point where 2 or more circuit elements join.
Loop	Closed Path. Start node = end node.
Mesh	Loop not enclosing another loop.
Path	Trace of adjoining elements, with no element included more than once.
Branch	Path that connects 2 nodes.
Essential Node	Node where 3 or more elements join. (n_e)
Essential Branch	Branch connecting 2 essential nodes. (b_e)

Simple Illustration



How many Nodes?

4

How many Branches?

5

What are the Essential Nodes and Branches?

2

1

Simultaneous Equations

- $\mathbf{b_e}$ = # of essential branches in the circuit
- $\mathbf{b_e}$ = # of independent unknown currents
- $\mathbf{n_e}$ = # of essential nodes in the circuit
- We can derive $\mathbf{(n_e-1)}$ independent equations with KCL applied to $\mathbf{(n_e-1)}$ nodes
- We can derive the other $\mathbf{b_e - (n_e-1)}$ independent equations with KVL applied to $\mathbf{b_e - (n_e-1)}$ loops or meshes

Example

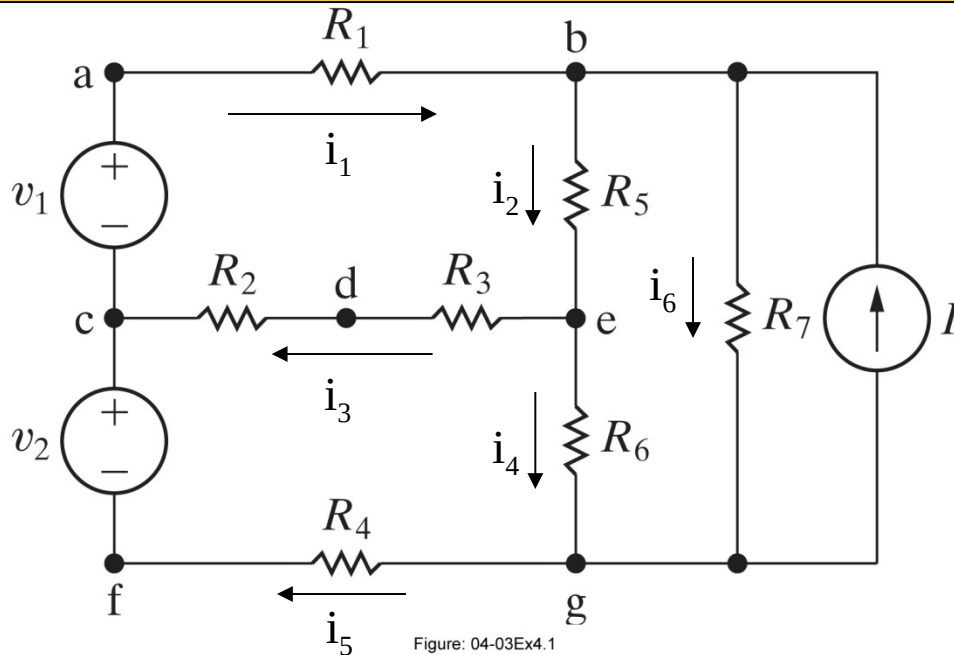


Figure: 04-03Ex4.1

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Essential Nodes } $n_e = 4$
b c e g
 3 or more elements connected

*Current is divided
at these nodes*

Essential Branches (Total of 6) } $b_e = 6$
 (c-a-b), (b-g), (c-d-e), (c-f-g), (b-e), (e-g)

Connects 2 essential nodes with none in between

There are six independent current variables } $b_e = 6$

Example (Contd.)

- KCL $\Rightarrow (n_e - 1) = 4 - 1 = 3$ Independent Equations
- KVL $\Rightarrow$ the remaining: $b_e - (n_e - 1) = 6 - (4 - 1)$
 $= 3$ Independent Equations

- Need $b_e = 6$ independent equations. $\left. \begin{array}{l} \sum i_{in} = \sum i_{out} \end{array} \right\}$ **KCL**

Six independent current variables

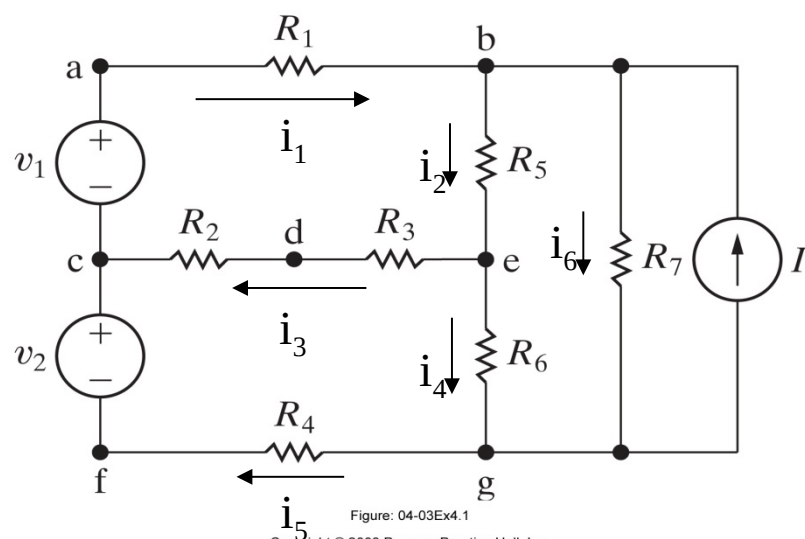


Figure: 04-03Ex4.1

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	In	Out	
KCL	(b) $i_1 + I$	$= i_2 + i_6$	(1)
	(c) $i_3 + i_5$	$= i_1$	(2)
	(e) $i_2 = i_3 + i_4$		(3)

Example (Contd.)

KVL: 4 meshes are in the circuit, but we will use only 3 meshes

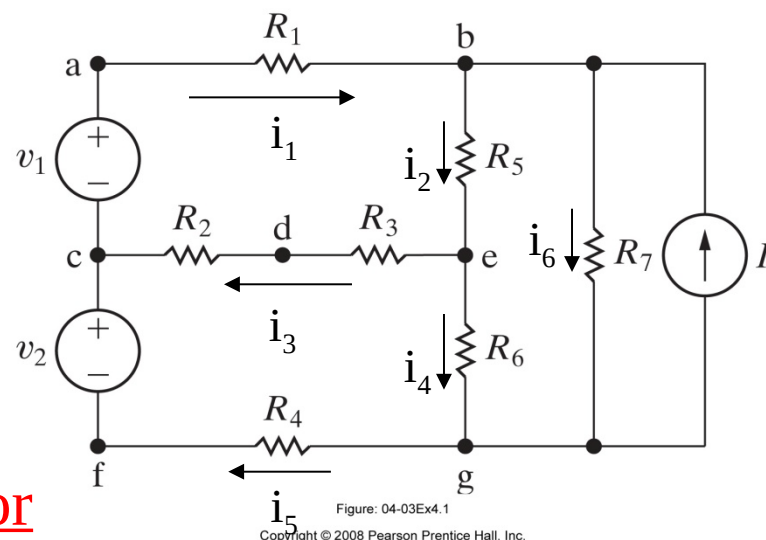
$$(a - b - e - d - c - a) \quad V_1 = i_1 R_1 + i_2 R_5 + i_3 (R_2 + R_3) \quad (4)$$

$$(c - d - e - g - f - c) \quad V_2 = -i_3 (R_2 + R_3) + i_4 R_6 + i_5 R_4 \quad (5)$$

$$(b - e - g - b) \quad i_2 R_5 + i_4 R_6 - i_6 R_7 = 0 \quad (6)$$

- Don't use $(b - g - b)$ across the I-source because we don't know the voltage drop

- We now have 6 equations to solve for 6 unknowns



Introduce New Techniques

If you set the problem up right, you don't need as many equations

Node Voltages Requires only $n_e - 1$ equations

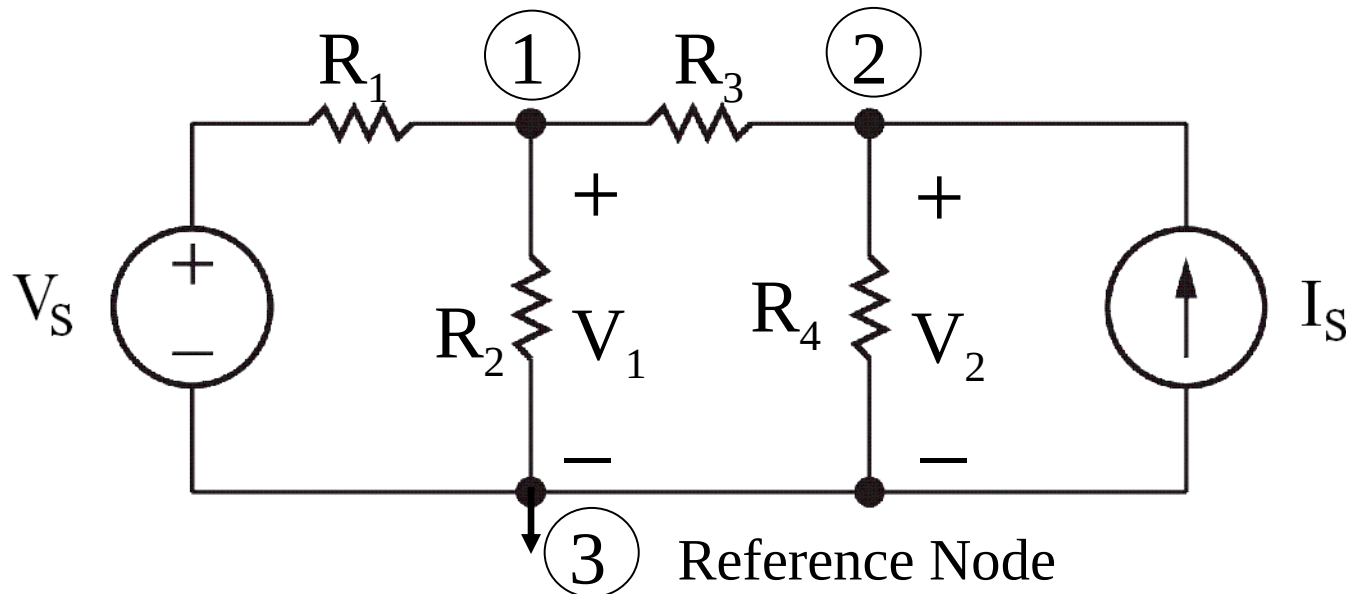
OR

Mesh Currents Requires only $b_e - (n_e - 1)$ equations

The Node-Voltage Method

Construct $(n_e - 1)$ Node Voltage Equations

n_e : Number of
Essential nodes



3 Essential nodes, $n_e = 3$ $(n_e - 1) = 3 - 1 = 2$ } Number of Equations

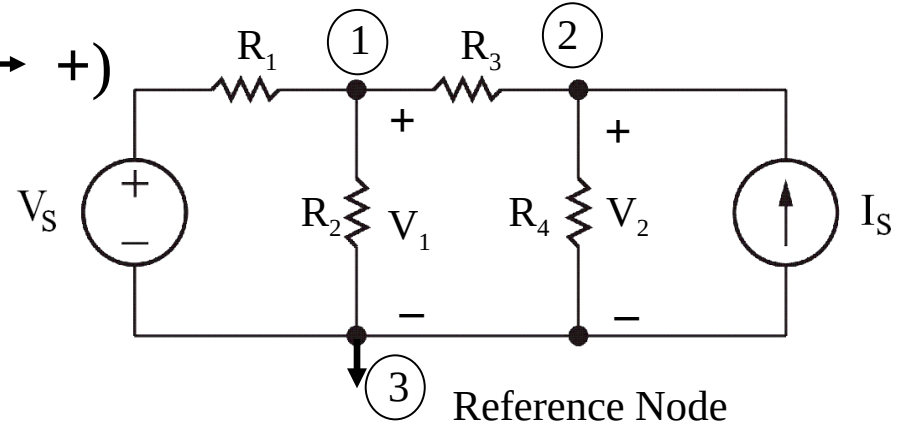
Choose reference node. (Usually node with most branches)

The Node-Voltage Method

Node Voltage: Voltage Rise ($- \rightarrow +$)

$\{V_1: \text{between 1 \& 3} \}$

$\{V_2: \text{between 2 \& 3} \}$

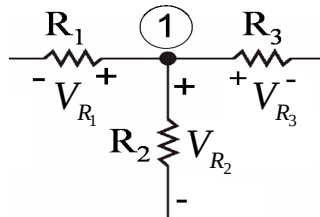


Node Equation for (1)

Step (1) Sum the currents Leaving Non-Reference Node = 0 } **KCL**

Step (2) Write Currents in Terms of Node Voltages

Step (3) $\frac{V_{R_1}}{R_1} + \frac{V_{R_2}}{R_2} + \frac{V_{R_3}}{R_3} = 0$ } **KCL**



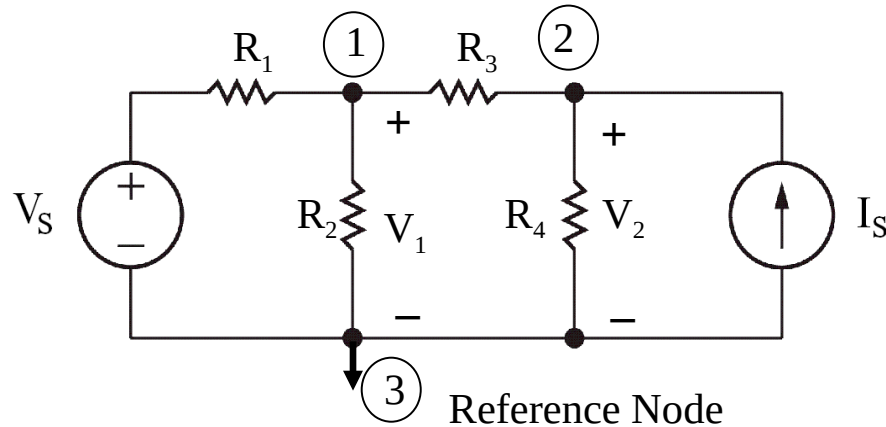
$$V_{R_1} = V_1 - V_S$$

$$V_{R_2} = V_1$$

$$V_{R_3} = V_1 - V_2$$

See Circuit

The Node-Voltage Method

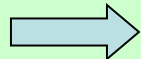


Current Leaving Node ②

$$\frac{V_2 - V_1}{R_3} + \frac{V_2}{R_4} - I_S = 0$$

Node Equation for ②

Current leaving node ② → opposite polarity across R_3 to that of node ①

Current source produces a current Flowing into ② ;  “-” sign

2 Equations and 2 Unknowns. Solve for V_1 and V_2

All other quantities can be computed from V_1 and V_2

Example With Numbers

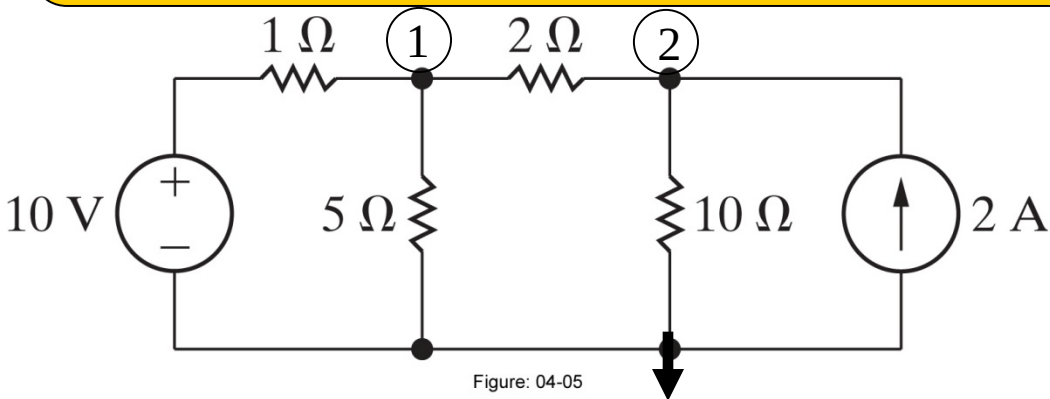


Figure: 04-05

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Find essential nodes and solve for node voltages

$$\frac{V_1 - 10}{1} + \frac{V_1}{5} + \frac{V_1 - V_2}{2} = 0 \quad (1)$$

$$\frac{V_2 - V_1}{2} + \frac{V_2}{10} - 2 = 0 \quad (2)$$

Multiply Both Equations by 10

$$10V_1 - 100 + 2V_1 + 5V_1 - 5V_2 = 0$$

$$17V_1 - 5V_2 = 100 \quad (1)$$

$$5V_2 - 5V_1 + V_2 - 20 = 0$$

$$-5V_1 + 6V_2 = 20 \quad (2)$$

Solve the 2 Equations Simultaneously

Example (Contd.)

Rearrange

②



$$V_2 = \frac{20 + 5V_1}{6}$$

Substitute into



①

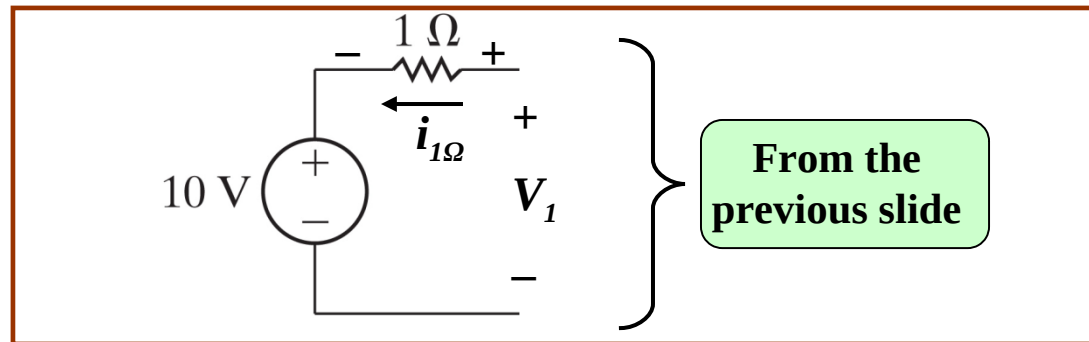


$$17V_1 - \frac{100 + 25V_1}{6} = 100$$

Solve (1) for V_1 , then (2) for V_2

$$V_1 = 9.09(V) \quad V_2 = 10.91(V)$$

Now that we know V_1 and V_2 , finding other quantities is straightforward



Example: $10 = -i_{1\Omega}1 + V_1 \quad \therefore i_{1\Omega} = \frac{-10 + 9.09}{1} = -0.91(A)$

Example (Contd.)

Previous example is much easier to solve using
the Nodal Method



Solve 2
Equations

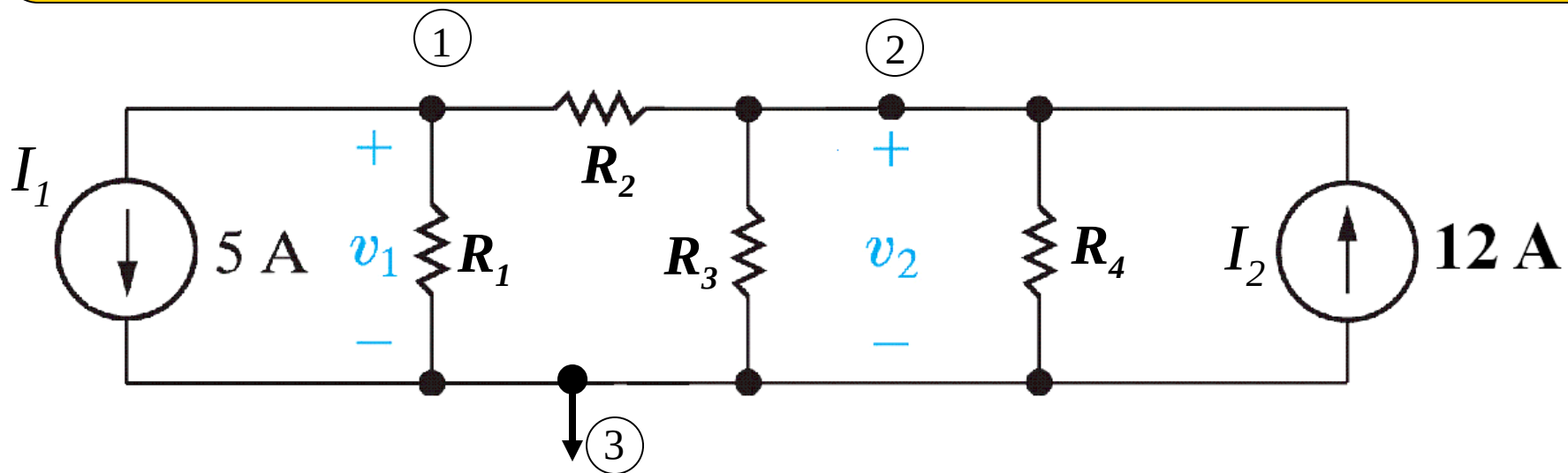
Compare

$$(n_e - 1) = (3 - 1) = 2 \text{ KCL Equations}$$
$$b_e - (n_e - 1) = 4 - 2 = 2 \text{ KVL Equations}$$

Solve 4 Equations for 4 Unknowns

Standard
Approach

Drill Exercise: Find the Node Voltages



$$R_1 = 16\Omega \quad R_2 = 2\Omega \quad R_3 = 20\Omega \quad R_4 = 80\Omega$$

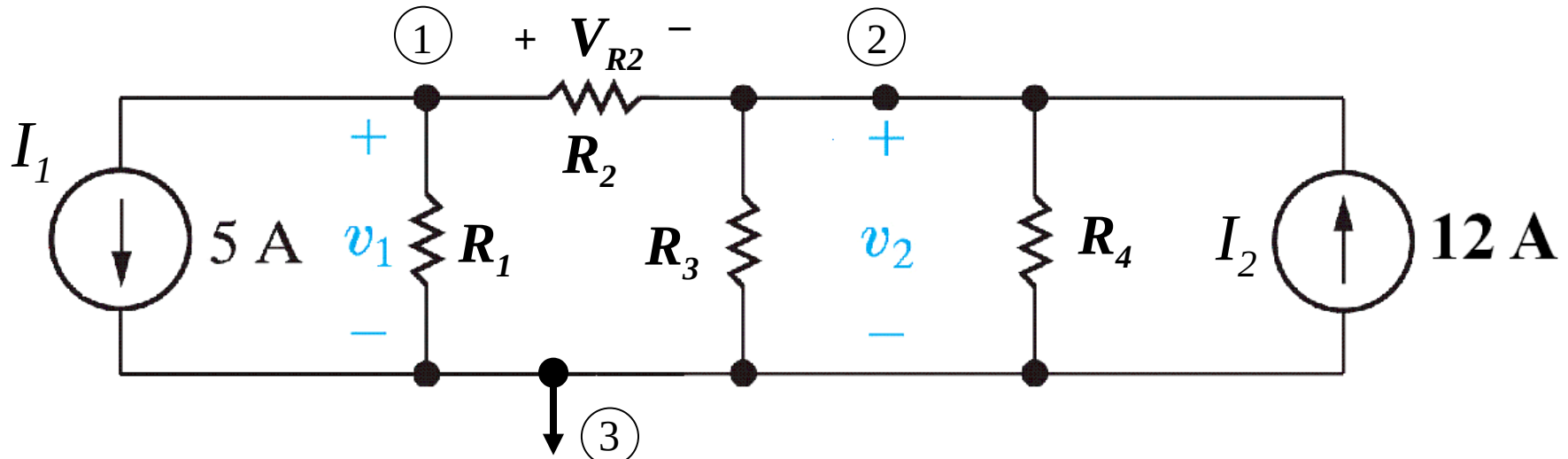
$n_e = 3$ Essential Nodes

$(n_e - 1)$ Node Voltage Equations = 2 Node Voltages

Node Voltages V_1 and V_2 Reference Node = ③

Example: Find the node equation for V_1

(Contd.)



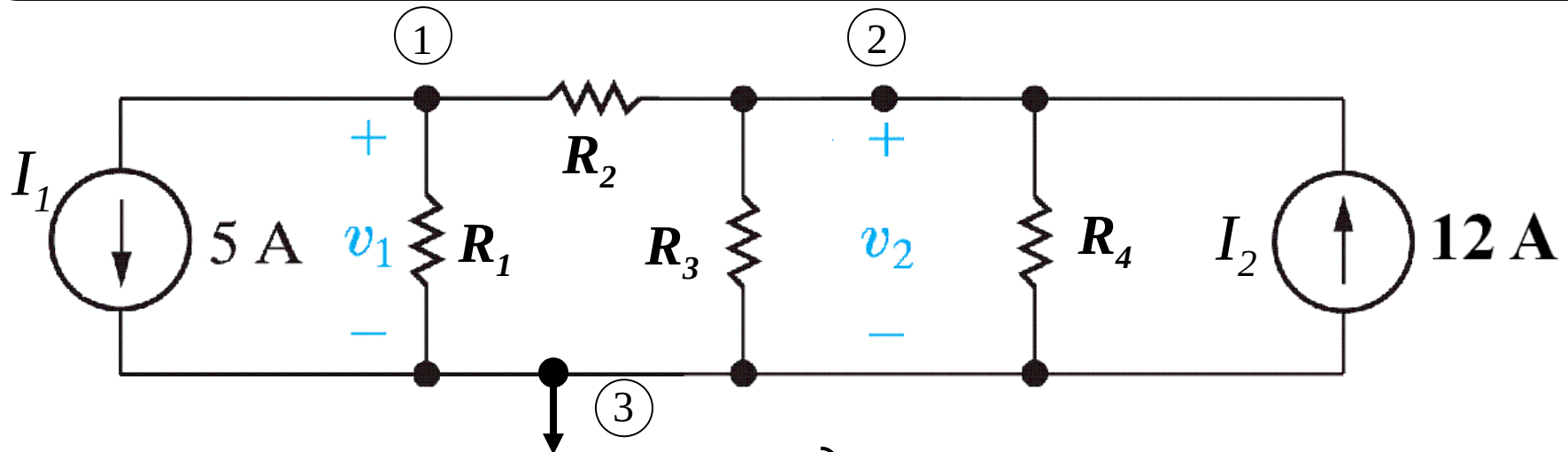
$$\textcircled{1} \quad 5 + \frac{V_1}{R_1} + \frac{V_{R_2}}{R_2} = 0 \quad \left. \vphantom{\frac{V_{R_2}}{R_2}} \right\} \text{Currents Out of Node } \textcircled{1}$$

$$5 + \frac{V_1}{R_1} + \frac{V_1 - V_2}{R_2} = 0 \quad \left. \vphantom{\frac{V_1 - V_2}{R_2}} \right\} V_{R_2} = V_1 - V_2$$

$$\textcircled{1} \quad \left(5 + \frac{V_1}{16} + \frac{V_1 - V_2}{2} = 0 \right) \quad \left. \vphantom{\frac{V_1 - V_2}{2}} \right\} \text{Node Voltage Equation}$$

Example: Find the node equation for V_2

(Contd.)



$$\textcircled{2} \quad \frac{V_2 - V_1}{R_2} + \frac{V_2}{R_3} + \frac{V_2}{R_4} - 12 = 0 \quad \left\{ \begin{array}{l} \text{Currents Out of Node } \textcircled{2} \end{array} \right.$$

Opposite to
Node $\textcircled{1}$ equation

Node 2 Connected to
4 Elements

$$\textcircled{2} \quad \left(\frac{V_2 - V_1}{2} + \frac{V_2}{20} + \frac{V_2}{80} - 12 = 0 \right) \quad \left\{ \begin{array}{l} \text{Node Voltage Equation} \end{array} \right.$$

Example (Contd.)

Solve 2 equations for 2 unknowns

After multiplying (1) by 16

$$\left\{ \begin{array}{l} V_1 + 8V_1 - 8V_2 = -80 \end{array} \right. \quad (1)$$

$$9V_1 - 8V_2 = -80 \quad (1)$$

After multiplying (2) by 80

$$\left\{ \begin{array}{l} 40V_2 - 40V_1 + 4V_2 + V_2 = -960 \end{array} \right. \quad (2)$$

$$-40V_1 + 45V_2 = 960 \quad (2)$$

Solution

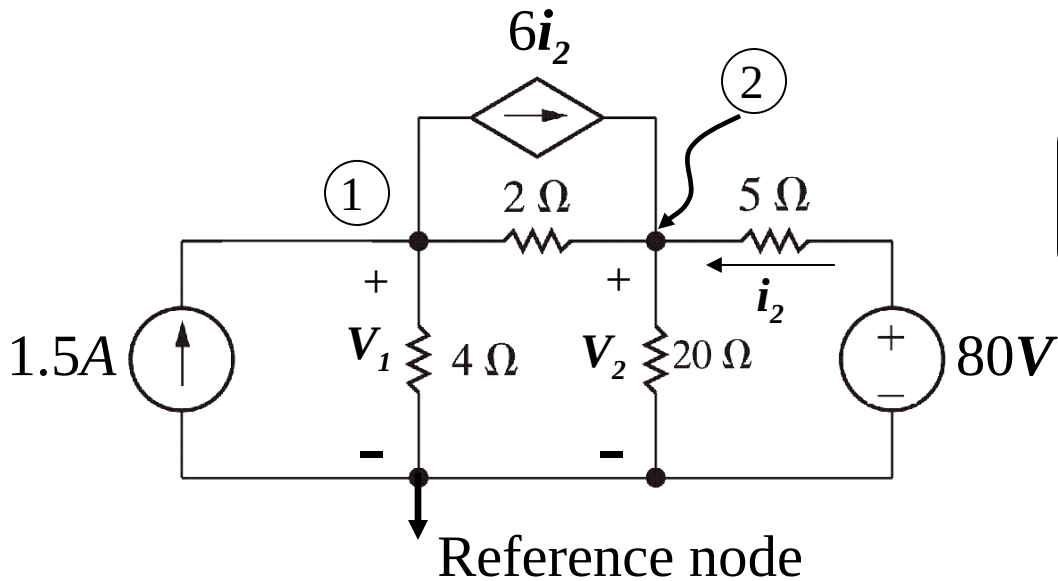
$$\left. \begin{array}{l} V_1 = 48(V) \\ V_2 = 64(V) \end{array} \right\}$$

Solve two equations
for two unknowns

Dependent Sources

- 1 How Many Node Voltage Equations?**
- 2 Choose “Best” Reference Node.**
- 3 Write Independent Node Voltage Equations.**
- 4 Find another equation for the dependent source**

Example with a Dependent Source



Need additional equation because of the dependent current source

KVL

$$i_2 = \frac{80 - V_2}{5} \quad (3)$$

$$(1) \quad -1.5 + \frac{V_1}{4} + \frac{V_1 - V_2}{2} + 6i_2 = 0$$

$$(2) \quad \frac{V_2 - V_1}{2} + \frac{V_2}{20} - 6i_2 + \frac{V_2 - 80}{5} = 0$$

Results

$$V_1 = 10(V)$$

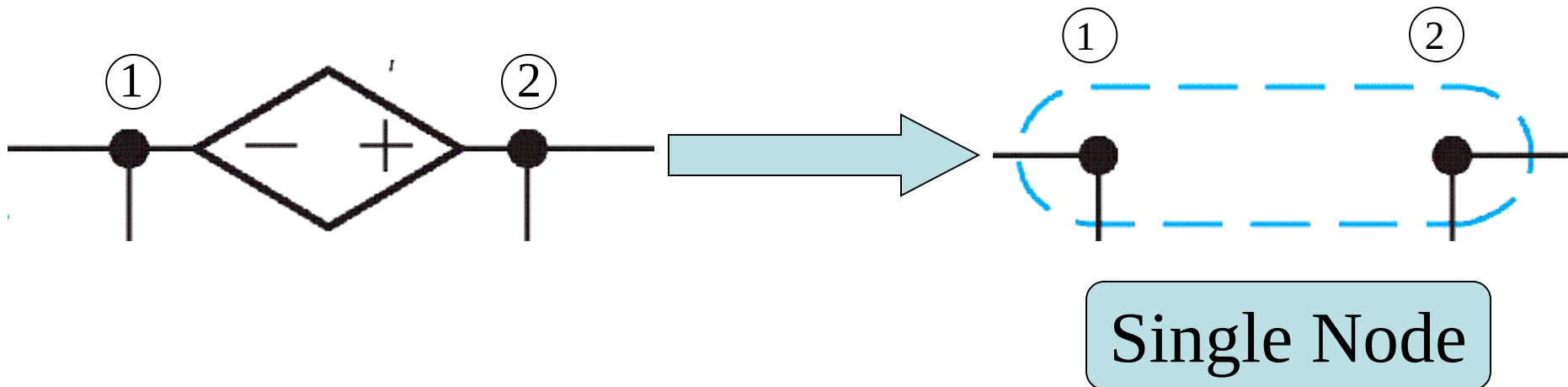
$$V_2 = 60(V)$$

$$i_2 = 4(A)$$

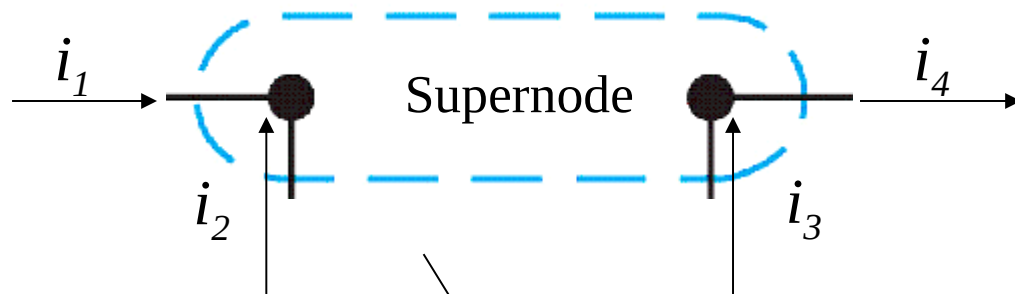
Solve 3 equations for 3 unknowns

Using the Supernode Concept to deal with a Dependent Voltage Source

Combine 2 Nodes into 1 “Supernode” when a Voltage Source is between them

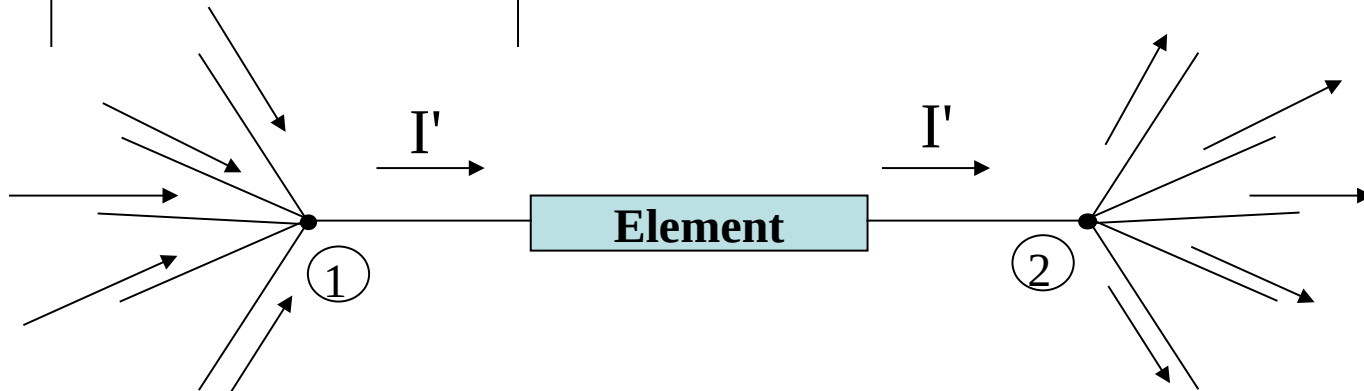


KCL Applies to the Supernode



$$i_1 + i_2 + i_3 = i_4$$

Why is this true?



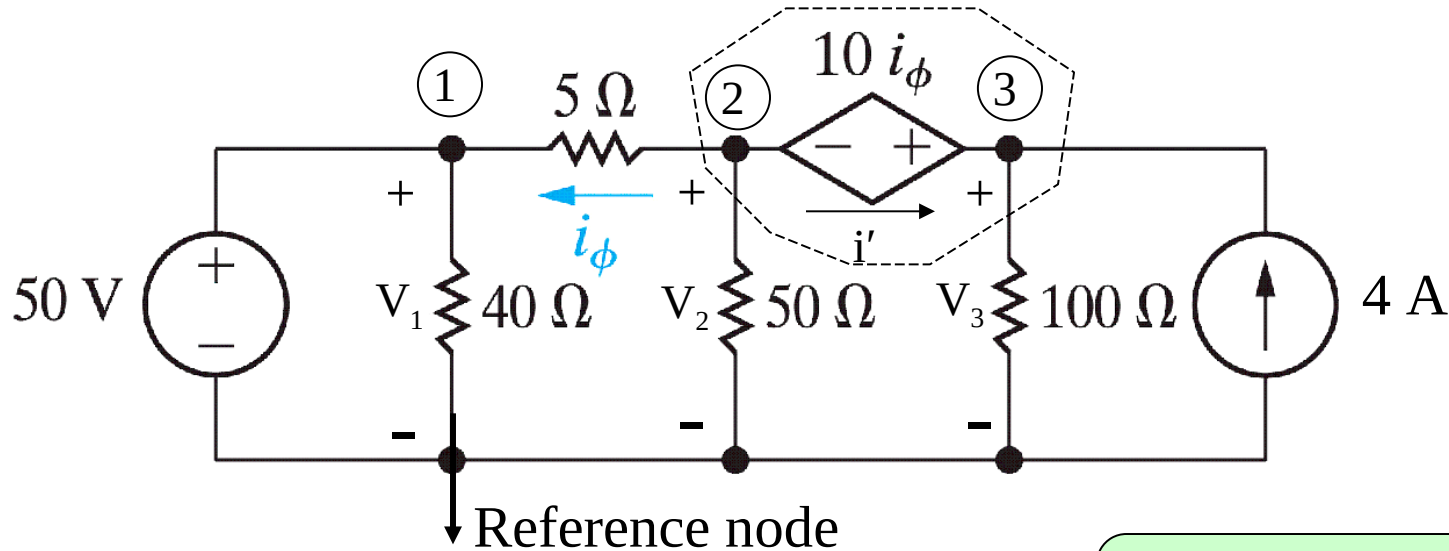
$$\sum \text{ of these currents} = I'$$

$$\sum \text{ of these currents} = I'$$

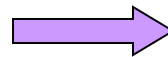
$$\sum \text{ entering } \textcircled{1} = \sum \text{ leaving } \textcircled{2}$$

Can combine nodes ① and ② into one supernode, which is treated like a node

Why use the Supernode concept?



Without using the Supernode



4 Essential Nodes
3 Node Voltage equations

Can't find the current i' in terms of the node voltages

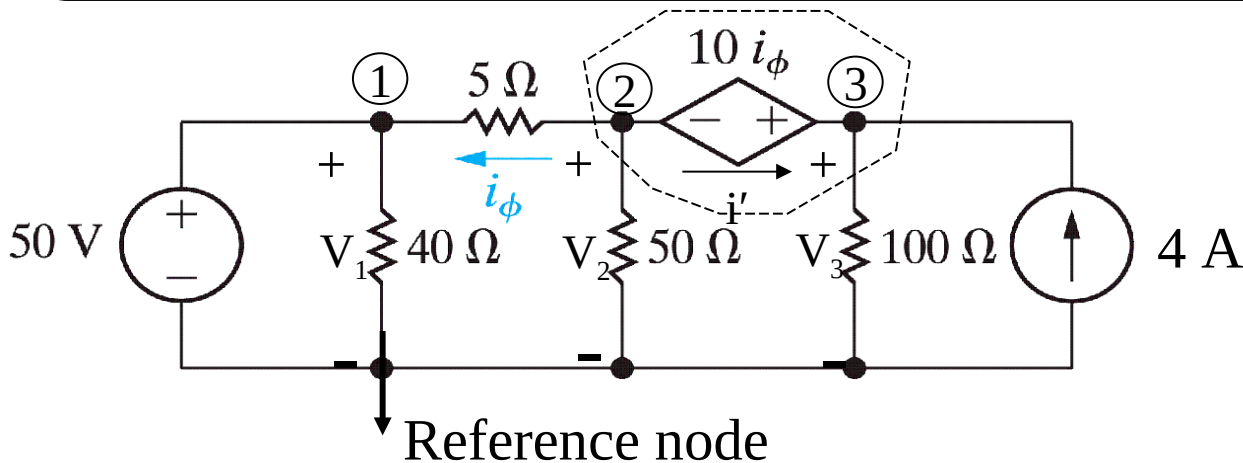
Node
Equation

②

$$\frac{V_2 - V_2}{5} + \frac{V_2}{50} + (?) = 0$$

Don't know what
to put here

Why use the Supernode concept? (Contd.)



Introduce Extra Variable and find extra equation

∴ Introduce new variable i'

$$\textcircled{2} \quad \frac{V_2 - V_1}{5} + \frac{V_2}{50} + i' = 0 \Rightarrow$$

$$\textcircled{3} \quad -i' + \frac{V_3}{100} - 4 = 0$$

$$\textcircled{3} \quad i' = \frac{V_3}{100} - 4$$

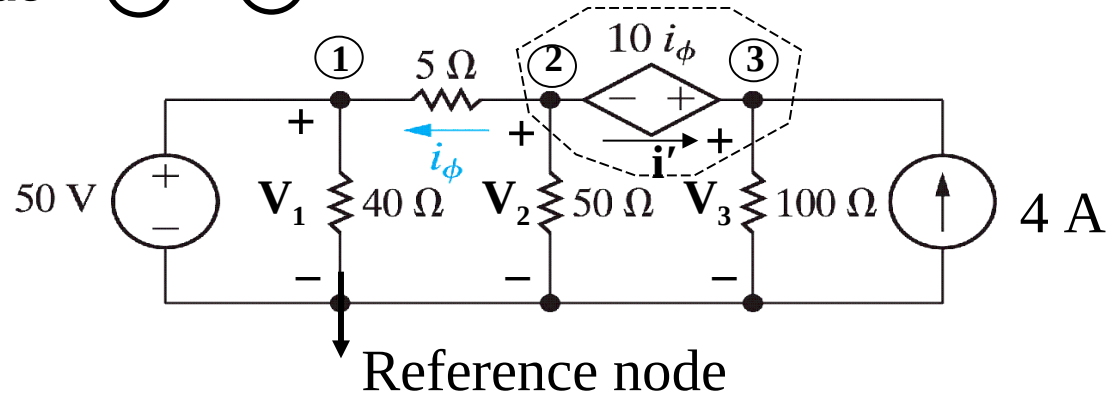
After substituting $\textcircled{3}$ into $\textcircled{2}$, becomes

$$\frac{V_2 - V_1}{5} - \frac{V_1}{5} + \frac{V_2}{50} + \frac{V_3}{100} - 4 = 0$$

Can simplify this process with Supernode

Why use the Supernode concept? (Contd.)

$\sum i$ Leaving Supernode (2) — (3)



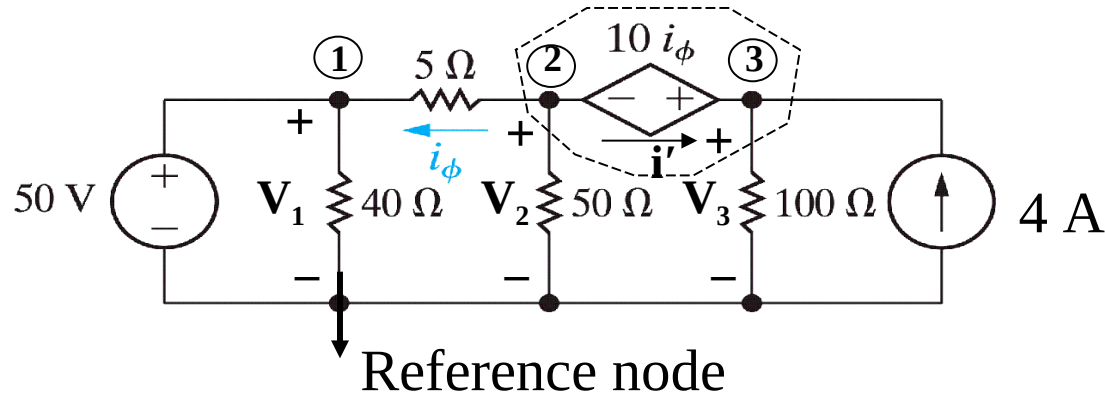
$$(2) - (3) \quad \frac{V_2 - V_1}{5} + \frac{V_2}{50} + \frac{V_3}{100} - 4 = 0 \quad \left. \vphantom{\frac{V_2 - V_1}{5}} \right\} \text{Went from 4 steps to just 1 step}$$

1 Equation and 2 Unknowns

Note $V_1 = 50$

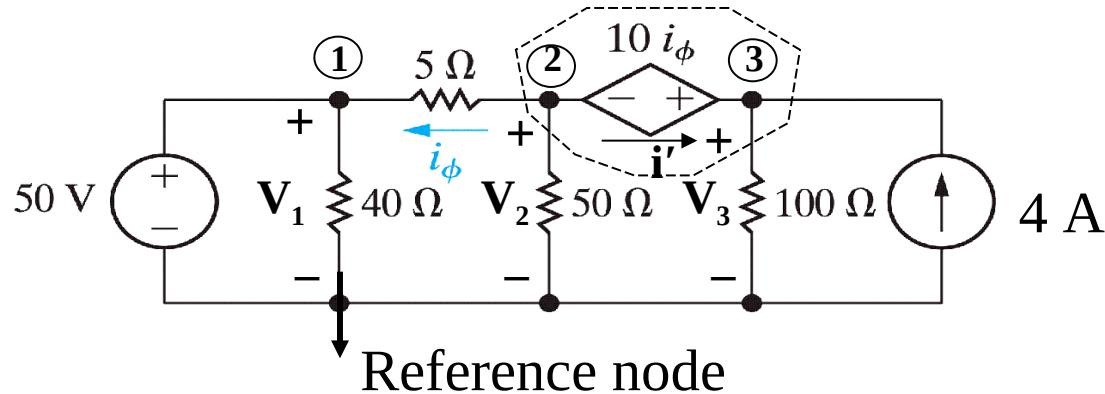
Need to find one more equation

Why use the Supernode concept? (Contd.)



$$\begin{array}{l}
 V_3 = 10i_\phi + V_2 \quad \left. \vphantom{V_3 = 10i_\phi + V_2} \right\} \text{Use KVL to find another equation} \\
 i_\phi = \frac{V_2 - V_1}{5} \quad \left. \vphantom{i_\phi = \frac{V_2 - V_1}{5}} \right\} \text{Use Ohm's Law to find expression for } i_\phi \\
 \rightarrow V_3 = 2V_2 - 2V_1 + V_2 \quad \left. \vphantom{V_3 = 2V_2 - 2V_1 + V_2} \right\} \text{After using expression for } i_\phi \\
 V_3 = 3V_2 - 2V_1 = 3V_2 - 2(50) \quad \left. \vphantom{V_3 = 3V_2 - 2V_1 = 3V_2 - 2(50)} \right\} \text{Substitute } V_1 = 50
 \end{array}$$

Why use the Supernode concept? (Contd.)

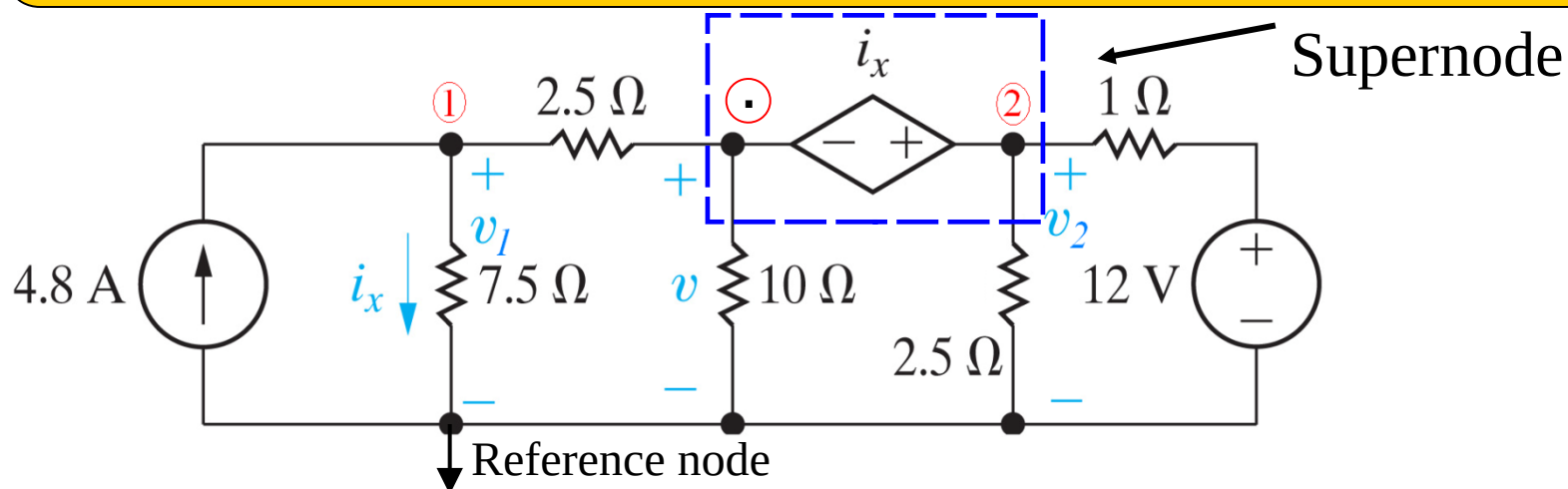


$$V_3 = 3V_2 - 100 \quad \left. \vphantom{V_3 = 3V_2 - 100} \right\} \text{New equation}$$

$$\textcircled{2} - \textcircled{3} \quad \left. \begin{array}{l} \text{Substitute } V_1 = 50 \\ \frac{V_2 - 50}{5} + \frac{V_2}{50} + \frac{3V_2 - 100}{100} - 4 = 0 \end{array} \right\} \begin{array}{l} \text{Solve} \\ 2 \text{ equations} \\ \text{for} \\ 2 \text{ unknowns} \end{array}$$

Results: $V_2 = 60(V)$ $i_\phi = 2(A)$ $V_3 = 80(V)$

Drill Exercise: Find v



$$n_e = 4$$

Find 3 equations

$$\textcircled{1} - 4.8 + \frac{v_1}{7.5} + \frac{v_1 - v}{2.5} = 0$$

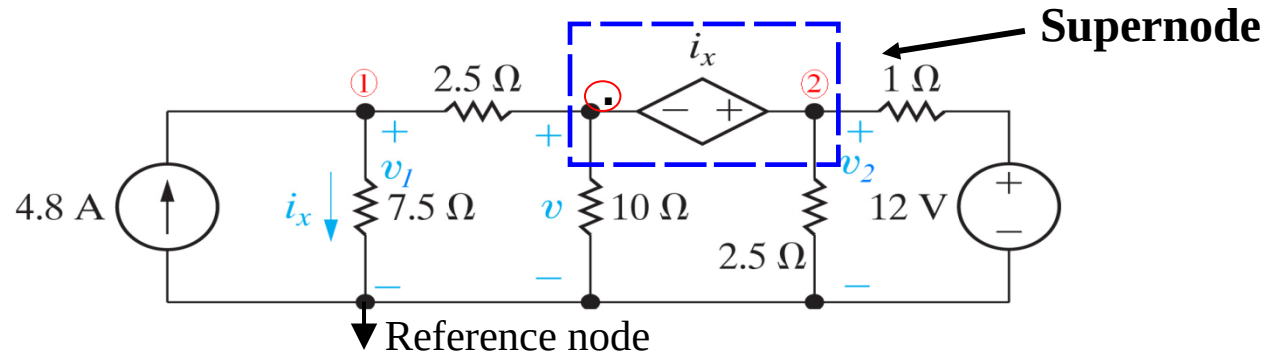
Supernode: 4 Currents Exiting The Supernode

$$\textcircled{1} - \textcircled{2} \quad \frac{v - v_1}{2.5} + \frac{v}{10} + \frac{v_2}{2.5} + \frac{v_2 - 12}{1} = 0$$

2 Equations, 3 Unknowns: v, v_1, v_2

(Contd...)

Drill Exercise (Contd.)



Relate i_x to Node Voltages

$$i_x = \frac{v_1}{7.5}$$

Use Ohm's Law

$$v_2 = \underbrace{i_x}_{\text{voltage}} + v$$

Use KVL to find new equation

③

$$v_2 = \frac{v_1}{7.5} + v$$

Eliminate i_x to write new equation
in terms of node voltages

Answer

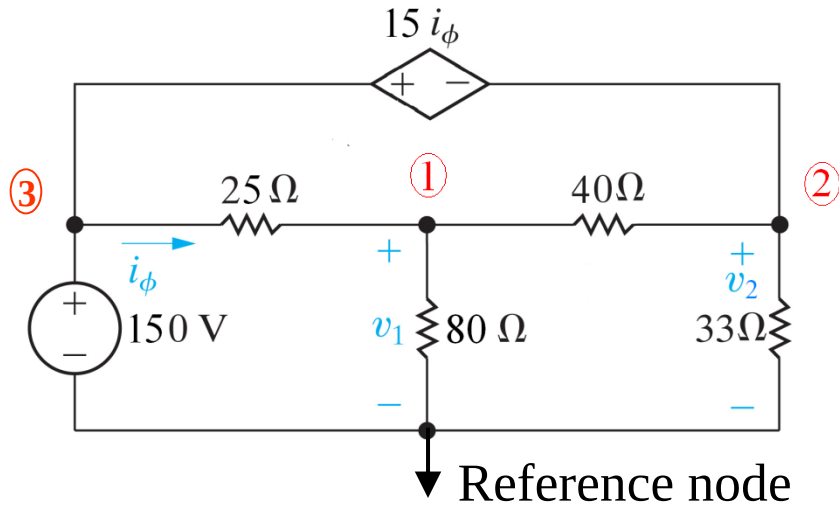


$$v = 8(V)$$

Solve 3 equations for 3 unknowns

①, ② and ③

Drill Exercise: Find v_1



$n_e = 4$ } 3 node equations
 { Dealing with a dependent source without using a supernode }

Notice: $v_3 = 150V \Rightarrow 2$ Unknowns

$\{ v_2 = -15i_\phi + 150 \}$ KVL around outer loop

$$i_\phi = \frac{150 - v_1}{25}$$

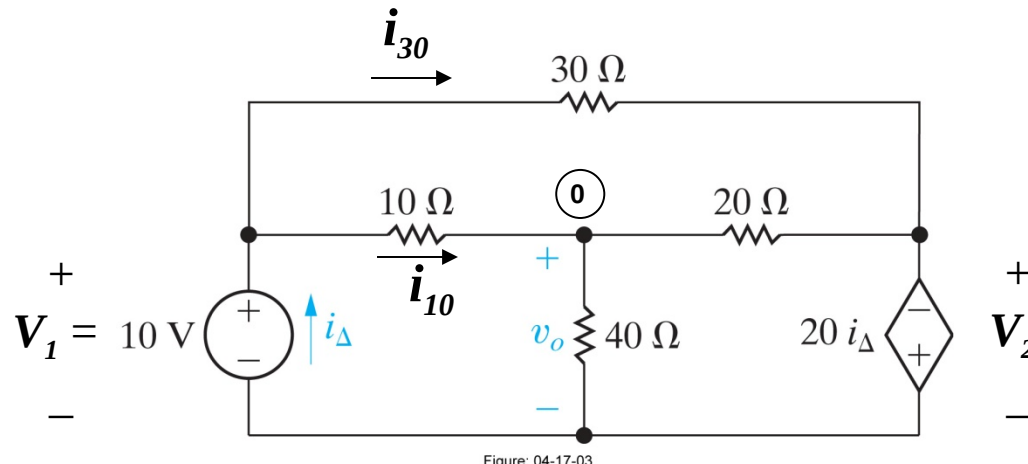
Ohm's Law at 25Ω resistor
 i_ϕ in terms of v_1

Node equation

Solved 1 equation for 1 unknown

$$\textcircled{1} \quad \frac{v_1 - 150}{25} + \frac{v_1}{80} + \frac{v_1 - v_2}{40} = 0 \Rightarrow \frac{v_1 - 150}{25} + \frac{v_1}{80} + \frac{v_1 - (150 - 15i_\phi)}{40} = 0 \Rightarrow v_1 = 120(V)$$

Drill Exercise: Find v_o



Need to account for the dependent source

Need equation for i_{Δ}

$n_e = 4$; 3 node voltages: v_1, v_2, v_o $v_1 = 10 \text{ V}$

$$i_{\Delta} = i_{10} + i_{30} \quad \text{KCL}$$

$$i_{\Delta} = \frac{10 - v_o}{10} + \frac{10 - v_2}{30}$$

$$v_2 = -20i_{\Delta} \quad (\text{dependent source})$$

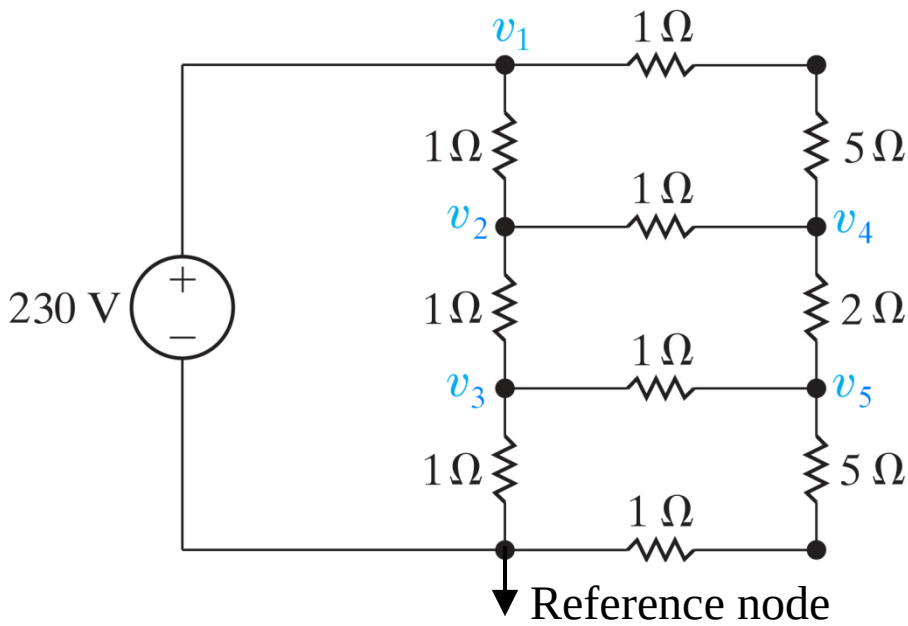
Write currents in terms of Node Voltages

Solve 3 equations for 3 unknowns

$$\frac{v_o - 10}{10} + \frac{v_o}{40} + \frac{v_o - v_2}{20} = 0$$

Node Equation at 0

Node Analysis Example



6 essential nodes; therefore, we must find 5 node voltage equations

Reference node at bottom is one of the node voltages

1) $v_1 = 230(V)$ } Due to voltage source

$$(2) \quad \frac{v_2 - 230}{1} + \frac{v_2 - v_4}{1} + \frac{v_2 - v_3}{1} = 0$$

$$(3) \quad \frac{v_3 - v_2}{1} + \frac{v_3 - v_5}{1} + \frac{v_3}{1} = 0$$

$$(4) \quad \frac{v_4 - 230}{(5+1)} + \frac{v_4 - v_2}{1} + \frac{v_4 - v_5}{2} = 0$$

$$(5) \quad \frac{v_5 - v_4}{2} + \frac{v_5 - v_3}{1} + \frac{v_5}{(5+1)} = 0$$

Simplifying and Putting Equations in Matrix Form

4 equations 4 unknowns

$$3v_2 - v_3 - v_4 + 0v_5 = 230$$

$$-v_2 + 3v_3 + 0v_4 - v_5 = 0$$

$$-6v_2 + 0v_3 + 10v_4 - 3v_5 = 230$$

$$0v_2 - 6v_3 - 3v_4 + 10v_5 = 0$$

$$A = \begin{vmatrix} 3 & -1 & -1 & 0 \\ -1 & 3 & 0 & -1 \\ -6 & 0 & 10 & -3 \\ 0 & -6 & -3 & 10 \end{vmatrix}$$

Put in Matrix Form

$$AX = Y$$

$$X = \begin{bmatrix} v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix} \quad Y = \begin{bmatrix} 230 \\ 0 \\ 230 \\ 0 \end{bmatrix}$$

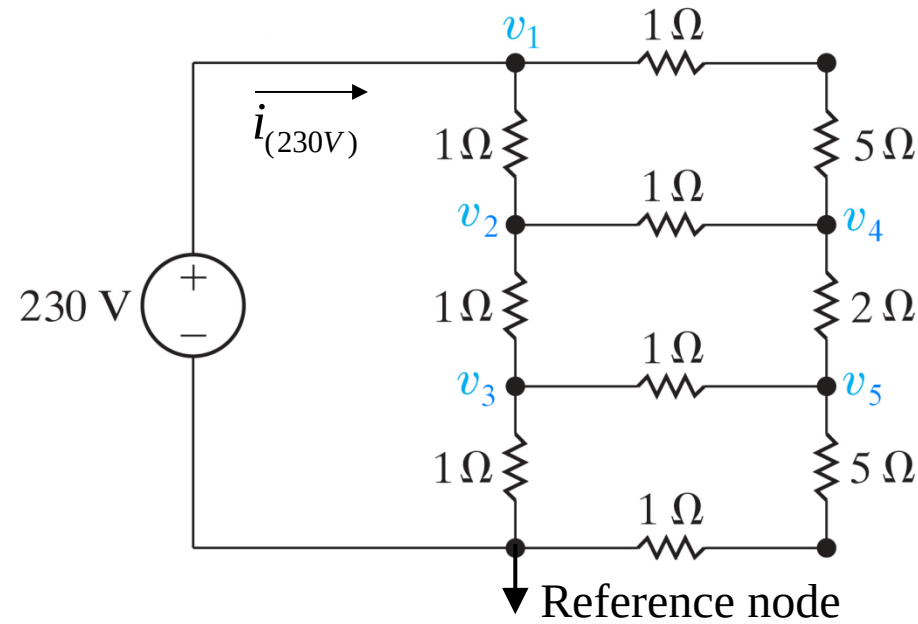
Solution $X = A^{-1}Y$

Calculator

$$X = \begin{bmatrix} 150 \\ 80 \\ 140 \\ 90 \end{bmatrix} \Rightarrow \left. \begin{array}{l} v_2 = 150 \\ v_3 = 80 \\ v_4 = 140 \\ v_5 = 90 \end{array} \right\}$$

**Node
Voltages**

Node Analysis Example (Contd.)



Find power supplied by the source

$i_{(230V)}$ = Current flowing into node (1)

$$i_{(230V)} = \frac{v_1 - v_2}{1} + \frac{v_1 - v_4}{5 + 1} \quad \left\{ \text{KCL} \right.$$

$$= \frac{230 - 150}{1} + \frac{230 - 140}{6} \quad \left\{ \begin{array}{l} v_2 = 150 \\ v_4 = 140 \end{array} \right.$$

$$= 80 + 15$$

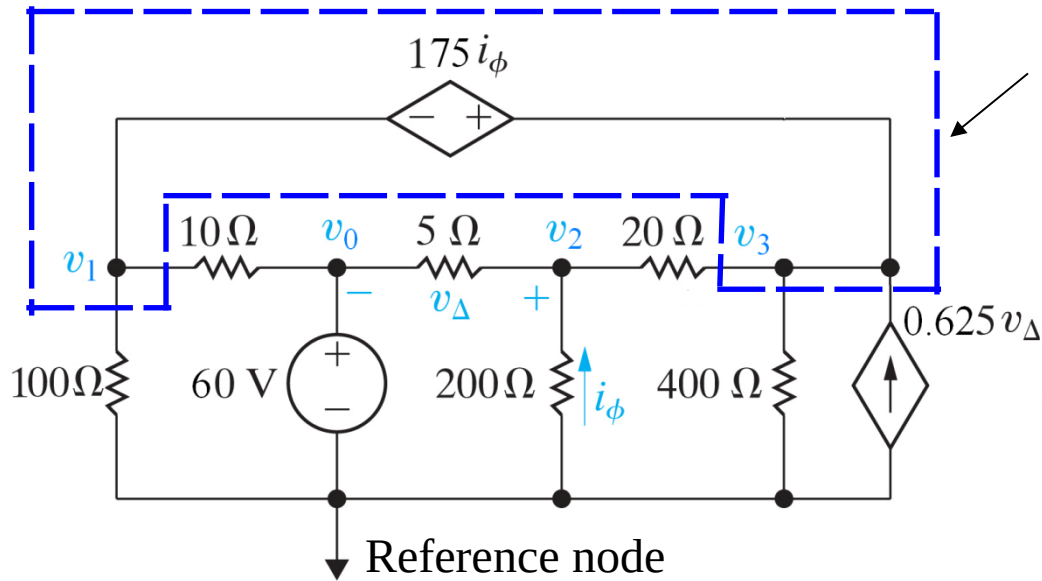
$$i_{(230V)} = 95(A)$$

$$p = Vi$$

$$p_{(230V)} = (230V)(95A) = 21,850 (W)$$

Supplied

Example: Find the Node Voltages



Supernode

5 Essential Nodes $\Rightarrow$ 4 Node Voltages

$v_0 = 60\text{ V}$; due to our choice of Reference Node.

Dependent voltage source between (1) and (3) creates a problem: make this a Supernode

Think of supernode as a “Blob”

$$\Sigma i \text{ out of the “blob”} = 0$$

Supernode

① — ③ Node Equation

$$\left\{ \frac{v_1}{100} + \frac{v_1 - 60}{10} + \frac{v_3 - v_2}{20} + \frac{v_3}{400} - 0.625v_\Delta = 0 \right.$$

Multiply by 400

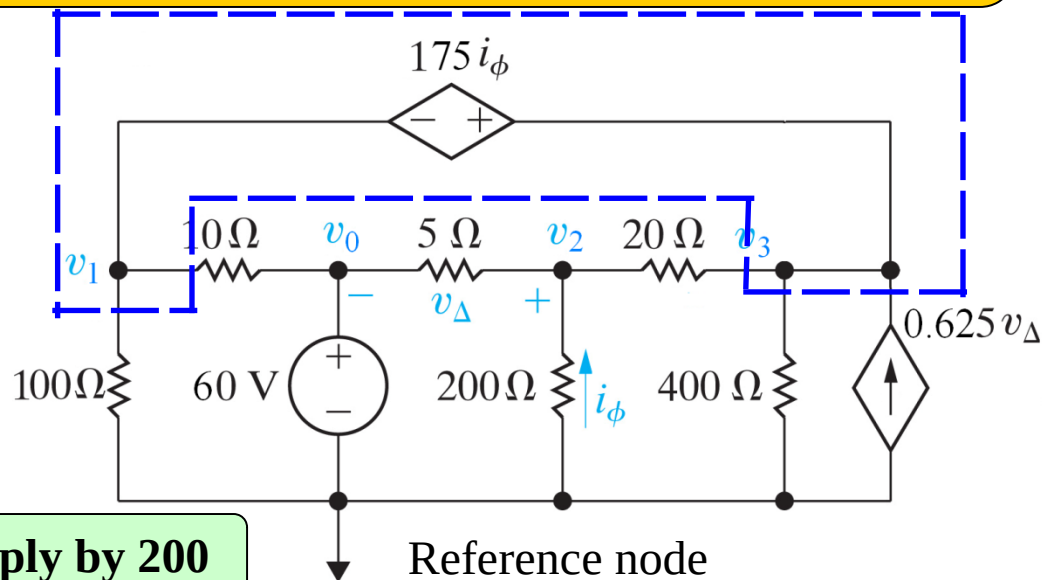
$$4v_1 + 40v_1 - 2400 + 20v_3 - 20v_2 + v_3 - 250v_\Delta = 0$$

Simplify

1

$$44v_1 - 20v_2 + 21v_3 - 250v_\Delta = 2400$$

Node Analysis Example (Contd.)



Node equation at (2)

$$\frac{v_2 - 60}{5} + \frac{v_2}{200} + \frac{v_2 - v_3}{20} = 0$$

Multiply by 200

$$40v_2 - 2400 + v_2 + 10v_2 - 10v_3 = 0$$

Simplify

$$51v_2 - 10v_3 = 2400 \quad [2]$$

$$44v_1 - 20v_2 + 21v_3 - 250v_\Delta = 2400 \quad [1]$$

$$v_\Delta = v_2 - 60 \quad [3] \quad \text{KVL}$$

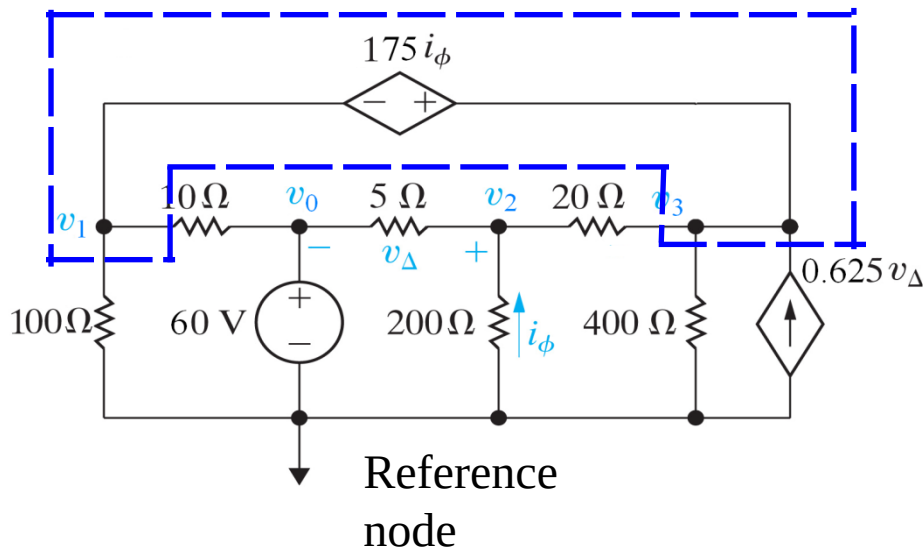
2 equations

4 unknowns

Need two more equations

Node Analysis Example (Contd.)

Since there is a dependent voltage source between ① and ③, we can relate v_1 to v_3



a $v_2 = -200i_\phi$ } Ohm's Law

b $v_1 = -175i_\phi + v_3$ } KVL

$$\therefore v_1 = \frac{7}{8}v_2 + v_3$$

4 } Eliminate i_ϕ from b

Plugging 3 and 4 into 1, we obtain

$$-231.5v_2 + 65v_3 = -12,600$$

5

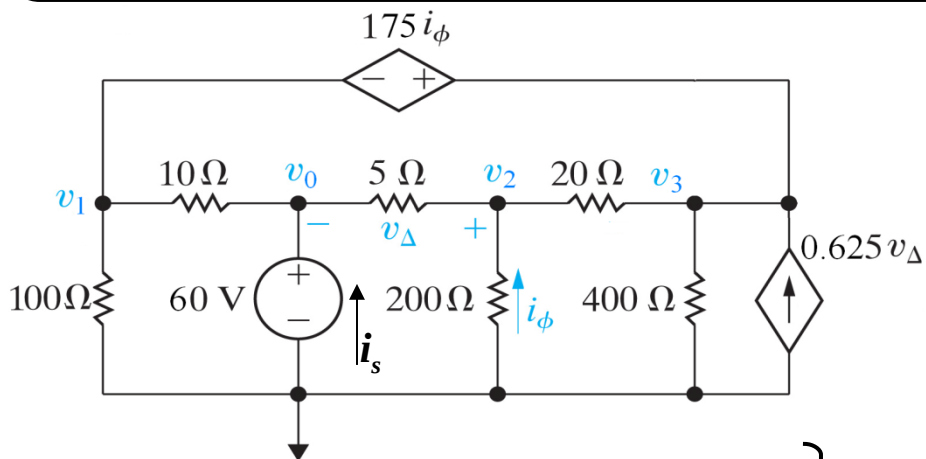
Solving 2 and 5 simultaneously:

$$v_2 = 30V$$

$$v_3 = -87V$$

$$4 \Rightarrow v_1 = -60.75(V)$$

Node Analysis Example (Contd.)



**Find Power of
the 60V source**

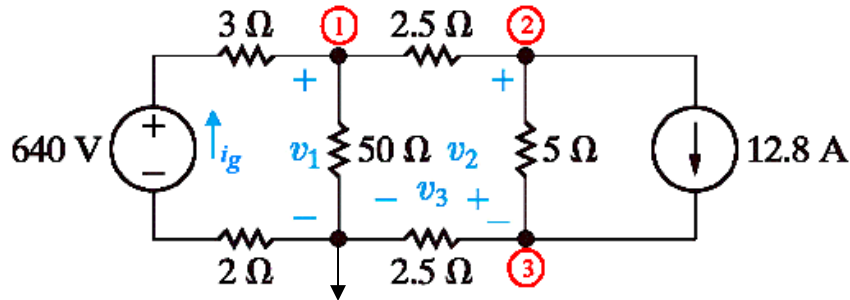
$$p_{60V} = 60V \cdot (i_s) \quad \left. \vphantom{p_{60V}} \right\} \text{Power Equation}$$

$$i_s = \frac{v_0 - v_1}{10} + \frac{v_0 - v_2}{5} \quad \left. \vphantom{i_s} \right\} \text{KCL } \Sigma i_{in} = \Sigma i_{out} \text{ and Ohm's Law}$$

$$i_s = \frac{60 - (-60.75)}{10} + \frac{60 - 30}{5} = 18.075 (A) \quad \left. \vphantom{i_s} \right\} \text{Plug in numbers}$$

$$p_{60V} = 60V \cdot (18.075) = 1084.5 (W) \quad \left. \vphantom{p_{60V}} \right\} \text{Power Delivered}$$

Drill Exercise: Find Node Voltages



$$n_e = 4$$

3 equations

$$\begin{aligned} \textcircled{1} \quad & \frac{v_1 - 640}{3 + 2} + \frac{v_1}{50} + \frac{v_1 - v_2}{2.5} = 0 \\ \textcircled{2} \quad & \frac{v_2 - v_1}{2.5} + \frac{v_2 - v_3}{5} + 12.8 = 0 \\ \textcircled{3} \quad & \frac{v_3}{2.5} + \frac{v_3 - v_2}{5} - 12.8 = 0 \end{aligned}$$

Simplify

$$\begin{aligned} \textcircled{1} \quad & 31v_1 - 20v_2 + 0v_3 = 6400 \quad \left\{ \begin{array}{l} \text{(multiply by 50)} \\ \text{(multiply by 5)} \\ \text{(multiply by 5)} \end{array} \right. \\ \textcircled{2} \quad & -2v_1 + 3v_2 - v_3 = -64 \\ \textcircled{3} \quad & 0v_1 - v_2 + 3v_3 = 64 \end{aligned}$$

Solution

$$v_1 = 380(V); \quad v_2 = 269(V); \quad v_3 = 111(V)$$

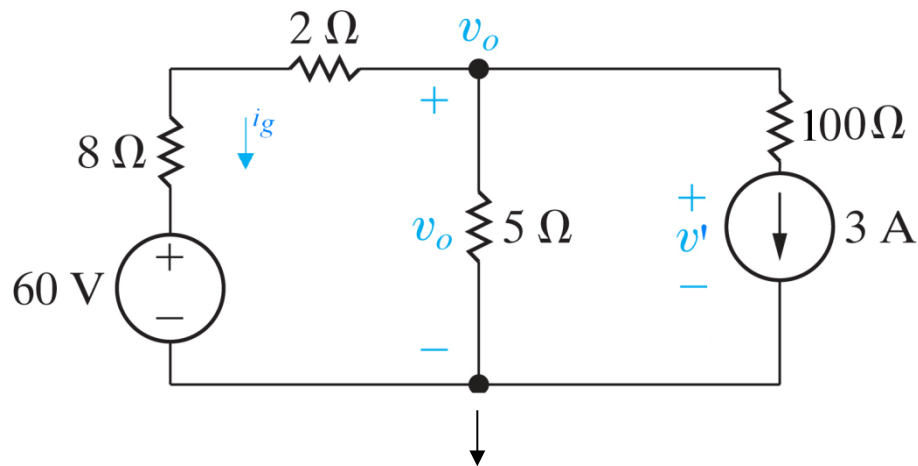
Find power of 640V source

$$i_g = \frac{640 - v_1}{3 + 2} = \frac{640 - 380}{5} = 52(A)$$

$$p = vi_g = 640(52) = 32,280(W)$$

Power Delivered

Drill Exercise: Find Power of the Sources



a) 1 node equation

$$\frac{v_o - 60}{8 + 2} + \frac{v_o}{5} + 3 = 0$$

$$v_o - 60 + 2v_o + 30 = 0$$

➡ **$v_o = 10(V)$**

b) Find p_{3A} (delivered) **Find v'**

$$\left\{ \begin{array}{l} v_o = v_{100\Omega} + v' = 300 + v' \\ v' = v_o - 300 = 10 - 300 \end{array} \right\} \begin{array}{l} \text{Substitute } v_{100\Omega} = 300(V) \\ \text{Simplify} \end{array}$$

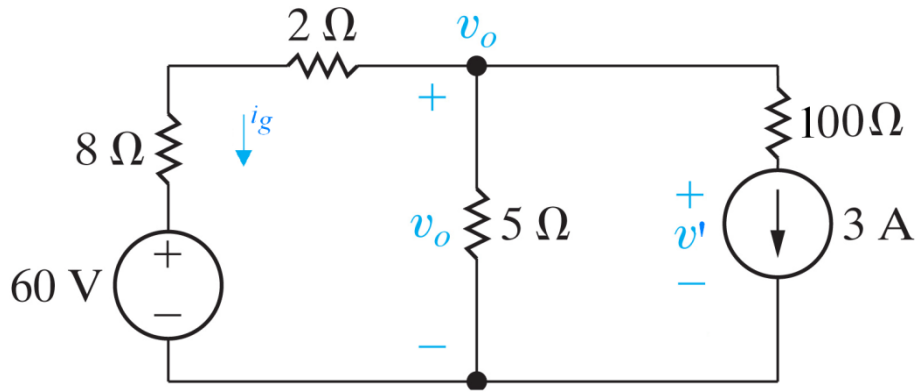
$v' = -290(V)$

$p_{3A} = v' I$ } **Passive sign convention**

$p_{3A} = -290(3) = -870(W)$ } $p < 0$ Power extracted from

$p_{3A} = +870(W)$ *Delivered*

Drill Exercise (Contd.)



c) Find $p_{60V} = (60)i_g$ } **Passive sign convention**

Find $i_g \Rightarrow i_g = \frac{v_o - 60}{8 + 2} = \frac{10 - 60}{8 + 2} = -\frac{50}{10} = -5(A)$

$p_{60V} = 60(-5) = -300(W)$ } $p < 0$ Power extracted from

$p_{60V} = 300(W)$ } **Delivered**

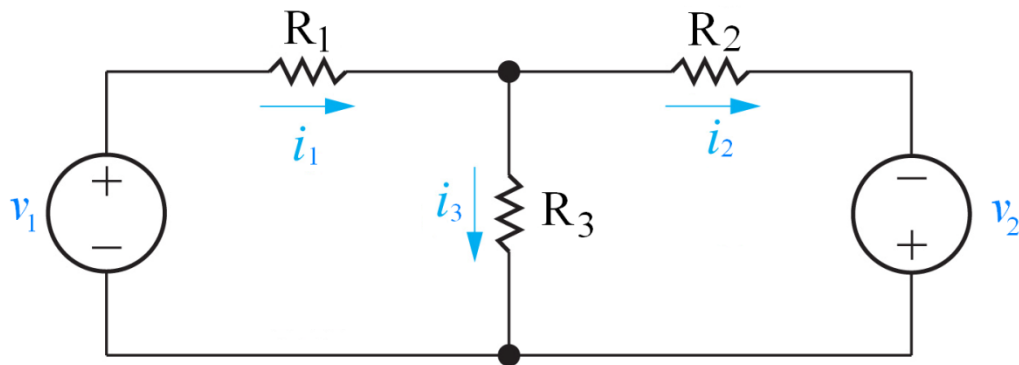
$\Sigma p_{dev} = 870 + 300 = 1170(W)$ } **Total Power Delivered**

Mesh Current Analysis: Planar Circuits

Mesh: Loop with no other loops inside

The circuit can be described by: $b_e - (n_e - 1)$ equations.

Don't worry about this definition.
Need an equation; therefore a mesh current, for each mesh



- i_1, i_2, i_3 branch currents
- 3 unknowns

2 equations 2 unknowns

KVL $\left\{ \begin{array}{l} v_1 = i_1 R_1 + i_3 R_3 \\ v_2 = i_2 R_2 - i_3 R_3 \end{array} \right\}$ **2 equations**
3 unknowns

KCL $\left\{ i_3 = i_1 - i_2 \right\}$ **Additional equation**

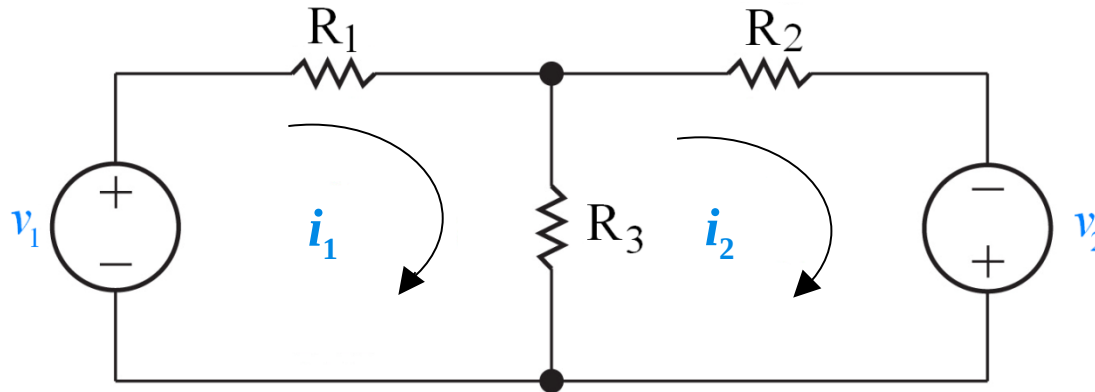
Mesh Equations

$$v_1 = i_1 (R_1 + R_3) - i_2 R_3 \quad (1)$$

$$v_2 = -i_1 R_3 + i_2 (R_2 + R_3) \quad (2)$$

Mesh Currents

Mesh Current: Current flowing only along the perimeter of a mesh

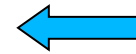
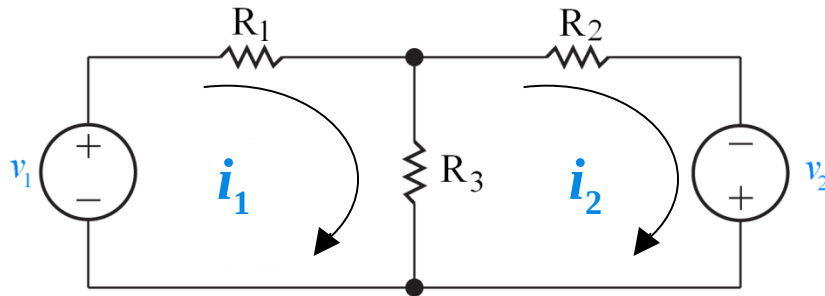


i_1 and i_2 are the mesh currents

m meshes $\Rightarrow$ **m** currents $\Rightarrow$ **m** equations!

Mesh Current

Current flowing only along the perimeter of a mesh



i_1, i_2 Mesh Currents

m meshes $\Rightarrow$ **m** currents $\Rightarrow$ **m** equations

Current through any branch is determined by considering the mesh currents flowing in every mesh in which the branch belongs

• R_1 only in Mesh 1

i_1 identified as branch current

• R_2 only in Mesh 2
current

i_2 identified as branch

• R_3 in both the Meshes

$(i_1 - i_2)$ or $(i_2 - i_1)$ is branch current

Mesh current into a node also flows out of it

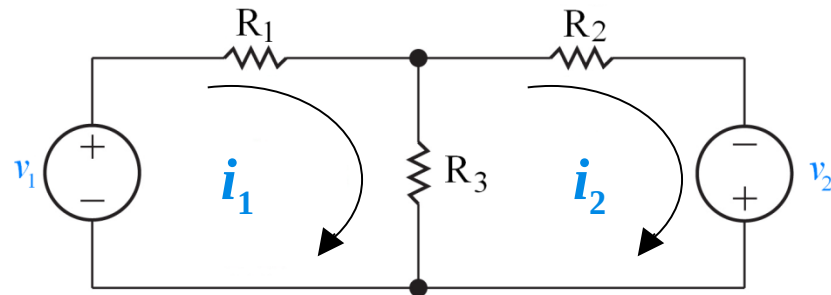
KCL is Automatically Satisfied

Mesh Analysis

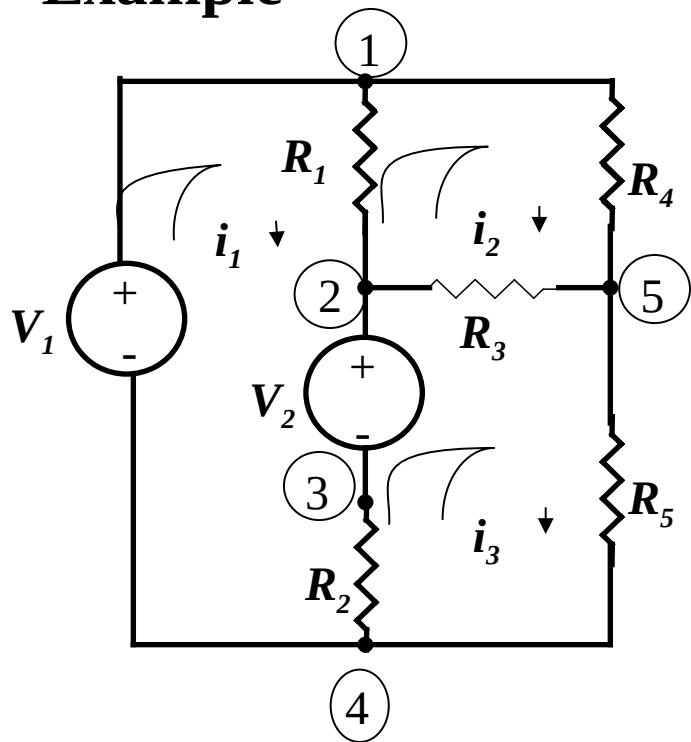
To analyze a circuit, use KVL

$$\begin{aligned} \textcircled{1} \quad V_1 &= i_1 R_1 + (i_1 - i_2) R_3 \\ \textcircled{2} \quad V_2 &= i_2 R_2 + (i_2 - i_1) R_3 \end{aligned}$$

*Mesh
equations
found using
KVL*

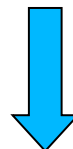


Example



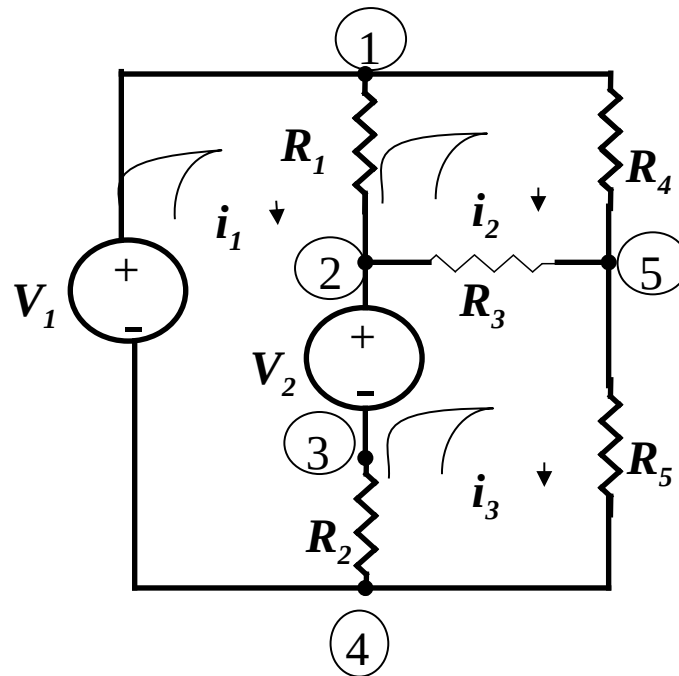
**How many meshes and
equations in this circuit?**

3 meshes



3 equations

Example: Mesh Equations (Contd.)



Mesh 1

$$V_1 = (i_1 - i_2)R_1 + V_2 + (i_1 - i_3)R_2$$

i_1 is positive in mesh 1

Mesh 2

$$i_2R_4 + (i_2 - i_3)R_3 + (i_2 - i_1)R_1 = 0$$

i_2 is positive in mesh 2

Mesh 3

$$V_2 = (i_3 - i_2)R_3 + i_3R_5 + (i_3 - i_1)R_2$$

i_3 is positive in mesh 3

Example: Mesh Equations

$$V_1 - V_2 = (i_1 - i_2)R_1 + (i_1 - i_3)R_2 \quad \boxed{1}$$

$$0 = (i_2 - i_1)R_1 + (i_2 - i_3)R_3 + i_2 R_4 \quad \boxed{2}$$

$$V_2 = (i_3 - i_1)R_2 + (i_3 - i_2)R_3 + i_3 R_5 \quad \boxed{3}$$

Note:

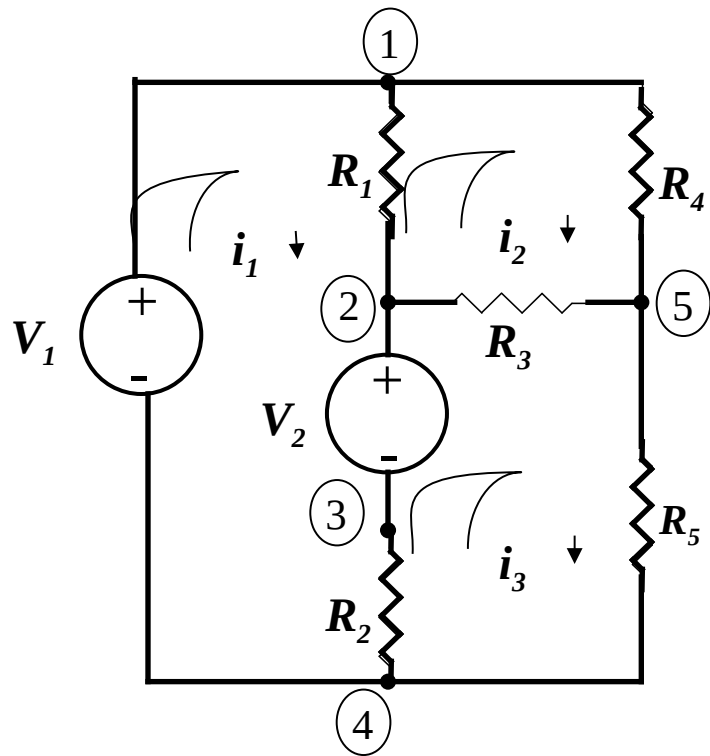
R_1 in meshes $\boxed{1}$ & $\boxed{2}$ $\longrightarrow i_1$ & i_2 R_2 in meshes $\boxed{1}$ & $\boxed{3}$ $\longrightarrow i_1$ & i_3

R_3 in meshes $\boxed{2}$ & $\boxed{3}$ $\longrightarrow i_2$ & i_3 R_4 in mesh $\boxed{2}$ *only* $\longrightarrow i_2$

R_5 in mesh $\boxed{3}$ *only* $\longrightarrow i_3$

Solving Simultaneous Equations

Previous Example with Numbers



$$V_1 = 7V$$

$$V_2 = 6V$$

$$R_1 = 1\Omega$$

$$R_2 = 2\Omega$$

$$R_3 = 3\Omega$$

$$R_4 = 2\Omega$$

$$R_5 = 1\Omega$$

$$V_1 = (i_1 - i_2)R_1 + V_2 + (i_1 - i_3)R_2$$

1

$$0 = i_2R_4 + (i_2 - i_3)R_3 + (i_2 - i_1)R_1$$

2

$$V_2 = (i_3 - i_2)R_3 + i_3R_5 + (i_3 - i_1)R_2$$

3

Put in matrix form

$$(R_1 + R_2)i_1 - R_1i_2 - R_2i_3 = V_1 - V_2$$

$$(-R_1)i_1 + (R_1 + R_3 + R_4)i_2 - R_3i_3 = 0$$

$$(-R_2)i_1 - R_3i_2 + (R_2 + R_3 + R_5)i_3 = V_2$$

Separate the currents out

Solving Simultaneous Equations

Previous Example with Numbers

Plug Numbers Into the Matrix Form

$$3i_1 - i_2 - 2i_3 = 1$$

$$-i_1 + 6i_2 - 3i_3 = 0$$

$$-2i_1 - 3i_2 + 6i_3 = 6$$

**3 Equations
3 Unknowns
How do we solve?**

Put in Matrix Notation

$$\begin{bmatrix} 3 & -1 & -2 \\ -1 & 6 & -3 \\ -2 & -3 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix}$$

**3x3 Square Matrix Of
Coefficients**

**Column
Variables**

**Column
Variables**

Solve The Matrix Equation

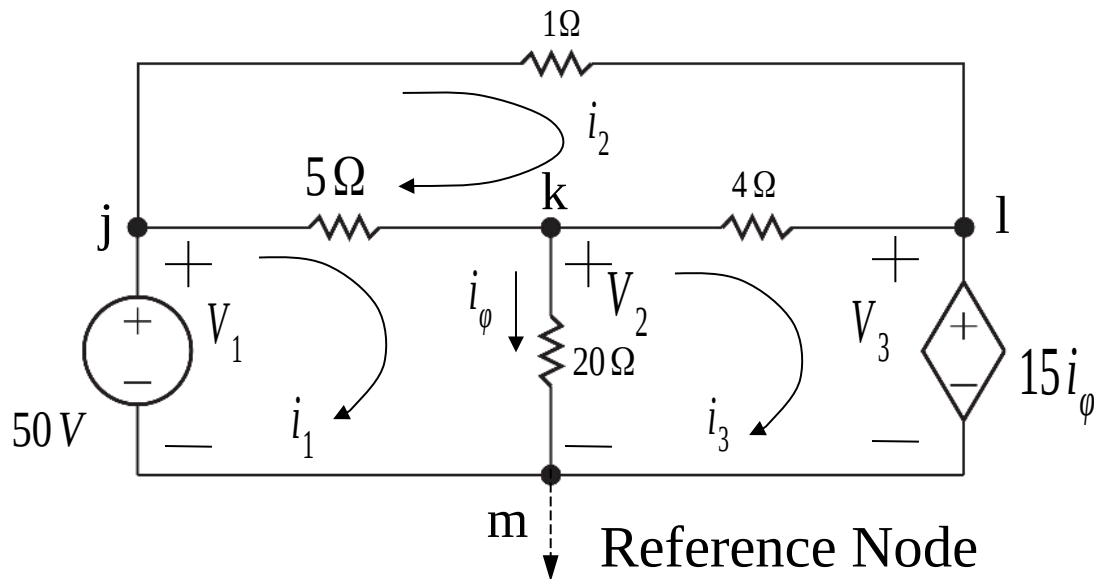
$$AX = Y \quad A = \begin{bmatrix} 3 & -1 & -2 \\ -1 & 6 & -3 \\ -2 & -3 & 6 \end{bmatrix} \quad X = \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 0 \\ 6 \end{bmatrix}$$

$$X = A^{-1}Y \quad \left. \vphantom{X = A^{-1}Y} \right\} \text{Use Calculator}$$

$$X = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix} \quad \left. \vphantom{X = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}} \right\} \text{Answer in vector form}$$

$$i_1 = 3 \text{ (A)} \quad i_2 = 2 \text{ (A)} \quad i_3 = 3 \text{ (A)}$$

Dependent Sources AND Choosing Nodal Or Mesh Method: Find $p_{4\Omega}$



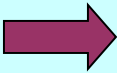
4 essential nodes

3 meshes

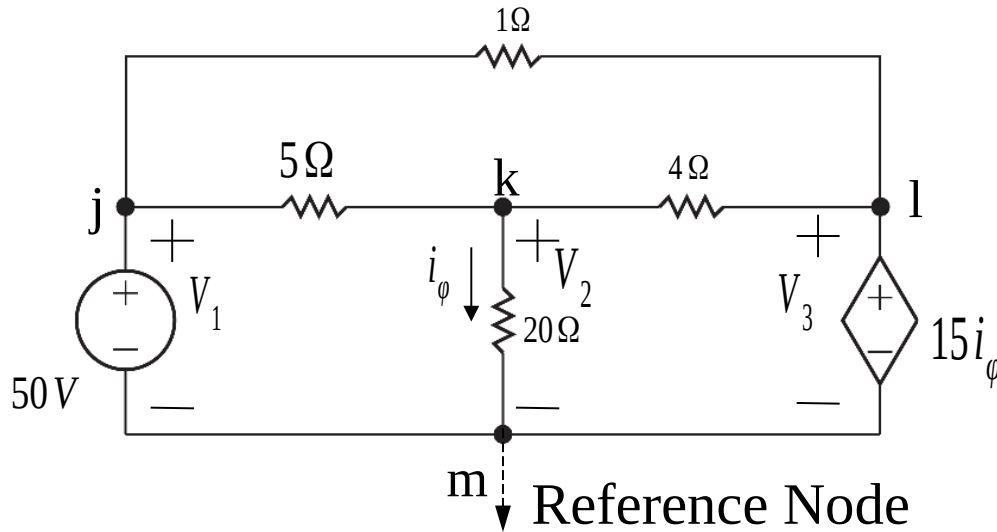
Dependent Voltage
Source

Book uses Mesh analysis



3 meshes  3 equations

Easier to use Node Voltage method



3 node voltages; however,
 $V_1 = 50(V)$ Independent Voltage Source
 $V_3 = 15i_\phi$ Dependent Voltage Source

node (k)

$$\left\{ \frac{V_2 - 50}{5} + \frac{V_2}{20} + \frac{V_2 - 15i_\phi}{4} = 0 \right\} \xrightarrow{\text{Multiply by 20}} 4V_2 - 200 + V_2 + 5V_2 - 75i_\phi = 0$$

$$10V_2 - 75i_\phi = 200 \quad \left\{ \begin{array}{l} \text{Simplified Equation} \end{array} \right.$$

Ohms Law

$$\left\{ i_\phi = \frac{V_2}{20} \Rightarrow 10V_2 - \frac{75}{20}V_2 = 200 \right.$$

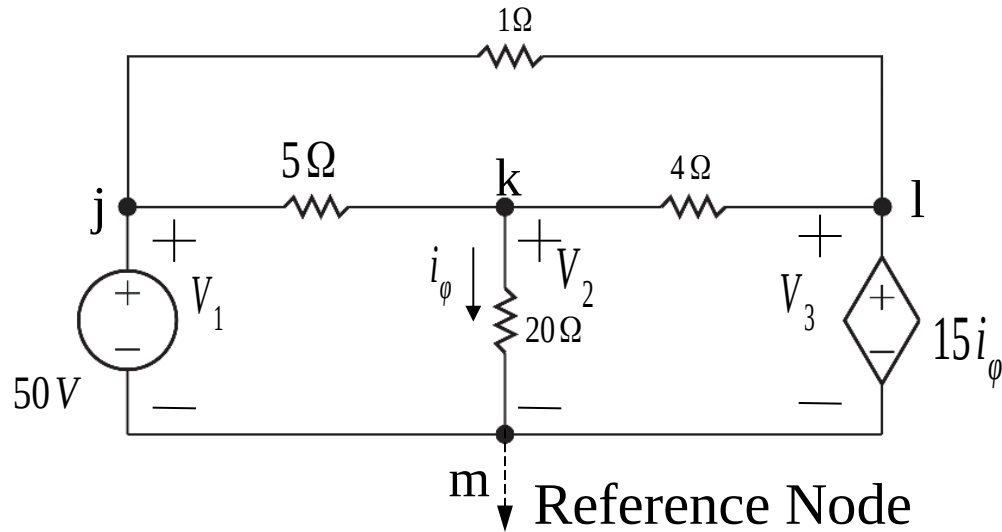
Eliminate i_ϕ

$$V_2 = 32(V)$$

$$i_\phi = 1.6(A)$$

$$V_3 = 15i_\phi = 24(V)$$

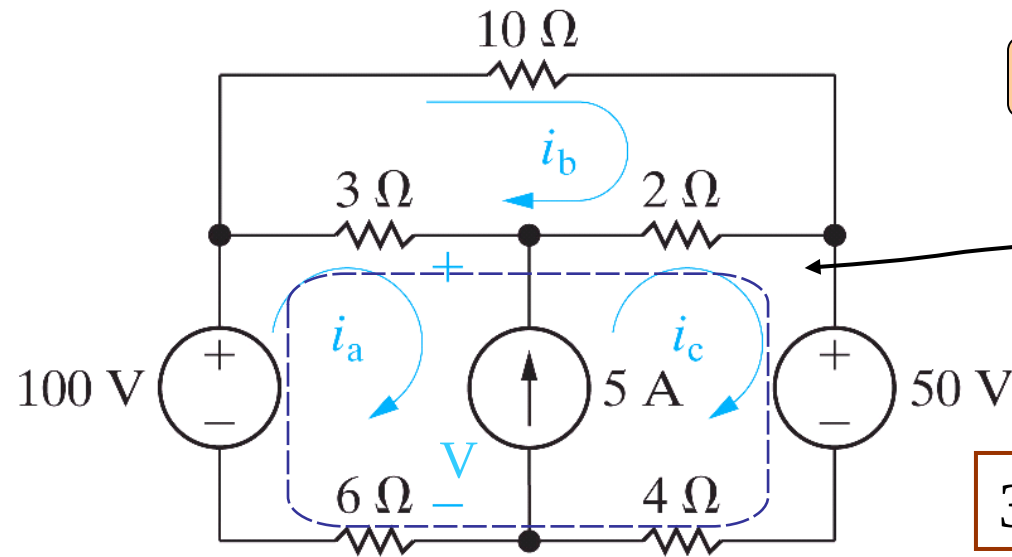
Easier to use Node Voltage method (Contd.)



$$p_{4\Omega} = \frac{V_{4\Omega}^2}{4} \quad \text{where } V_{4\Omega} = V_2 - V_3 \quad \left. \vphantom{p_{4\Omega}} \right\} \text{KVL}$$

$$p_{4\Omega} = \frac{(V_2 - V_3)^2}{4} = \frac{(32 - 24)^2}{4} \Rightarrow p_{4\Omega} = 16(W)$$

Mesh analysis with a Current Source



Supermesh

3 “ways” to handle this situation

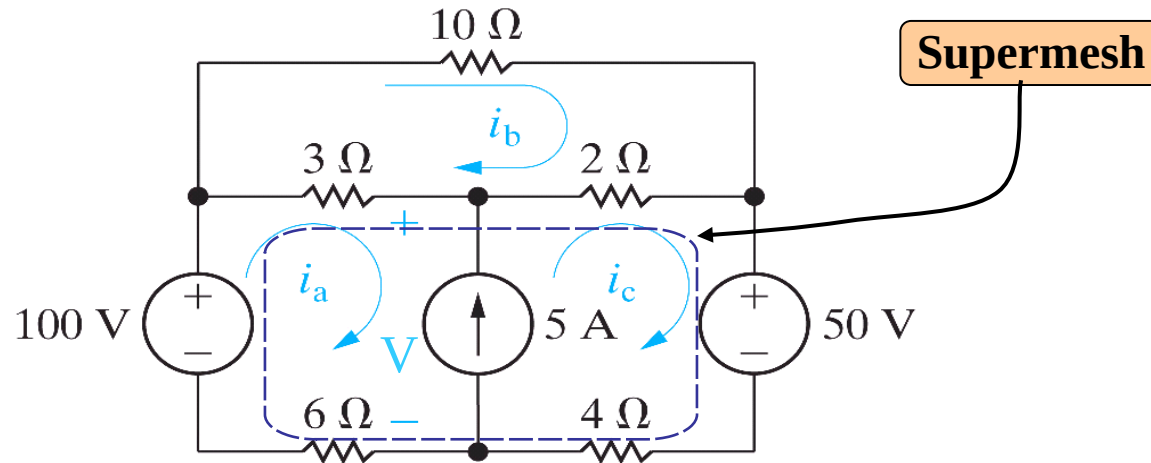
- ① Mesh analysis via KVL; however, we must

Introduce unknown voltage, V , across the current source.

KCL

Note that there is a constraint on i_a and i_c ($i_c = 5(A) + i_a$)

Mesh analysis with a Current Source (Contd.)



**M
E
S
H**

$$\left\{ \begin{array}{l} \text{a} \quad 100 = 3(i_a - i_b) + V + 6i_a \\ \text{c} \quad -50 = 2(i_c - i_b) + 4i_c - V \\ \text{b} \quad 0 = 10i_b + 2(i_b - i_c) + 3(i_b - i_a) \\ \quad i_c = 5(\text{A}) + i_a \end{array} \right. \quad \text{Constraint}$$

Solve 4 Equations for 4 unknowns

Can solve these equations for i_a , i_b , i_c and V

Mesh analysis with a Current Source

(Contd.)

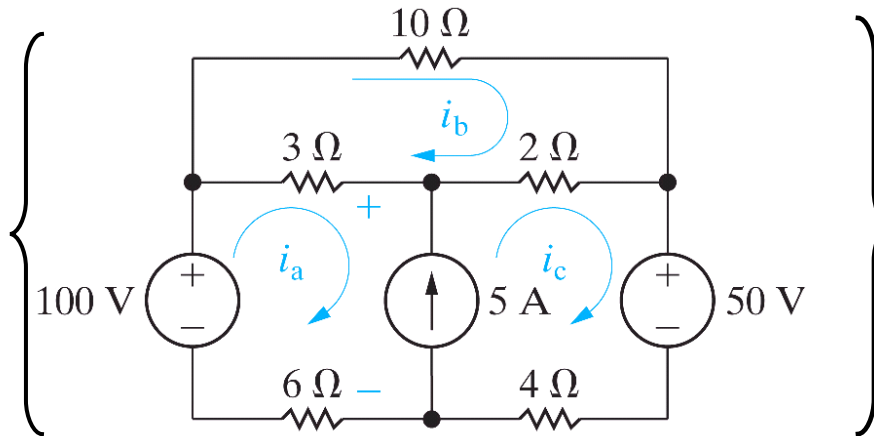
- ② **Supermesh:** Use KVL around supermesh, in terms of original mesh currents.

- ③ **Just Sum Voltages Around the Outer “Loop”**

This is equivalent to a Supermesh

Mesh analysis with a Current Source (Contd.)

New approach
to solve the
same problem



Write KVL around
loops that excludes
the current source

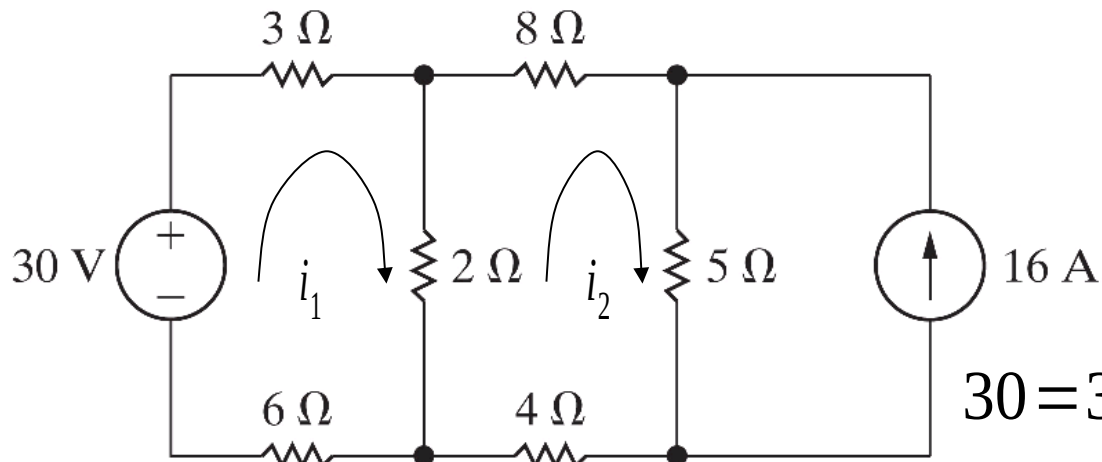
$$100 = 3(i_a - i_b) + 2(i_c - i_b) + 50 + 4i_c + 6i_a \quad \text{KVL around supermesh}$$

$$10i_b + 2(i_b - i_c) + 3(i_b - i_a) = 0 \quad \text{KVL around mesh b}$$

$$i_c = 5 + i_a \quad \text{Constraint equation}$$

3 Equations, 3 unknowns, easy to solve

Drill Exercise: Find $p_{2\Omega}$



$$\left. \begin{aligned} 30 &= 3i_1 + 2(i_1 - i_2) + 6i_1 \\ 30 &= 11i_1 - 2i_2 \end{aligned} \right\} \begin{array}{l} \text{Mesh } i_1 \\ \text{Simplified equation} \end{array}$$

$$\left. \begin{aligned} 8i_2 + 5(i_2 + 16) + 4i_2 + 2(i_2 - i_1) &= 0 \\ -80 &= -2i_1 + 19i_2 \end{aligned} \right\} \begin{array}{l} \text{Mesh } i_2 \\ \text{Simplified equation} \end{array}$$

$$\left. \begin{array}{l} \text{Find Power} \\ p_{2\Omega} = (i_1 - i_2)^2 2 = (2 + 4)^2 2 = 72 \text{ (W)} \end{array} \right\}$$

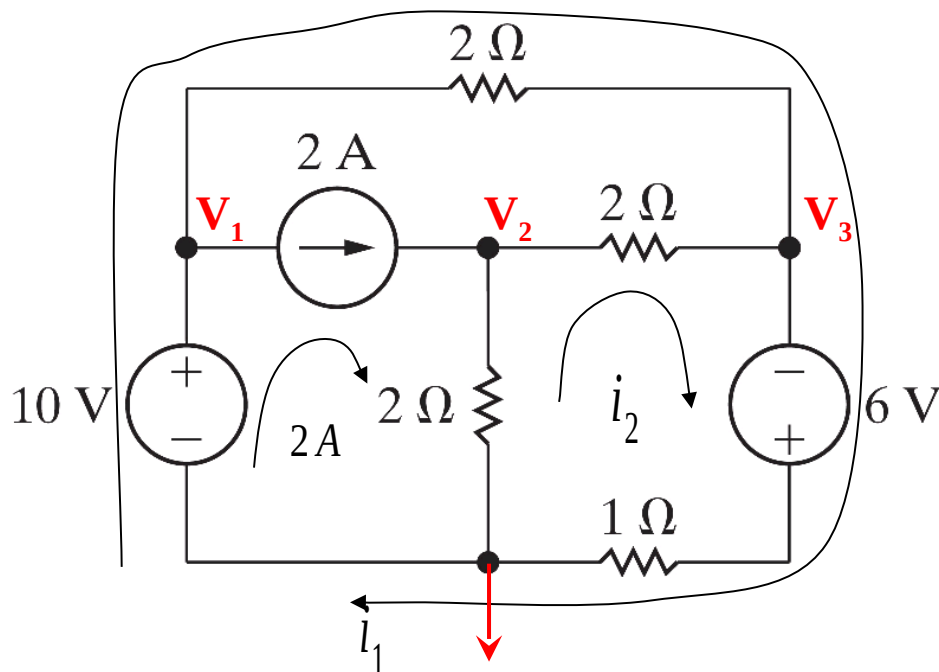
$$\begin{aligned} i_1 &= 2 \text{ (A)} \\ i_2 &= -4 \text{ (A)} \end{aligned}$$

Solve
2 equations
for
2 unknowns

Nodal Versus Mesh Analysis

- 1) Which method yields the least number of equations?
- 2) Supernodes: Voltage source between essential nodes
- 3) Supermesh: Use loop excluding any current source which is part of 2 meshes

Drill Exercise



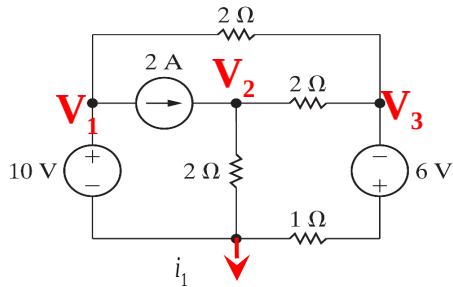
**Use Node
or
Mesh**

Only 2 mesh currents unknown. <3 meshes.>

However, 1 node voltage is also known.

How Would You Setup Nodal Equations?

Drill Exercise (Contd.)



(a)

$$-2 + \frac{V_2}{2} + \frac{V_2 - V_3}{2} = 0$$

Node Equation at V_2

(a)

$$-4 + V_2 + V_2 - V_3 = 0$$

Multiply by 2

(a)

$$2V_2 - V_3 = 4$$

Simplify

(b)

$$\frac{V_3 - V_2}{2} + \frac{V_3 - 10}{2} + \frac{V_3 + 6}{1} = 0$$

Node Equation at V_3

(b)

$$V_3 - V_2 + V_3 - 10 + 2V_3 + 12 = 0$$

Multiply by 2

(b)

$$-V_2 + 4V_3 = -2 \quad \longrightarrow \quad V_2 = 4V_3 + 2$$

Simplify

Drill Exercise (Contd.)

$$2(4V_3 + 2) - V_3 = 4 \quad \left. \vphantom{2(4V_3 + 2) - V_3 = 4} \right\} \text{Substitute } \textcircled{b} \text{ into } \textcircled{a}$$

$$8V_3 + 4 - V_3 = 4 \quad \left. \vphantom{8V_3 + 4 - V_3 = 4} \right\} \text{Simplify}$$

$$7V_3 = 4 - 4 \quad \left. \vphantom{7V_3 = 4 - 4} \right\} \text{Simplify}$$

$$V_3 = 0$$

$$V_2 = 4(0) + 2 \quad \left. \vphantom{V_2 = 4(0) + 2} \right\} \text{Substitute } V_3 = 0 \text{ into } \textcircled{b}$$

$$V_2 = 2(V)$$

Drill Exercise

Solution Manual Method for the same problem

- Supermesh (outer loop) current $\equiv i_1$
- Mesh currents $\equiv i_2$ and 2A source
- No mesh current defined in “Top” loop

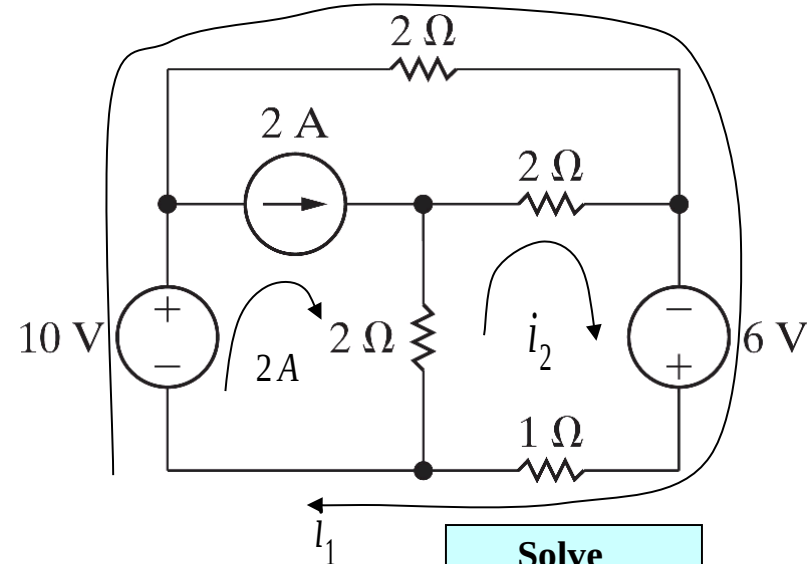
Outer loop (Supermesh)

$$(1) \quad 10 = 2i_1 - 6 + 1(i_1 + i_2) \quad \left. \vphantom{10 = 2i_1 - 6 + 1(i_1 + i_2)} \right\} \text{KVL}$$

$$(1) \quad 3i_1 + i_2 = 16 \quad \left. \vphantom{3i_1 + i_2 = 16} \right\} \text{Simplify}$$

$$(2) \quad 6 = 1(i_1 + i_2) + 2(i_2 - 2) + 2i_2 \quad \left. \vphantom{6 = 1(i_1 + i_2) + 2(i_2 - 2) + 2i_2} \right\} \text{KVL for Mesh 2}$$

$$(2) \quad i_1 + 5i_2 = 10 \quad \left. \vphantom{i_1 + 5i_2 = 10} \right\} \text{Simplify}$$



Solve
2 equations
for
2 unknowns

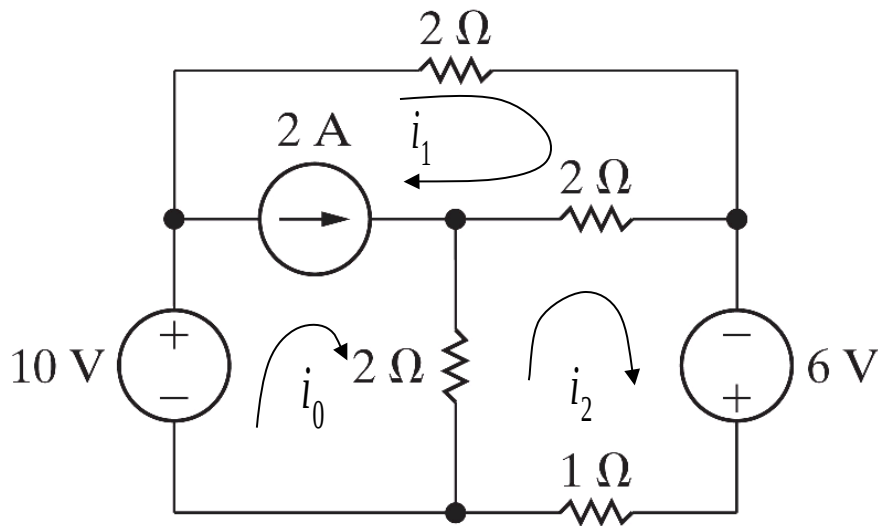
Solution

$$i_1 = 5(A)$$

$$i_2 = 1(A)$$

Drill Exercise

Third Method for the same problem



2A current source presents
a difficulty

Use “Supermesh” or “outer”
loop

Define 3 mesh currents, as shown above

Write KVL Loop Equation Around “Outer Loop”

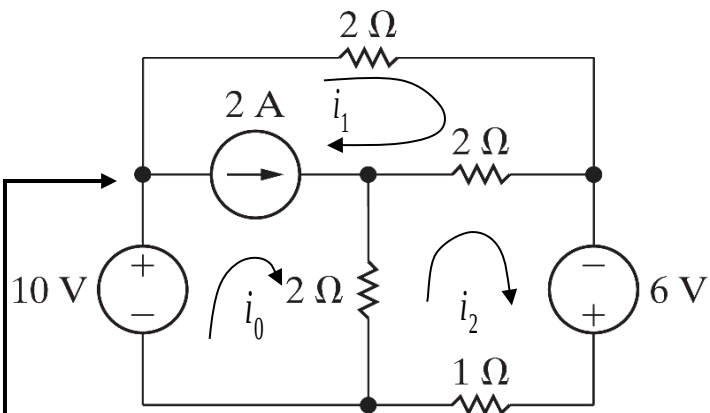
$$(1) \quad 10 = 2i_1 - 6 + 1i_2$$

$$(1) \quad 2i_1 + 1i_2 = 16$$

} **Simplify**

Drill Exercise (Contd.)

Third Method for the same problem



KVL Around Mesh “2”

$$(2) \quad 6 = 1i_2 + 2(i_2 - i_0) + 2(i_2 - i_1) \quad \left. \vphantom{6 = 1i_2 + 2(i_2 - i_0) + 2(i_2 - i_1)} \right\} \text{KVL}$$

$$(2) \quad 6 = 5i_2 - 2i_0 - 2i_1 \quad \left. \vphantom{6 = 5i_2 - 2i_0 - 2i_1} \right\} \text{Simplify}$$

$$(2) \quad 2i_0 + 2i_1 - 5i_2 = -6 \quad \left. \vphantom{2i_0 + 2i_1 - 5i_2 = -6} \right\} \text{Simplify}$$

Since we don't know the voltage across the 2A current source, we can not write a KVL for Mesh “0”.

2 Equations, 3 Unknowns. Need another equation.
Use **KCL** to relate Mesh Currents.

$$(3) \quad i_0 = 2 + i_1 \quad \left. \vphantom{i_0 = 2 + i_1} \right\} \text{Constraint}$$

Solve
3 equations
for
3 unknowns

Drill Exercise (Contd.)

Third Method for the same problem

So, our equations are

$$2i_1 + i_2 = 16 \quad \textcircled{1}$$

$$2i_0 + 2i_1 - 5i_2 = -6 \quad \textcircled{2}$$

$$i_0 = i_1 + 2 \quad \textcircled{3}$$

$$\textcircled{3} \longrightarrow \textcircled{2} \rightarrow 4i_1 - 5i_2 = -10 \quad \textcircled{4}$$

Solving $\textcircled{1}$ and $\textcircled{4}$ simultaneously

$$i_1 = 5(A)$$

$$i_2 = 6(A)$$

$$i_0 = i_1 + 2 = 5 + 2$$

$$i_0 = 7(A)$$

Drill Exercise Comparison

Third Method Versus Solution Manual Method

$$i_1 = 5(A)$$

$$i_2 = 1(A)$$

- Same

- Different

This is OK

Note: Current Through 10(V) Source is 7(A)

Solution

Manual: $\underline{I_{10V}} = i_1 + 2A = 5 + 2 = \underline{7(A)}$

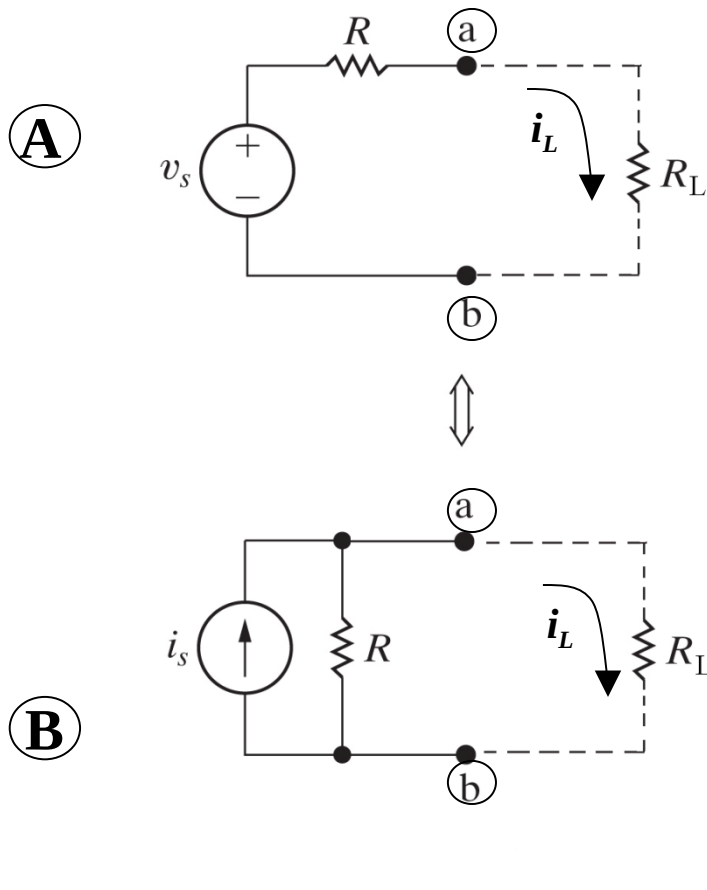
Third

Method: $\underline{I_{10V}} = i_0 = \underline{7(A)}$

All Net Currents Are
Consistent

Source Transformation

- Another Simplification Technique
- Analogous to Combining Resistors

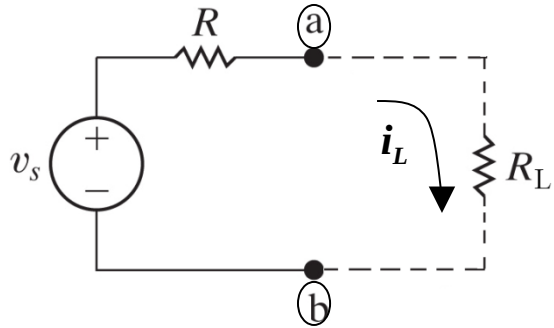


Equivalence:

If R_L “see’s” the same current flow and voltage drop in either circuit.

What is the Relationship Between V_s and i_s ?

Circuit (A)

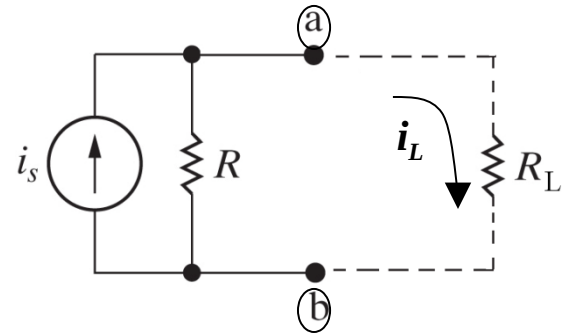


$$V_s = i_L (R + R_L) \quad \left. \vphantom{V_s = i_L (R + R_L)} \right\} \text{KVL}$$

$$i_L = \frac{V_s}{R + R_L}$$

Solve for i_L

Circuit (B)



$$i_L = i_s \frac{R}{R + R_L}$$

Solve for i_L by
Current Division

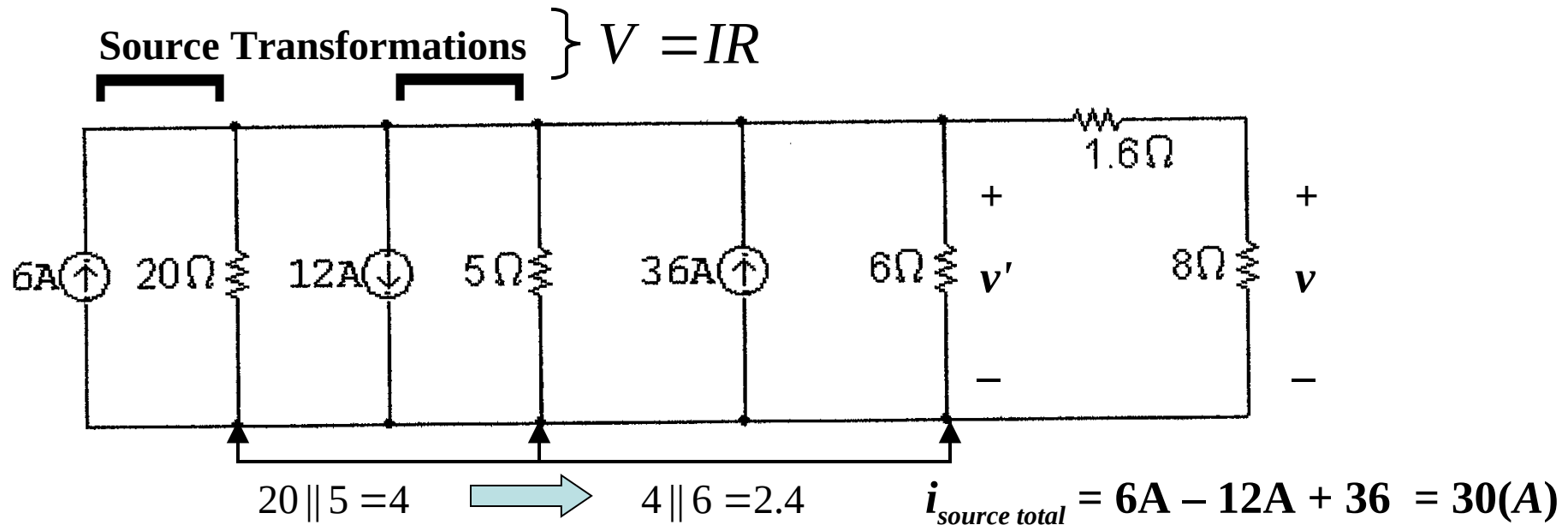
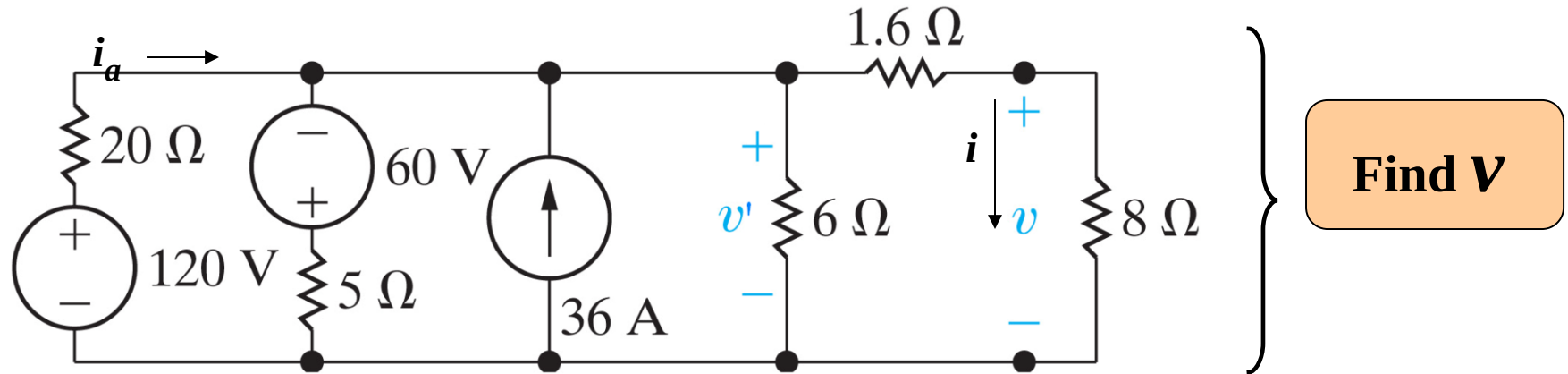
$$\textcircled{A} \equiv \textcircled{B} \quad \text{For Equivalence} \quad i_L = i_L$$

$$\cancel{\frac{V_s}{R + R_L}} = i_s \frac{R}{\cancel{R + R_L}}$$

$$\Rightarrow V_s = i_s R \quad \Rightarrow i_s = \frac{V_s}{R}$$

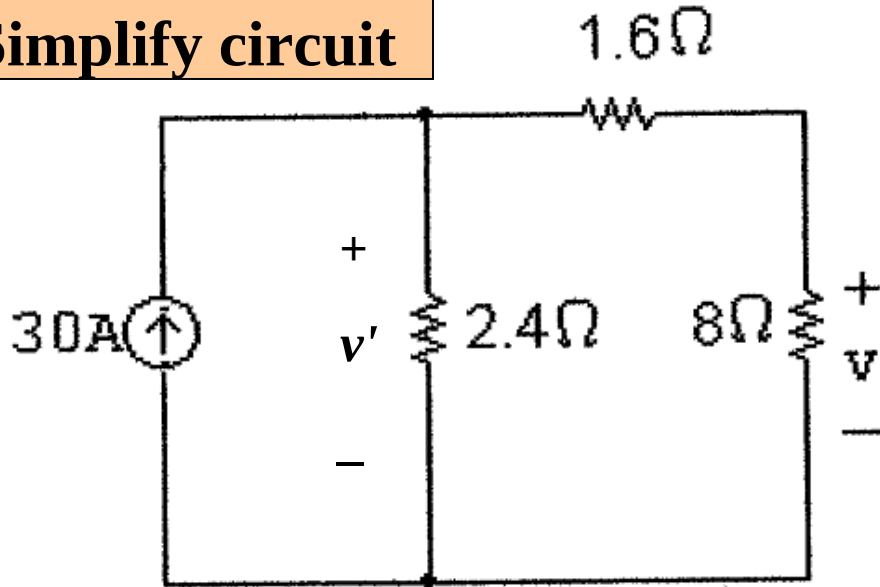
Drill Exercise

Finding V across the 8Ω Resistor

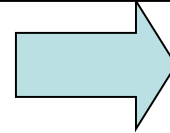


Drill Exercise (Contd.)

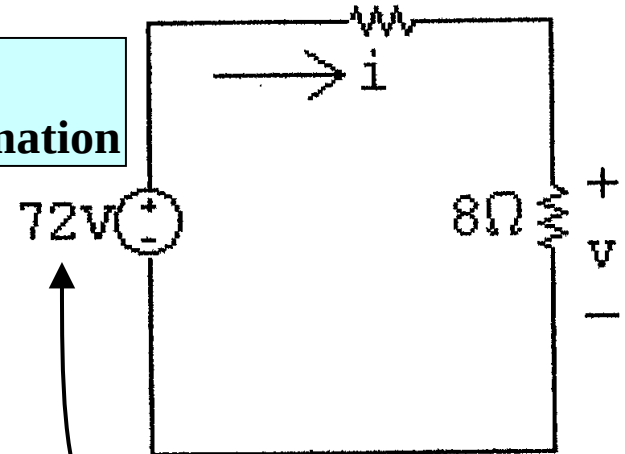
Simplify circuit



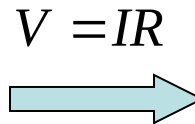
Source Transformation



Add resistors } $4\Omega = 2.4 + 1.6$



(Source Transformation)



$$V = IR$$

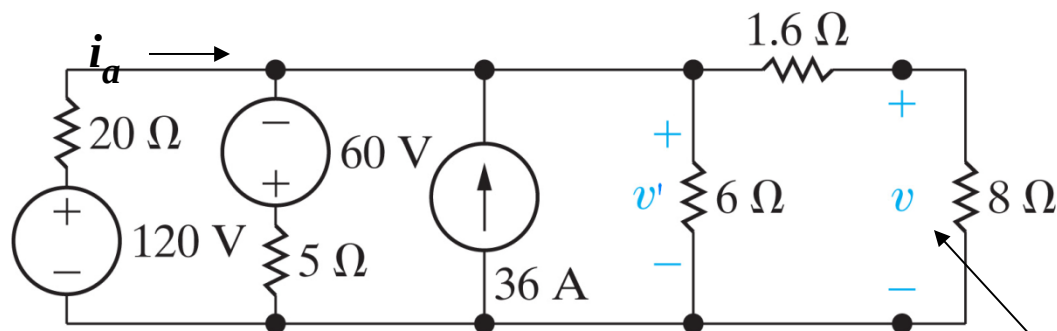
$$V = 30(2.4) = 72(V)$$

**Find V by
Voltage
Division**

$$V = 72 \frac{8}{8 + 4}$$

$$V = 48 (V)$$

Find the Power of the 120 V Source



Find i_a

Go back to the original circuit. Work from right to left.

$$i_{8\Omega \text{ resistor}} = \frac{v}{8} = \frac{48}{8} = 6(A)$$

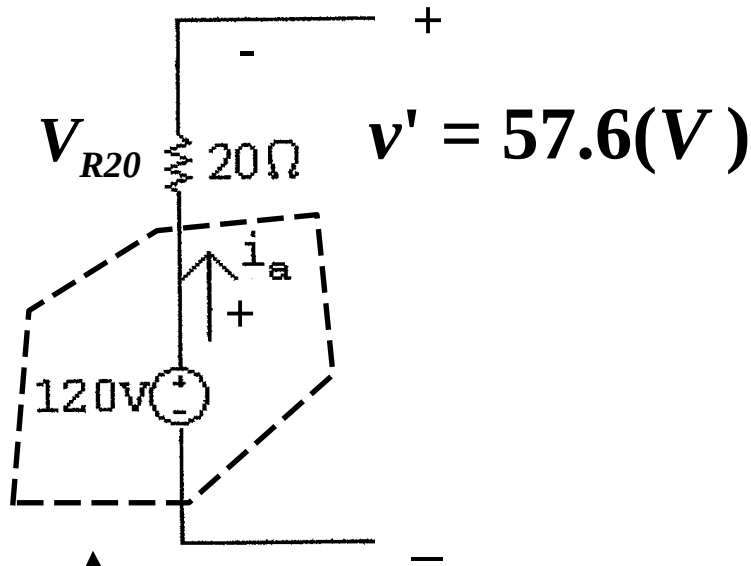
$$v = 48(V)$$

Find the Voltage across 6Ω Resistor

$$v' = i(1.6) + v = 6(1.6) + 48 = 57.6 (V) \quad \left. \vphantom{v'} \right\} \text{KVL}$$

v' is Dropped Across Each Essential Branch To The Left

Find the Power of the 120 V Source (Contd.)



$$v' = 120 - V_{R20} \quad \left. \vphantom{v'} \right\} \text{KVL}$$

$$V_{R20} = 120 - 57.6 = 62.4(V)$$

$$i_a = \frac{V_{R20}}{20} = \frac{62.4}{20} = 3.12(A) \quad \left. \vphantom{i_a} \right\} \text{Ohm's Law}$$

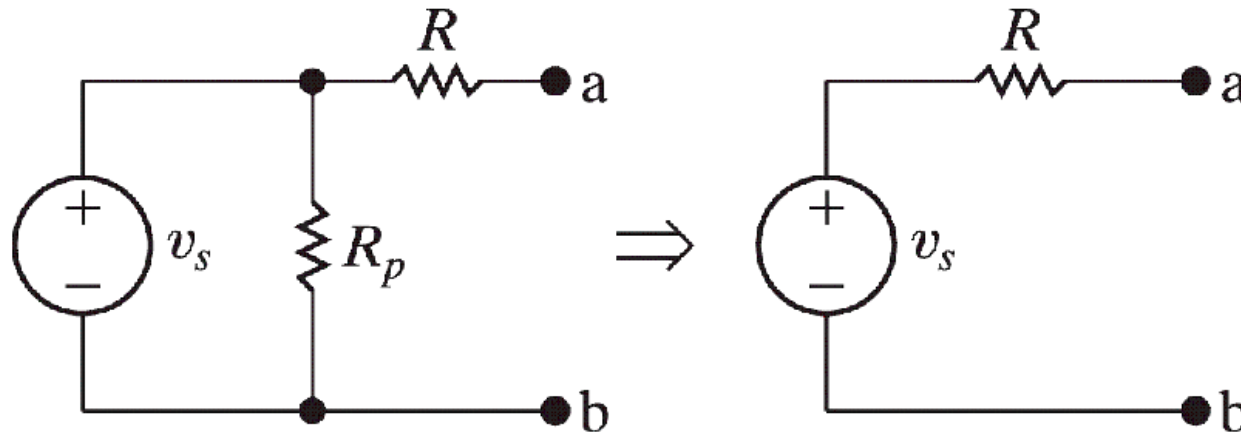
Passive sign convention

$$p_{120} = -120i_a = -120(3.12) \quad \left. \vphantom{p_{120}} \right\} \text{Power extracted}$$

$$p_{120} = 374.4 (W)$$

Delivered

Resistor in Parallel With a Voltage Source

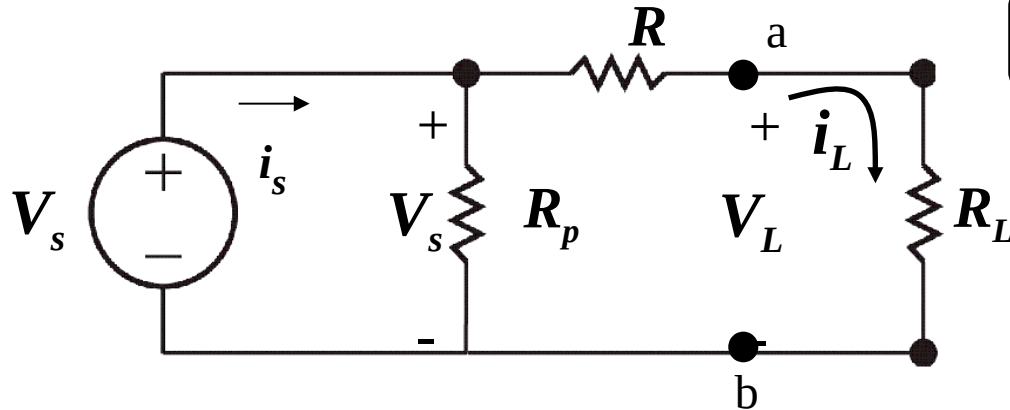


i.e., Simply Remove R_p . R_p has NO effect on the terminal V and i

WHY?

Either circuit $\Rightarrow$ same V_L and i_L for a load R_L

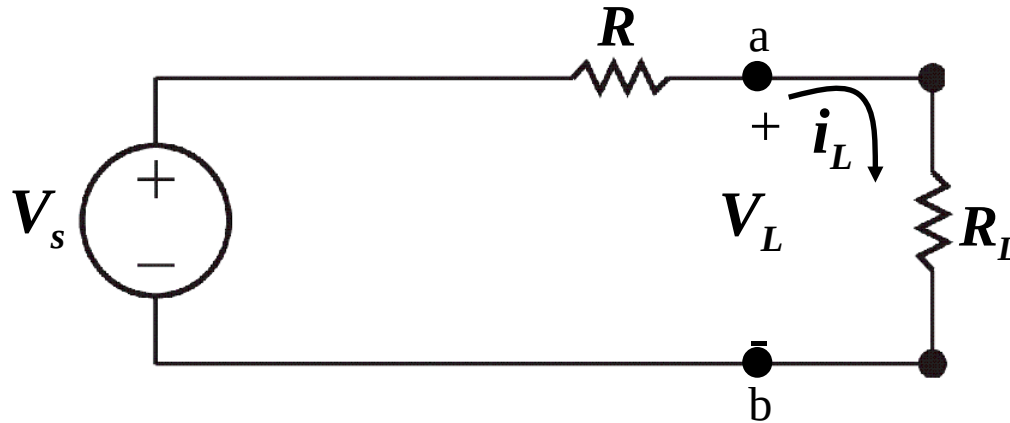
Resistor in Parallel With a Voltage Source (Contd.)



V_s Dropped Across R_p

$$V_L = V_s \frac{R_L}{R_L + R} = i_L R_L \quad \left. \vphantom{V_L = V_s \frac{R_L}{R_L + R} = i_L R_L} \right\} \text{Voltage Division}$$

$$i_L = \frac{V_s}{R_L + R} \quad \left. \vphantom{i_L = \frac{V_s}{R_L + R}} \right\} \text{Solve for } i_L$$



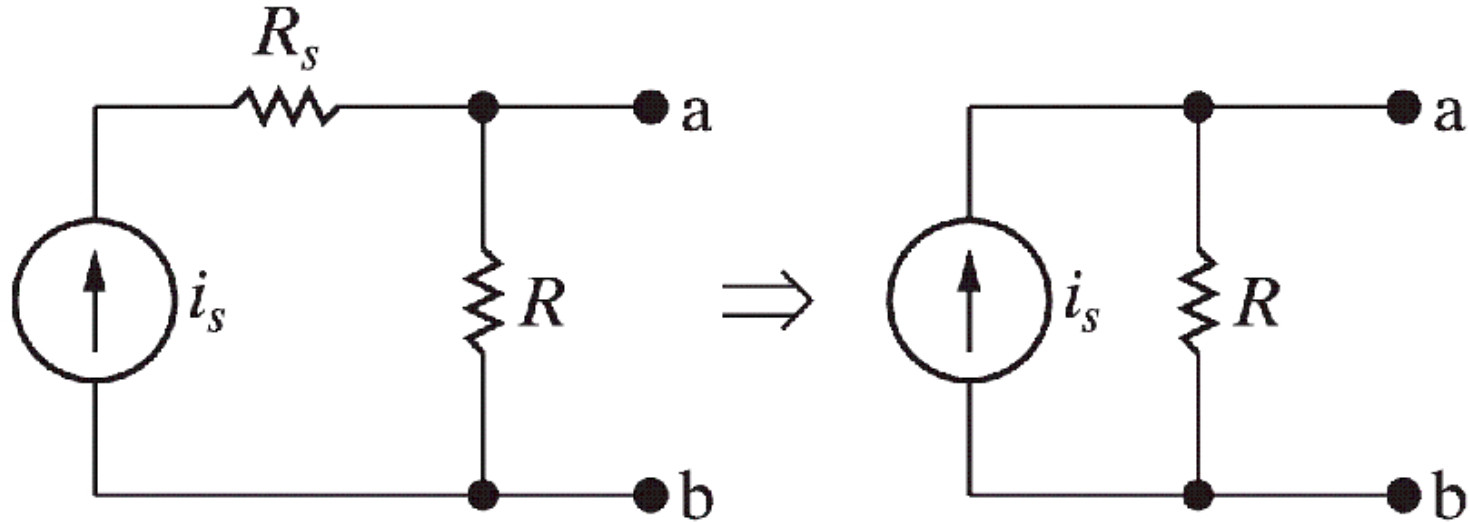
$$V_L = V_s \frac{R_L}{R_L + R} = i_L R_L \quad \left. \vphantom{V_L = V_s \frac{R_L}{R_L + R} = i_L R_L} \right\} \text{Voltage Division}$$

$$i_L = \frac{V_s}{R_L + R} \quad \left. \vphantom{i_L = \frac{V_s}{R_L + R}} \right\} \text{Solve for } i_L$$

Results are the same!

R_p Doesn't Matter for Equivalent Circuits

Resistor in Series With a Current Source

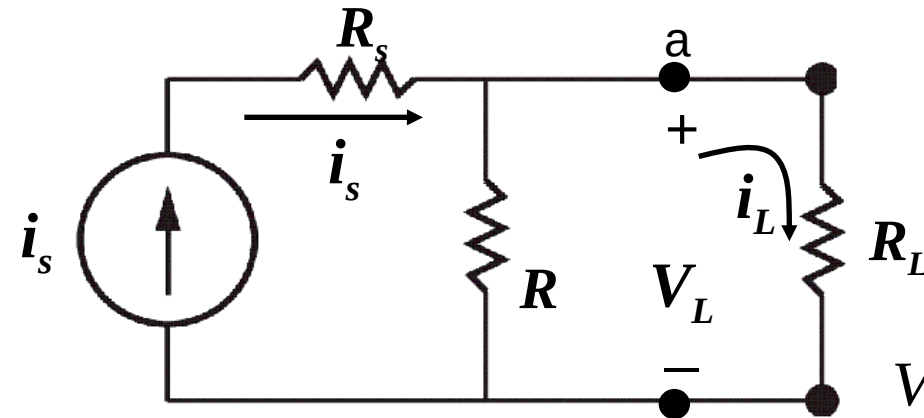


i.e., Simply Remove R_s . R_s has NO effect on the terminal V and i

WHY?

Either circuit $\Rightarrow$ same V_L and i_L for a load R_L

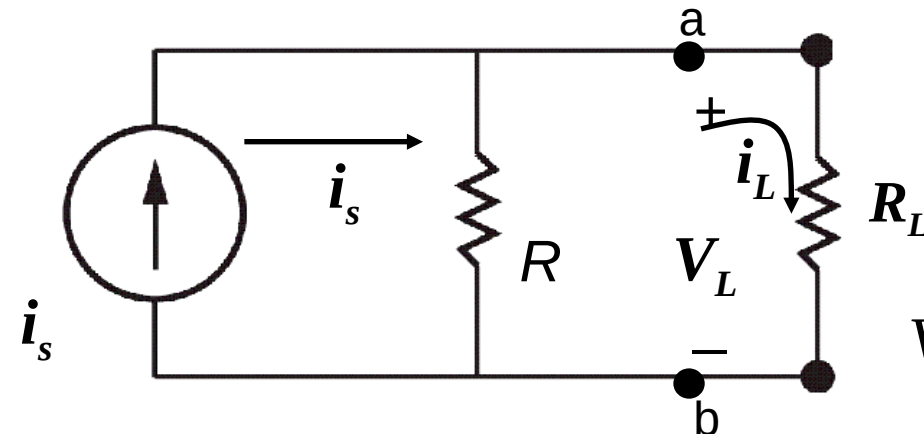
Resistor in Series With a Current Source (Contd.)



i_s flows into R_s and $R \parallel R_L$

$$i_L = i_s \frac{R}{R_L + R} = \frac{V_L}{R_L} \quad \left. \vphantom{i_L} \right\} \text{Current Division}$$

$$V_L = i_s \frac{R \cdot R_L}{R_L + R} = i_s (R \parallel R_L) \quad \left. \vphantom{V_L} \right\} \text{Ohm's Law}$$



$$i_L = i_s \frac{R}{R_L + R} = \frac{V_L}{R_L} \quad \left. \vphantom{i_L} \right\} \text{Current Division}$$

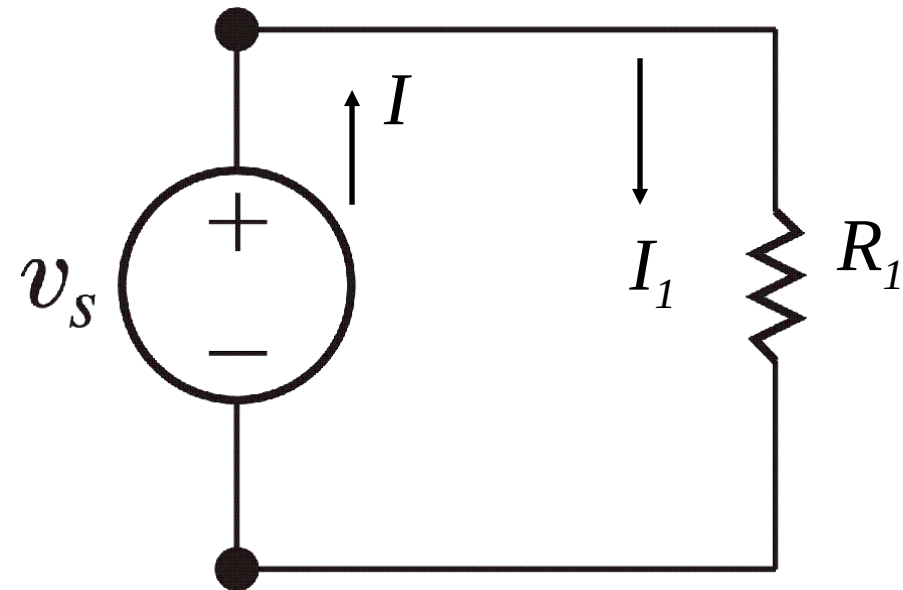
$$V_L = i_s \frac{R \cdot R_L}{R_L + R} = i_s (R \parallel R_L) \quad \left. \vphantom{V_L} \right\} \text{Ohm's Law}$$

Results are the same!

R_s Doesn't Matter for Equivalent Circuits.

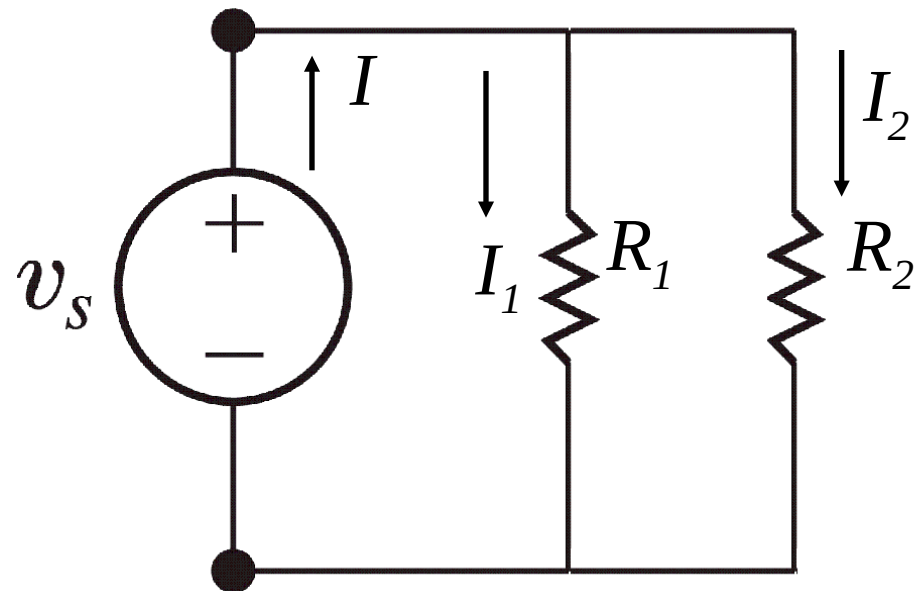
Supplemental Note on Parallel Resistors

One Resistor



$$I_1 = \frac{v_s}{R_1} \equiv I$$

Two Resistors



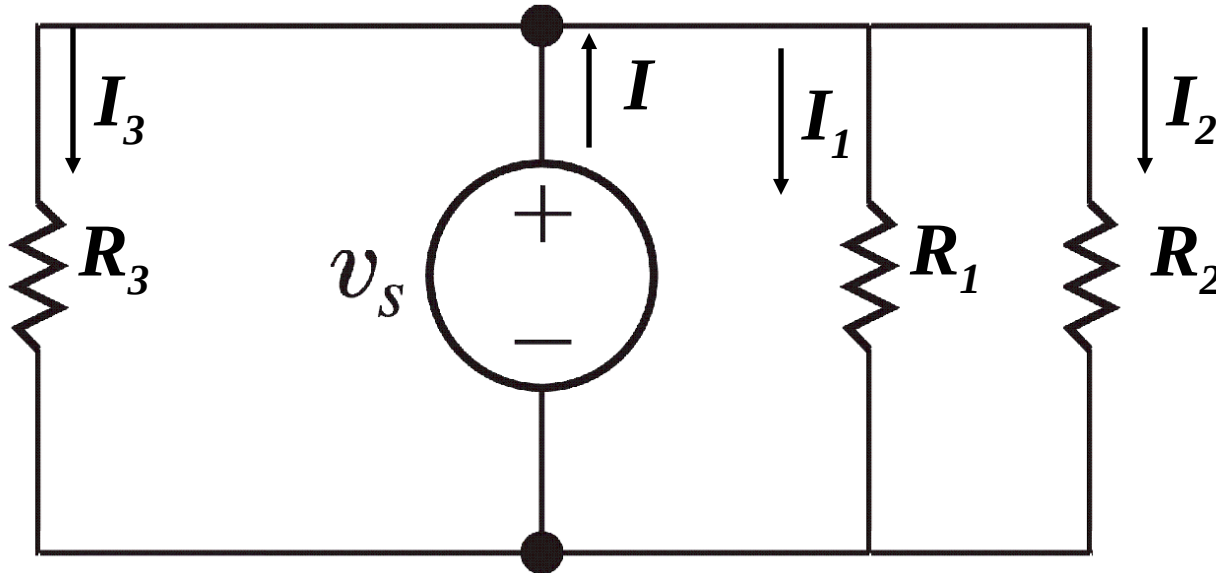
$$I_1 = \frac{v_s}{R_1} \quad I_2 = \frac{v_s}{R_2}$$

$$I = I_1 + I_2 \quad \left. \vphantom{I = I_1 + I_2} \right\} \text{KCL}$$

Supplemental Note on Parallel Resistors

(Contd.)

Three
Resistors



$$I_1 = \frac{v_s}{R_1} \quad I_2 = \frac{v_s}{R_2} \quad I_3 = \frac{v_s}{R_3}$$

$$I = I_1 + I_2 + I_3$$

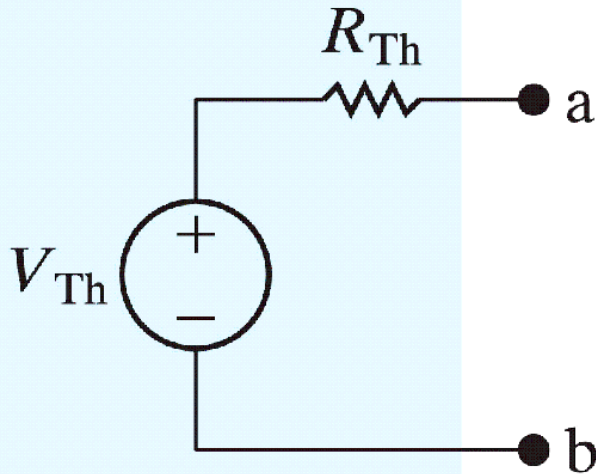
I increases with each added R .

More Resistors Cause More Current
To Be Drawn From The Voltage Source

Thevenin Equivalents

Sometimes we want to simplify a circuit at a particular pair of nodes in a circuit.

Example: Power supply attached to a load



$V_{Th} \equiv$ Thevenin Voltage

$R_{Th} \equiv$ Thevenin Resistance

- Thevenin Equivalent of some arbitrary circuit.
- Load R_L will “See” the same V and I in either the original or equivalent circuit.
- For ANY value of R_L

Thevenin Equivalents (Contd.)

- A Thevenin Equivalent is really a generalization of the Source Transformation.
- In fact, you could obtain a Thevenin Equivalent using Source Transformations, but it might take a long time. (Dependent sources might make it tough as well.)
- How do we determine V_{Th} and R_{Th} ? {

Look at the Two
Extreme Cases

Thevenin Equivalents (Contd.)

Find the

V_{Th}

If $R_L \equiv \infty$, Terminals a & b are open

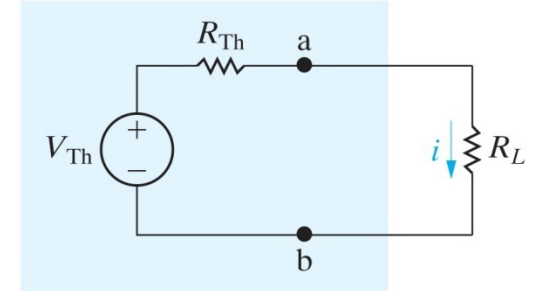


Figure: 04-59
Copyright © 2008 Pearson Prentice Hall, Inc.

$V_{oc} \equiv V_{Th}$ in Thevenin Equivalent.

No Current Through R_{Th}

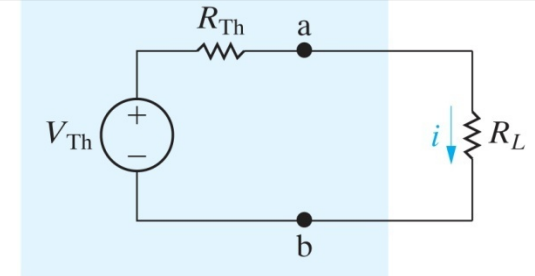
Must be the same for the original circuit.

$\therefore$ Calculate V_{oc} in original circuit – This is the V_{Th}

Thevenin Equivalents (Contd.)

Find the R_{Th}

If $R_L = 0$, Terminals (a) and (b) are Shorted.



$$i_{sc} = \frac{V_{Th}}{R_{Th}}$$

Thevenin Equivalent with (a) shorted to (b).

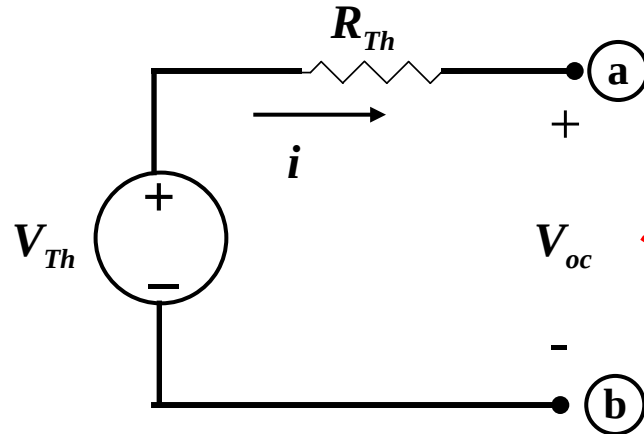


$$R_{Th} = \frac{V_{Th}}{i_{sc}}$$

Must be the same for original circuit,
with (a) shorted to (b)

$$R_{Th} \equiv \frac{\text{Open circuit voltage}}{\text{Short circuit current}}$$

Thevenin Illustration

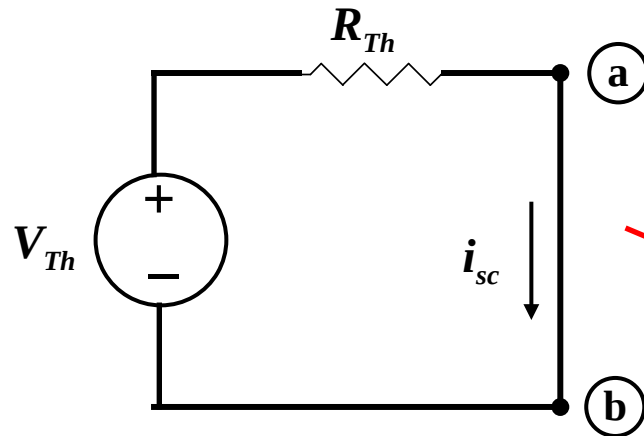


$$V_{oc} \equiv V_{Th}$$

since $i \equiv 0$

No closed paths

$$R_L \equiv \infty \quad [\text{Open}]$$



$$V_{Th} = i_{sc} R_{Th}$$

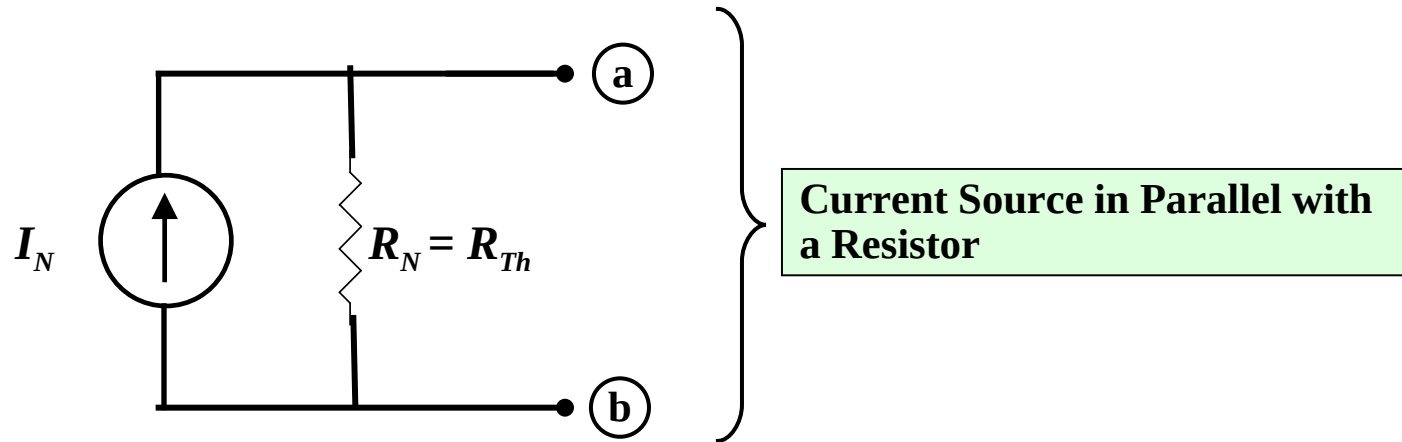
Ohm's Law

$$R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{V_{oc}}{i_{sc}}$$

$$R_L \equiv 0 \quad [\text{Short}]$$

Norton Equivalent

Dual of Thevenin



$$I_N = \frac{V_{Th}}{R_{Th}}$$

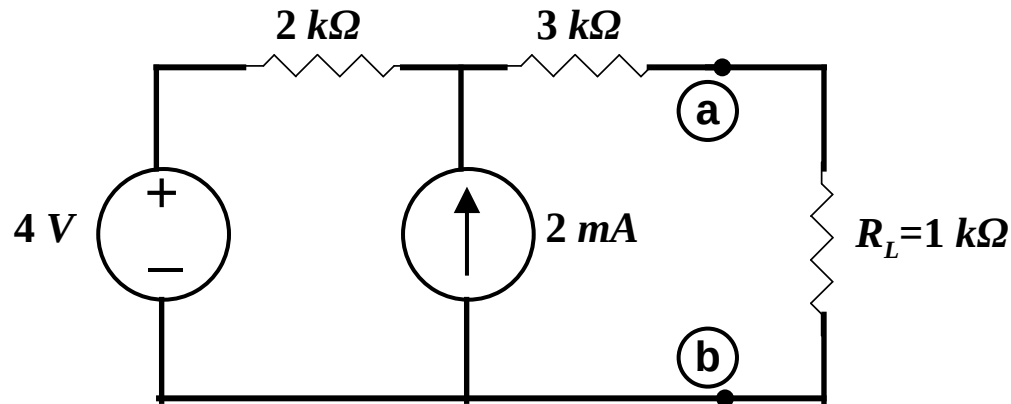
Source Transformation

$$R_N \equiv R_{Th}$$

Source Transformation

Example

Find Thevenin and Norton Equivalent at Terminals (a) and (b)

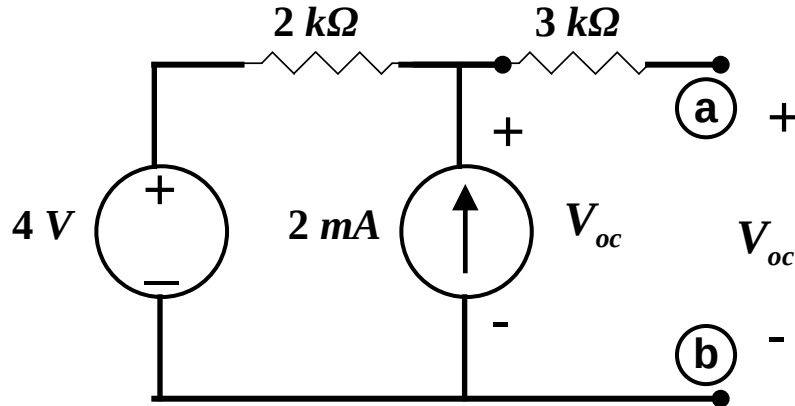


Do it three ways ...

Example

Find Thevenin and Norton equivalent

Method 1: Calculate $V_{oc} = V_{Th}$ and $i_{sc} = V_{Th}/R_{Th}$



Current in 3 kΩ Resistor is Zero

Open Circuit



Write Nodal Equation



$$\left. \frac{V_{oc} - 4}{2(K\Omega)} - 2(mA) = 0 \right\} \text{Node Equation}$$

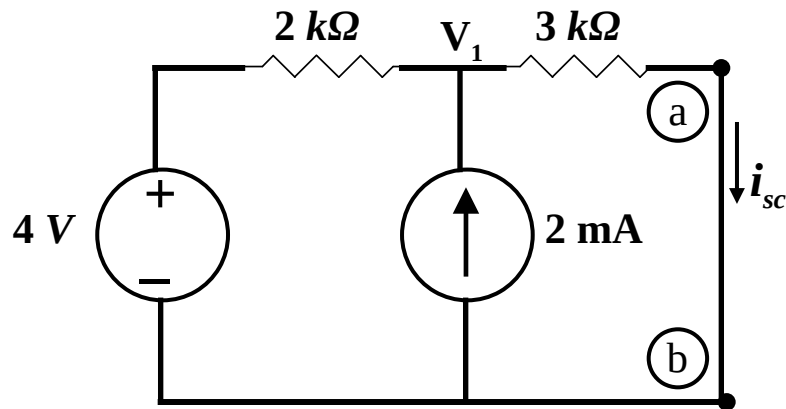
Solve for V_{oc} $\rightarrow V_{oc} - 4 = 2(mA)(2(K\Omega)) = 4(V)$

Simplify $\rightarrow V_{oc} = 4 + 4 = 8(V)$

$$V_{Th} = 8(V)$$

Example (Contd.)

Find Thevenin and Norton equivalent



Short Circuit



Write Nodal Equation

$$\left. \frac{V_1 - 4}{2K} - 2(mA) + \frac{V_1}{3K} = 0 \right\} \text{Node Equation}$$

$$\left. 3V_1 - 12 - 12 + 2V_1 = 0 \right\} \text{Simplify}$$

$$V_1 = 4.8(V)$$

$$\left. i_{sc} = \frac{V_1}{3K} = \frac{4.8}{3K} \right\} \text{Ohm's Law}$$

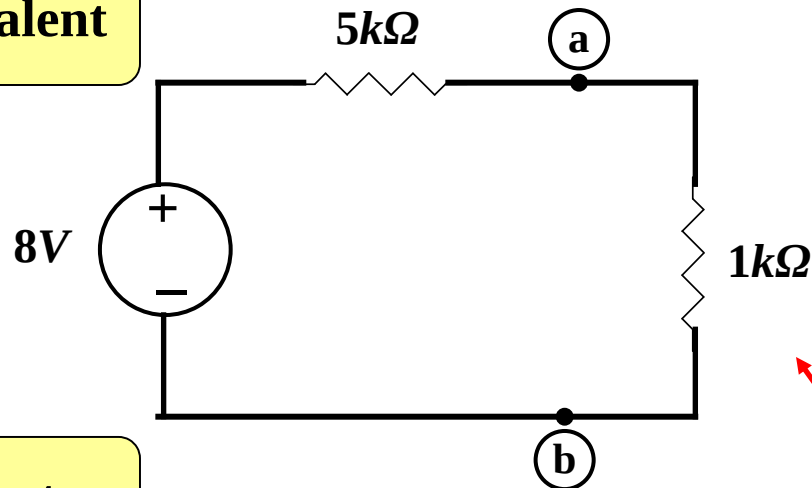
$$i_{sc} = 1.6(mA)$$

$$\left. R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{8}{1.6(mA)} = 5(k\Omega) \right\} \text{Find } R_{Th}$$

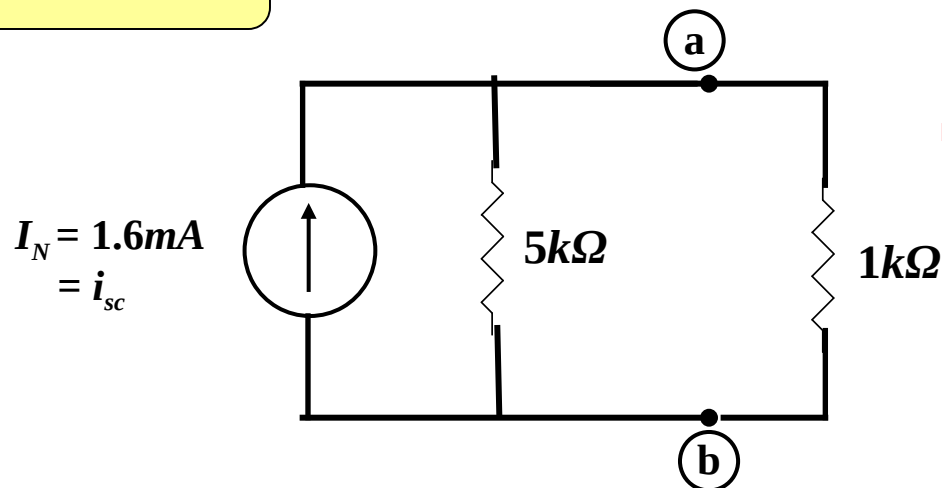
Example

Find Thevenin and Norton equivalent

Thevenin Equivalent



Norton Equivalent

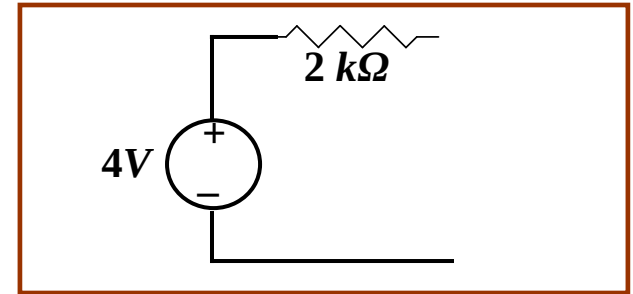
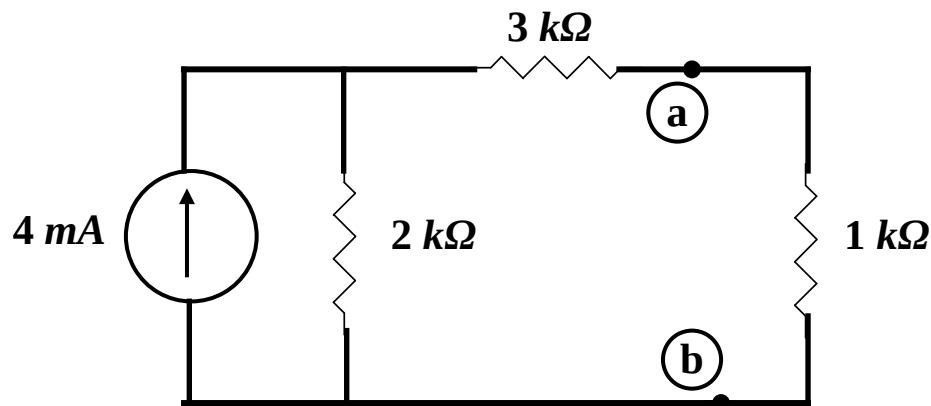
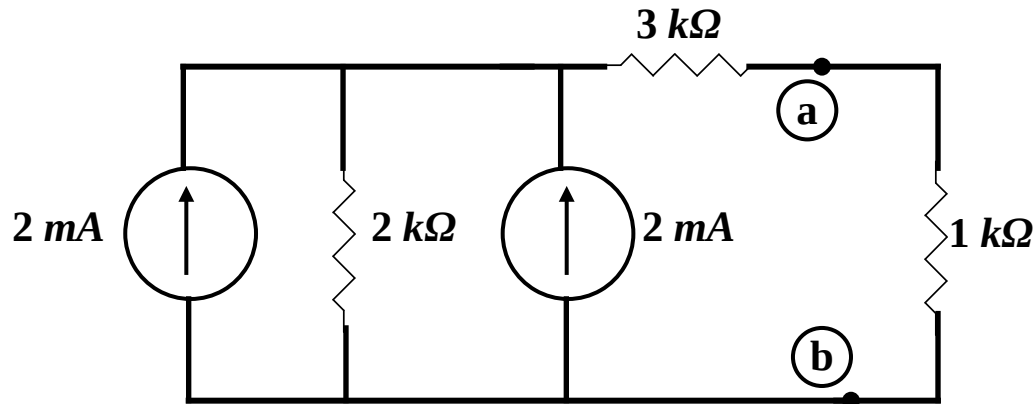


The $1k\Omega$ Resistor is the “Load” and NOT part of the Thevenin or Norton Equivalent

Example

Find Thevenin and Norton equivalent

Method 2: Use Source Transformations



Already changed 4 (V)
source to current source

$$4\text{ V} / 2\text{ K} = 2\text{ (mA)}$$

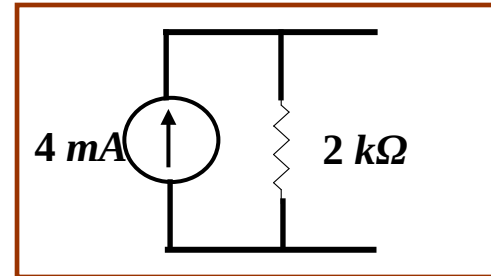
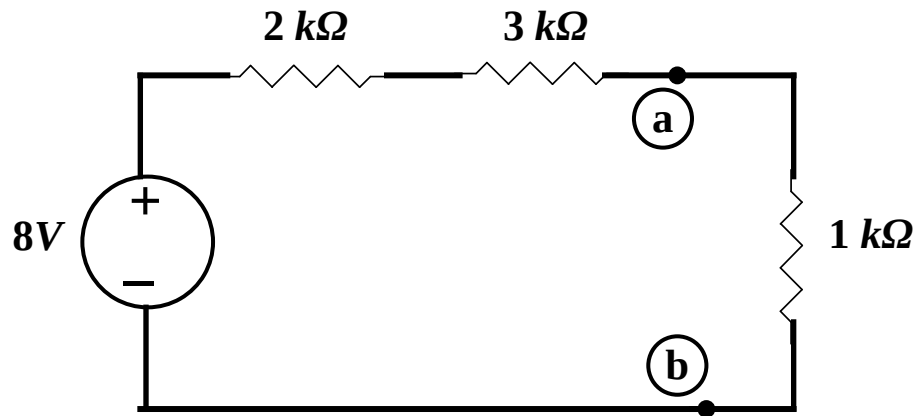


Sum Current
Sources

(Contd..)

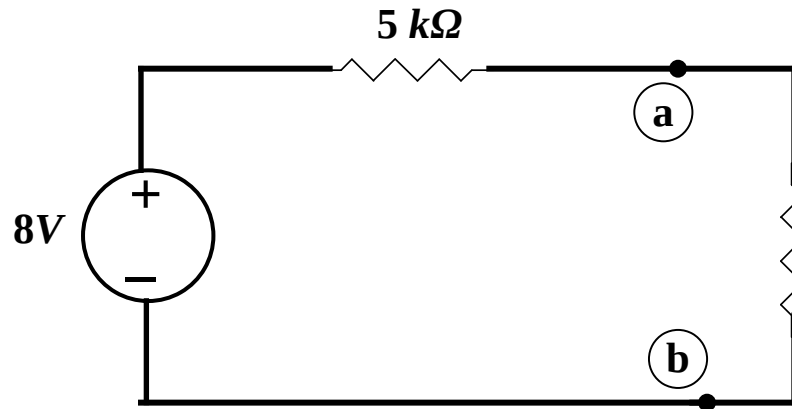
Example

Find Thevenin and Norton equivalent



Change 4(mA) source to voltage source

$$8(V) = (4(mA))(2(K\Omega))$$



NOT part of R_{Th}

$1\text{ k}\Omega = R_L$

$$V_{Th} = 8\text{ (V)}$$

$$R_{Th} = 5\text{ (k}\Omega\text{)} = R_N$$

$$I_N = 8/5K = 1.6\text{ (mA)}$$

Example

Find Thevenin and Norton equivalent

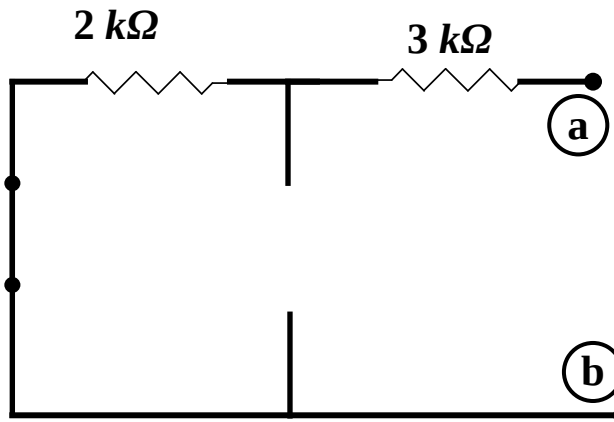
Method 3: [a] Independent Source Deactivation.

[b] Linear Superposition

Short Voltage Sources
Open Current Sources

Deactivate the Independent Sources

Original circuit becomes ...



$R_{Th} \equiv R$ "Looking" into terminals a and b

R 'Looking Into' (a) & (b)

What R_L "sees"

$$R_{Th} = 2K + 3K = 5(K\Omega)$$

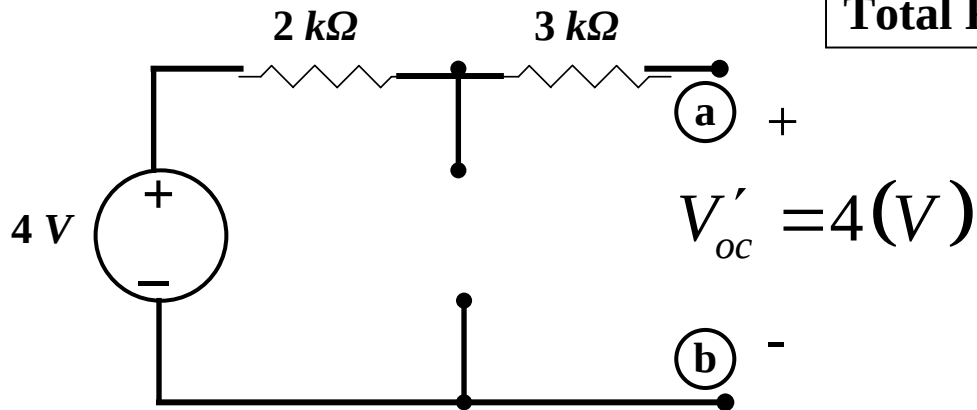
Example

Find Thevenin and Norton equivalent

Superposition

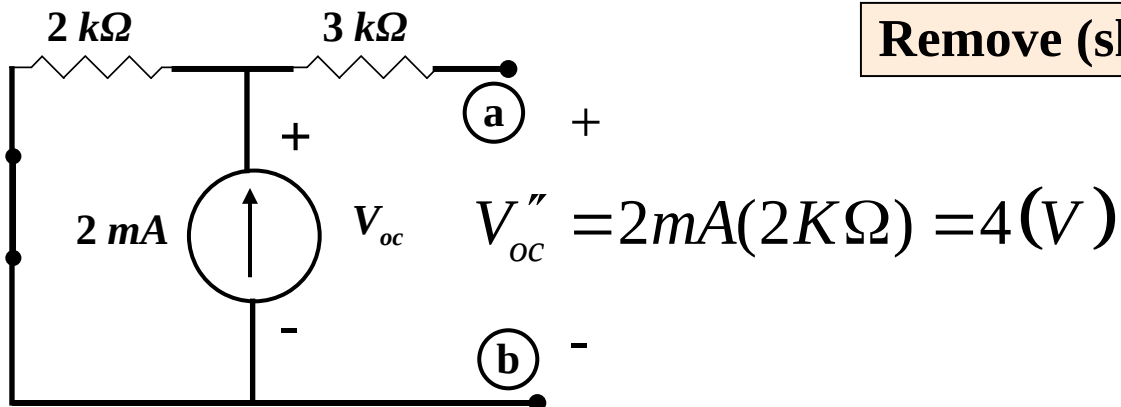
In a **linear** system, with independent sources:

Total Response $\equiv$ Sum of Individual Responses



Remove (open) current source

**No current flows
in resistors**



Remove (short) voltage source

**No current in 3K
resistor**

(Contd..)

Example

Find Thevenin and Norton equivalent

$$V_{oc} = V_{Th} = V_{oc}'' + V_{oc}' = 4(V) + 4(V) = 8(V)$$

$$V_{Th} = 8(V)$$

$$\text{Recall } R_{Th} = 5(K\Omega)$$

Same as
method 1 and 2

Get Norton from Source Transformation

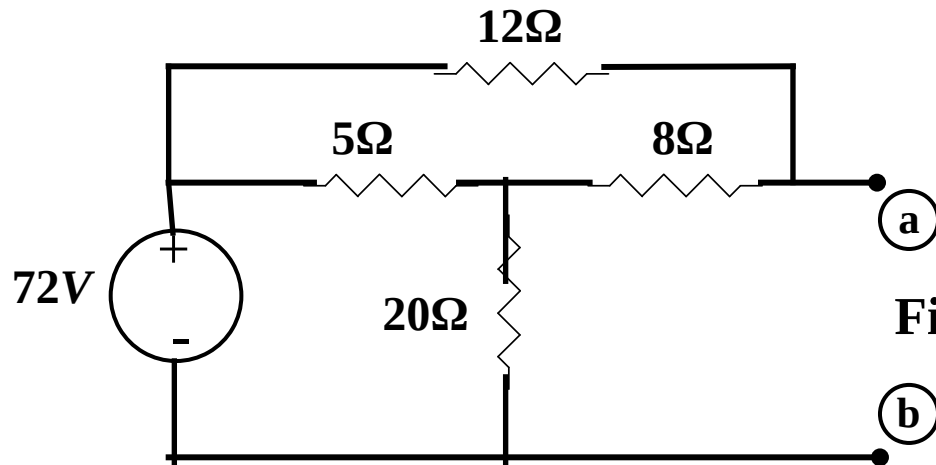
Most Common Way to Obtain Thevenin Equivalent

We could use Method 3 to obtain R_{Th} by Deactivating Sources,
and then

Obtain $V_{oc} = V_{Th}$ as in Method 1 to obtain V_{Th}

This is often the best method

Drill Exercise



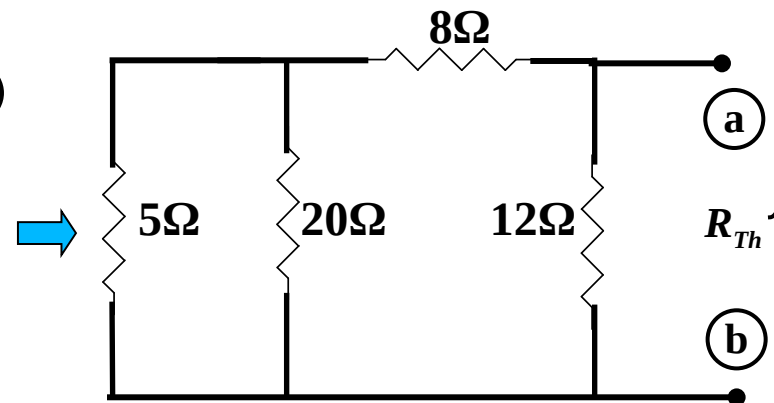
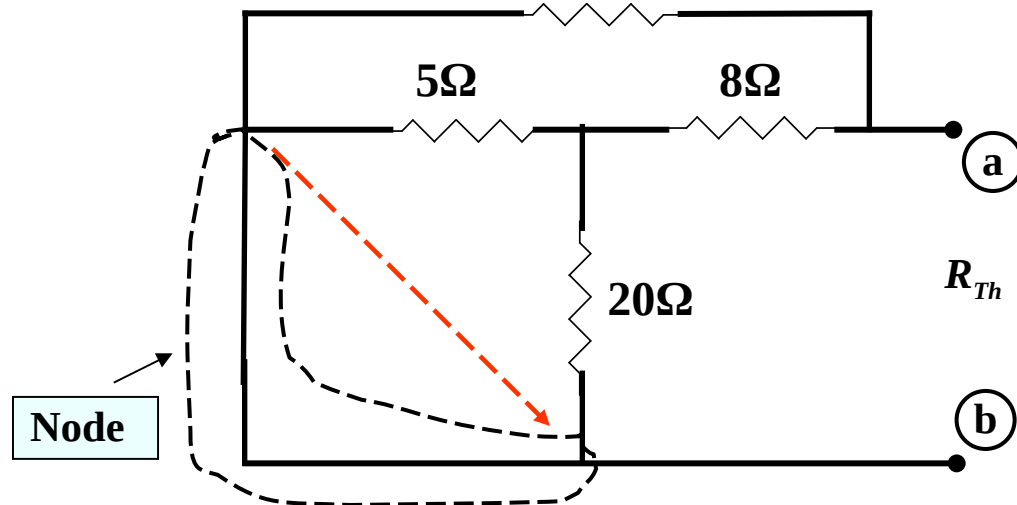
Find Thevenin Equivalent at (a) – (b)

Thevenin Resistance:
Short Voltage Source



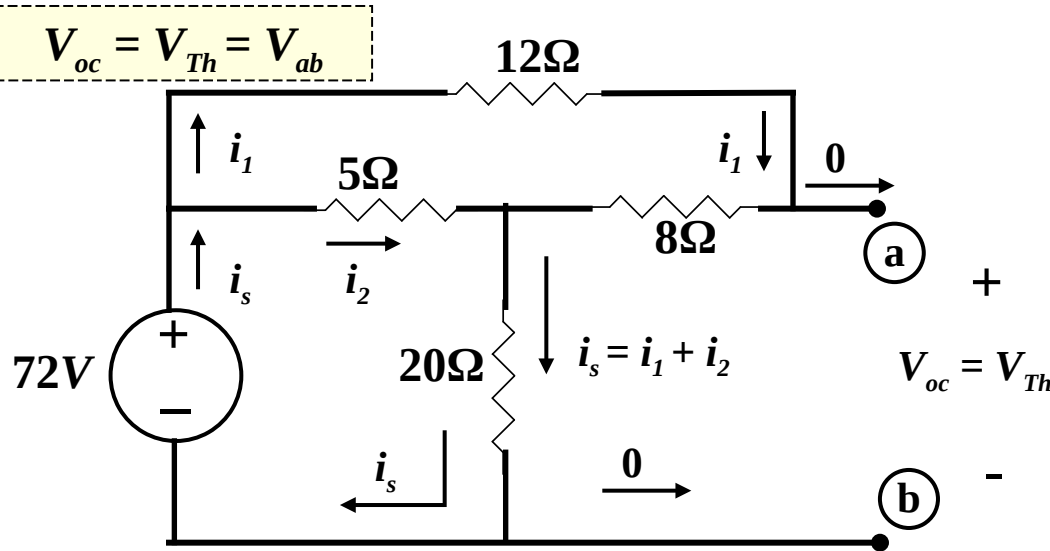
$$R_{Th} = [(5 \parallel 20) + 8] \parallel 12$$

$$= 6 (\Omega)$$

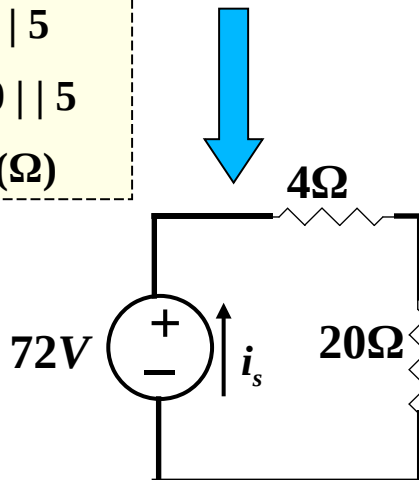


Drill Exercise (Contd.)

Find Thevenin Equivalent at (a) – (b)



$$\begin{aligned}
 R_{eq} &= (12 + 8) \parallel 5 \\
 &= 20 \parallel 5 \\
 &= 4 (\Omega)
 \end{aligned}$$



$$i_s = \frac{72(V)}{4 + 20} = 3(A) \quad \left. \vphantom{\frac{72(V)}{4 + 20}} \right\} \text{Ohm's Law}$$

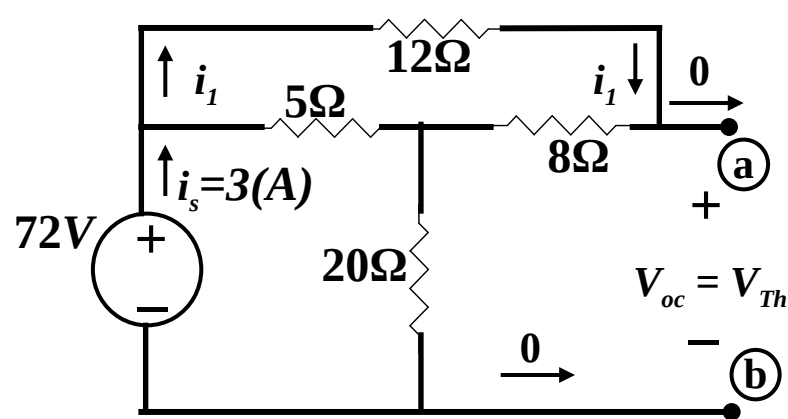
Current Supplied by Voltage Source

Lost Definition of V_{oc} ,

But OK For Now

Drill Exercise (Contd.)

Find Thevenin Equivalent at (a) – (b)



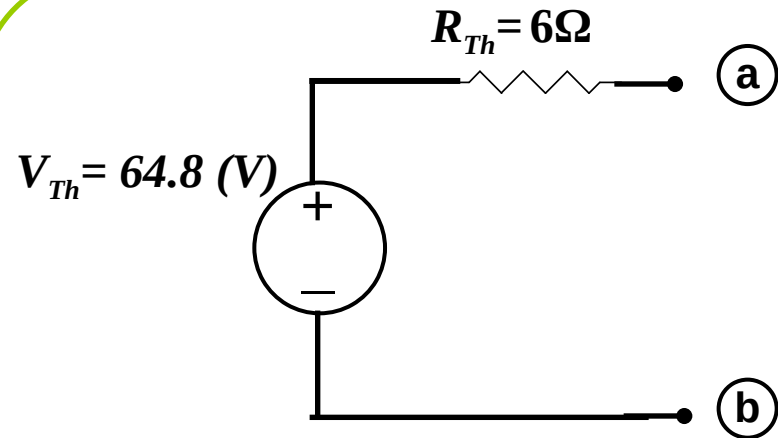
$$i_1 = i_s \frac{5}{5 + (12 + 8)} = 3 \frac{5}{5 + 20}$$

$$i_1 = 0.6(A)$$

Find i_1 by Current Division

$$\begin{aligned} V_{Th} &= 72 - V_{12\Omega} \\ &= 72 - 12i_1 \\ &= 72 - 12(0.6) \\ &= 72 - 7.2 \end{aligned}$$

$$V_{Th} = 64.8(V)$$



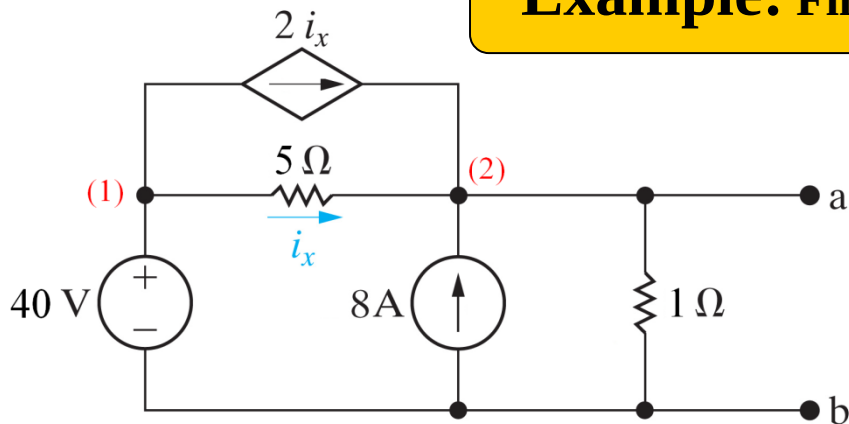
Thevenin Equivalent

THEVENIN WITH DEPENDENT SOURCES

To compute R_{Th} :

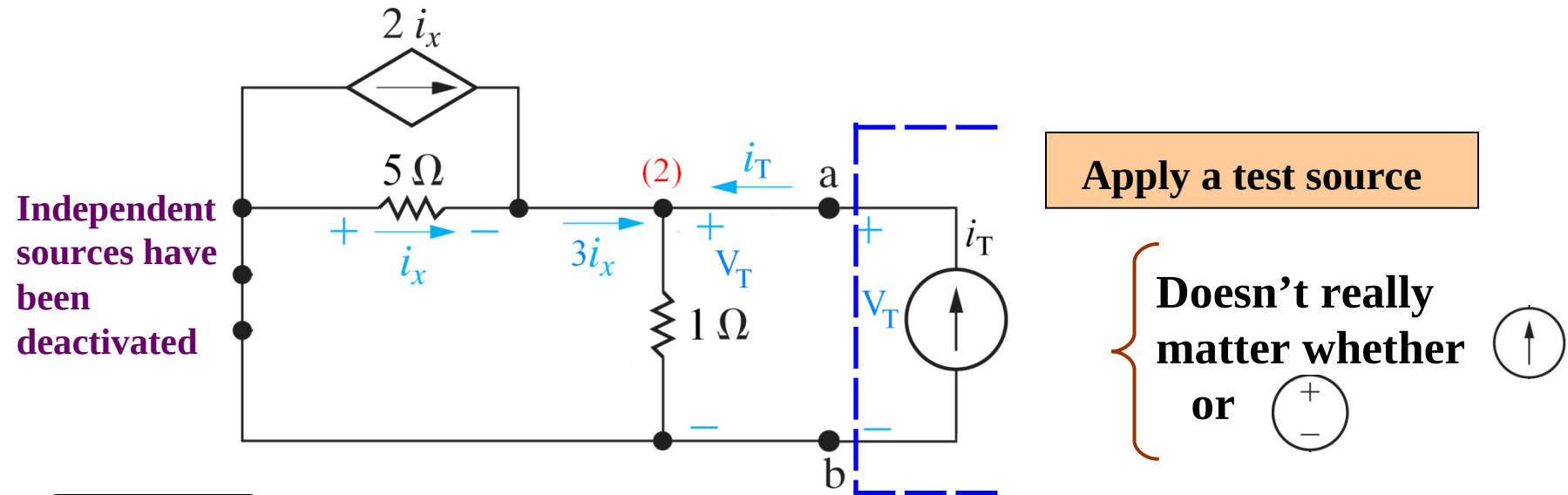
1. Short Independent Voltage Sources
2. Open Independent Current Sources
3. Apply Test Voltage or Current Source (a)-(b) } **Dummy Variable to deal with Dependent Sources**
4. $R_{Th} = V_T / i_T \longrightarrow$ {Also, you could use $R_{Th} = V_{OC} / i_{SC}$ }

Example: Find R_{Th} and V_{Th}



- Short 40V source
- Open 8A source

Example (Contd.)



Find R_{Th}

$$\textcircled{1} \quad 3i_x + i_T = \frac{V_T}{1} \quad \left. \vphantom{\frac{V_T}{1}} \right\} \text{KCL at node (2)}$$

$$\textcircled{2} \quad \underbrace{V_T = -5i_x}_{\text{KVL}} \quad \rightarrow \quad i_x = -\frac{V_T}{5}$$

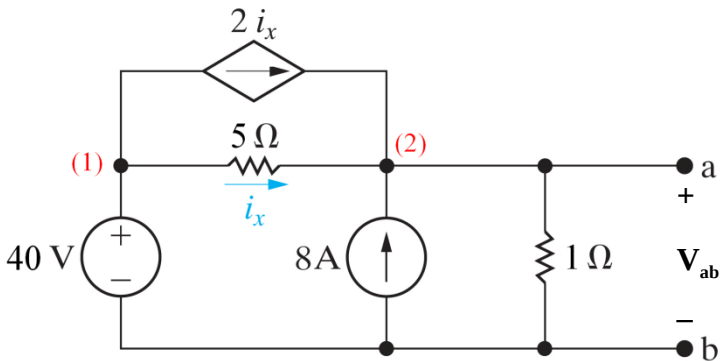
(Contd..)

Example: Find R_{Th} (Contd.)

Don't know V_T or i_T , but we can solve for the ratio

$$\textcircled{3} \quad 3 \left(-\frac{\overbrace{V_T}^{i_x}}{5} \right) + i_T = V_T \quad \left\{ \text{Substitute } \textcircled{2} \text{ into } \textcircled{1} \right.$$
$$\textcircled{3} \quad i_T = V_T \left[1 + \frac{3}{5} \right] = \frac{8}{5} V_T \quad \left\{ \text{Simplify} \right.$$
$$\therefore R_{Th} \equiv \frac{V_T}{i_T} = \frac{5}{8} \Rightarrow R_{Th} = 0.625(\Omega)$$

Example: Find V_{Th} (Contd.)



$V_{Th} \equiv V_{oc} = V_{a-b}$: Go back to the Original Circuit

2 Node Voltages: $V_1 = 40 V$

$V_2 = V_{ab} = V_{oc} = V_{Th}$

$$\left. \frac{V_{Th} - 40}{5} - 8 - 2i_x + \frac{V_{Th}}{1} = 0 \right\} \text{Node equation at } \textcircled{2}$$

$$i_x + 2i_x + 8 = \frac{V_{Th}}{1} \Rightarrow \left. i_x = \frac{V_{Th} - 8}{3} \right\} \text{KCL @ } \textcircled{2}$$

RESULT: $V_{Th} = 20 (V)$

Solve
2 equations
for
2 unknowns

MAXIMUM POWER TRANSFER THEOREM

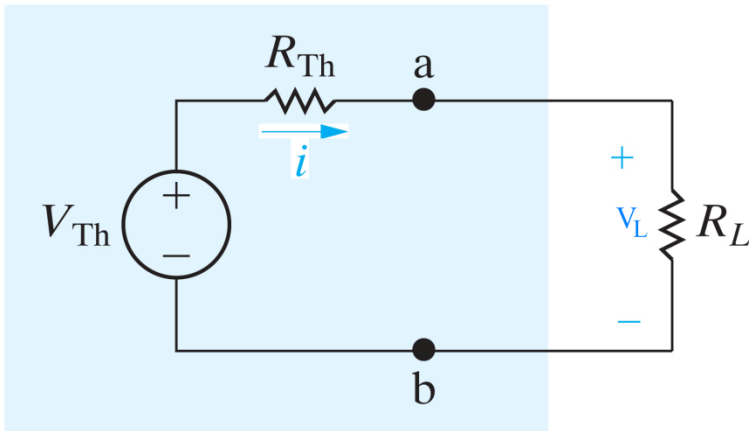
Maximum Power is delivered by a circuit to a load when:

$$R_{Th} \text{ (Circuit)} \equiv R_L \text{ (load)}$$

Important for efficient transmission of power.

Related to impedance matching.

Proof: - Any circuit can be reduced to a Thevenin Equivalent
- Resistive Load modeled by R_L



What value of R_L gives the Maximum Power Transfer?

For a given circuit, we assume that V_{Th} and R_{Th} are fixed

MAXIMUM POWER TRANSFER THEOREM (Contd.)

$$\textcircled{1} \quad p_L = i \cdot V_L = i^2 R_L = \underbrace{V_L^2 / R_L}_{\text{Power absorbed at the Load}}$$

$$\textcircled{2} \quad V_L = V_{Th} \frac{R_L}{R_L + R_{Th}}$$

Voltage Division

$$\textcircled{3} \quad p_L = \frac{V_{Th}^2 \frac{R_L^2}{(R_L + R_{Th})^2}}{R_L} \equiv \frac{V_{Th}^2}{(R_L + R_{Th})^2} \cdot R_L \quad \left. \begin{array}{l} \text{Substitute } \textcircled{1} \\ \text{into } \textcircled{2} \end{array} \right\}$$

To find $R_L \rightarrow$ Maximize p_L

$$\frac{\partial p_L}{\partial R_L} = 0 = V_{Th}^2 \left[\frac{(1)(R_L + R_{Th})^2 - R_L 2(R_L + R_{Th})(1)}{(R_L + R_{Th})^4} \right] \quad \left. \begin{array}{l} \text{Take partial} \\ \text{derivative and} \\ \text{set to zero} \end{array} \right\}$$

(Contd...)

MAXIMUM POWER TRANSFER THEOREM (Contd.)

$$(R_L + R_{Th})^2 = 2R_L (R_L + R_{Th}) \quad \left. \vphantom{(R_L + R_{Th})^2} \right\} \text{Simplify}$$

$$R_L^2 + \cancel{2R_L R_{Th}} + R_{Th}^2 = 2R_L^2 + \cancel{2R_L R_{Th}} \quad \left. \vphantom{R_L^2 + \cancel{2R_L R_{Th}} + R_{Th}^2} \right\} \text{Simplify}$$

$$R_L^2 = R_{Th}^2$$

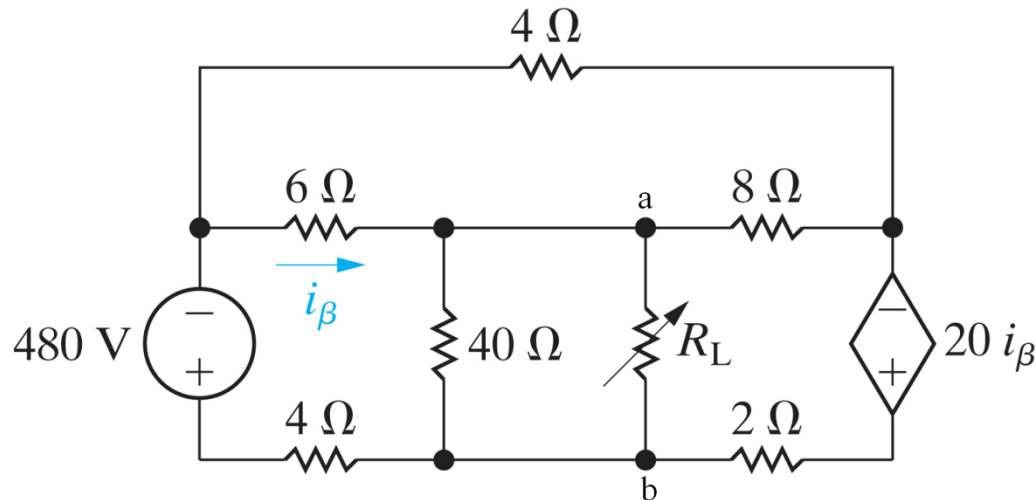
$$R_L = R_{Th} \quad \text{For Maximum Power Transfer}$$

$$\textcircled{3} \quad p_{L \max} = \frac{V_{Th}^2}{(R_L + R_L)^2} \cdot R_L \quad \xrightarrow{\text{Since } R_L = R_{Th}}$$

$$p_{L \max} = \frac{V_{Th}^2}{4R_L}$$

**Maximum
Power**

Example



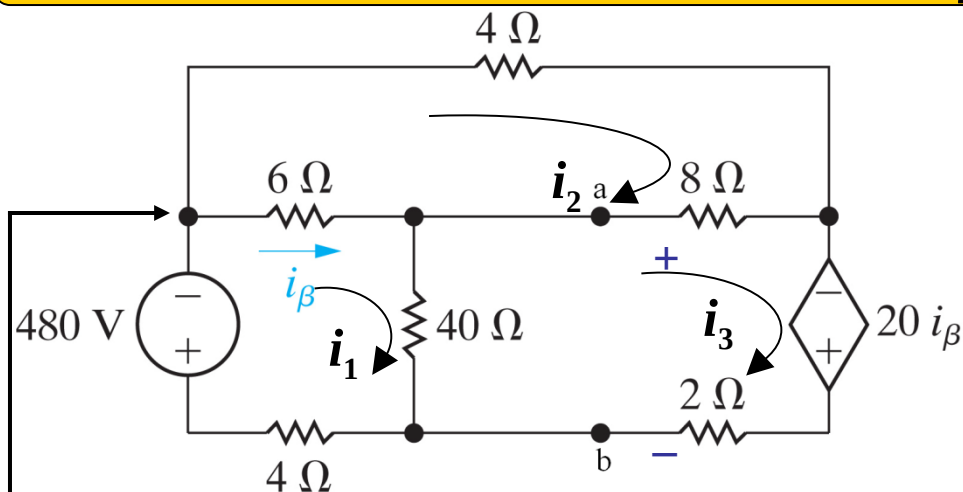
Find R_L for Maximum Power Transfer

- Need to find Thevenin Equivalent between nodes a and b
- So remove R_L and find:

$$V_{ab} \equiv V_{oc} \equiv V_{Th}$$

Example (Contd.)

Use Mesh Analysis



Find V_{Th} at (a)-(b)

①

$$-480 = 6(i_1 - i_2) + 40(i_1 - i_3) + 4i_1$$

Mesh 1

①

$$-480 = 50i_1 - 6i_2 - 40i_3$$

Simplify

②

$$0 = -6i_1 + 18i_2 - 8i_3$$

Mesh 2

③

$$8(i_3 - i_2) - 20(i_1 - i_2) + 2i_3 + 40(i_3 - i_1) = 0$$

Mesh 3

$$i_\beta = i_1 - i_2$$

③

$$0 = -60i_1 + 12i_2 + 50i_3$$

Simplify

$$i_1 = -99.60 \text{ (A)}$$

$$i_3 = -100.80 \text{ (A)}$$

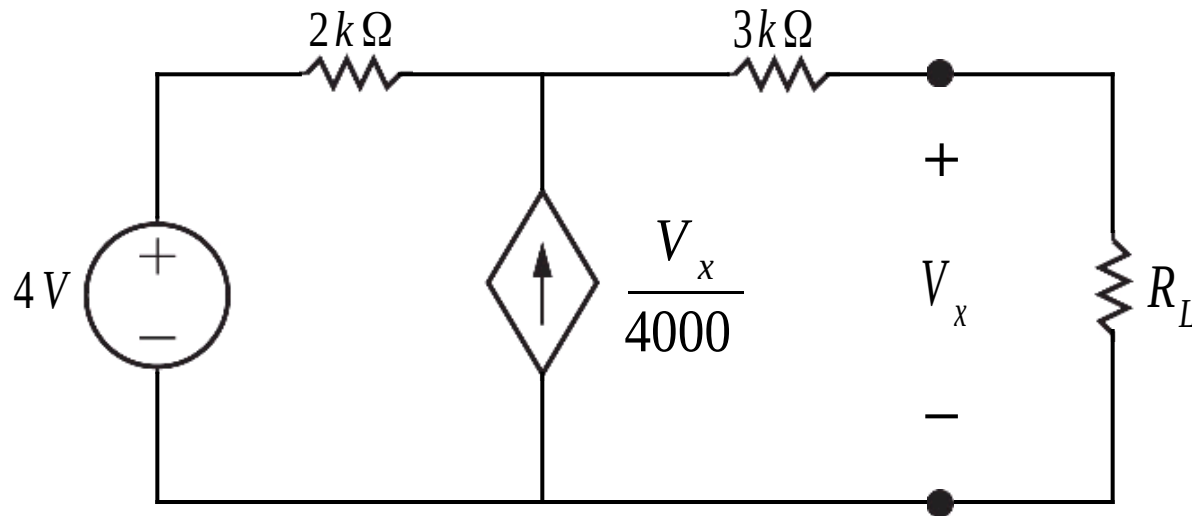
Solve 3 equations for 3 unknowns

$$V_{ab} \equiv V_{40} = 40(i_1 - i_3) = 48 \text{ (V)}$$

Ohm's Law

$$V_{Th} = 48 \text{ (V)}$$

Thevenin Equivalent Example and Maximum Power Transfer



a) Find R_L for Max Power Transfer

b) Find Max Power Transferred to R_L

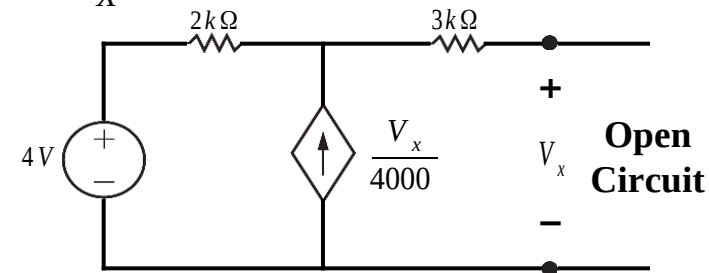
Example (Contd.)

a) Find Thevenin Equivalent seen by R_L

$$V_{Th} = V_{oc} \equiv V_X$$

With terminals open, no current in $3k\Omega$ resistor

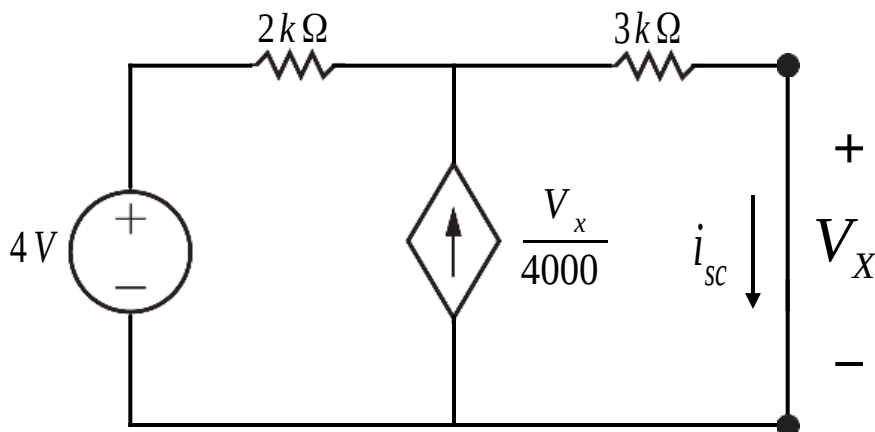
All the current, $V_X / 4000$, goes into the $2k\Omega$ resistor



$$\text{KVL} \left\{ 4 = -2000 \left[\frac{V_X}{4000} \right] + V_X \right.$$

$$\text{Simplify} \left\{ 4 = -\frac{1}{2} V_X + V_X = \frac{1}{2} V_X \Rightarrow V_X \equiv V_{oc} \equiv V_{Th} = 8(V) \right.$$

Find R_{Th} Find i_{sc} Note $V_x = 0$



$$4(V) = (2K + 3K) i_{sc} \quad \left. \vphantom{4(V)} \right\} \text{Ohm's Law}$$

$$i_{sc} = \frac{4}{5K} \quad \left. \vphantom{i_{sc}} \right\} \text{Simplify}$$

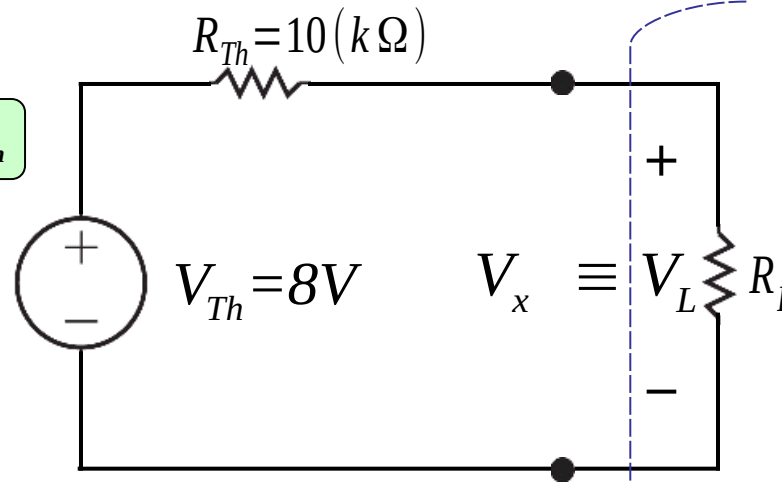
$$i_{sc} = 0.8(mA)$$

Example (Contd.)

$$R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{8}{0.8 \times 10^{-3}}$$

Find R_{Th}

$$R_{Th} = 10(k\Omega)$$



Thevenin
Equivalent
Circuit

$$R_L = R_{Th} = 10(k\Omega)$$

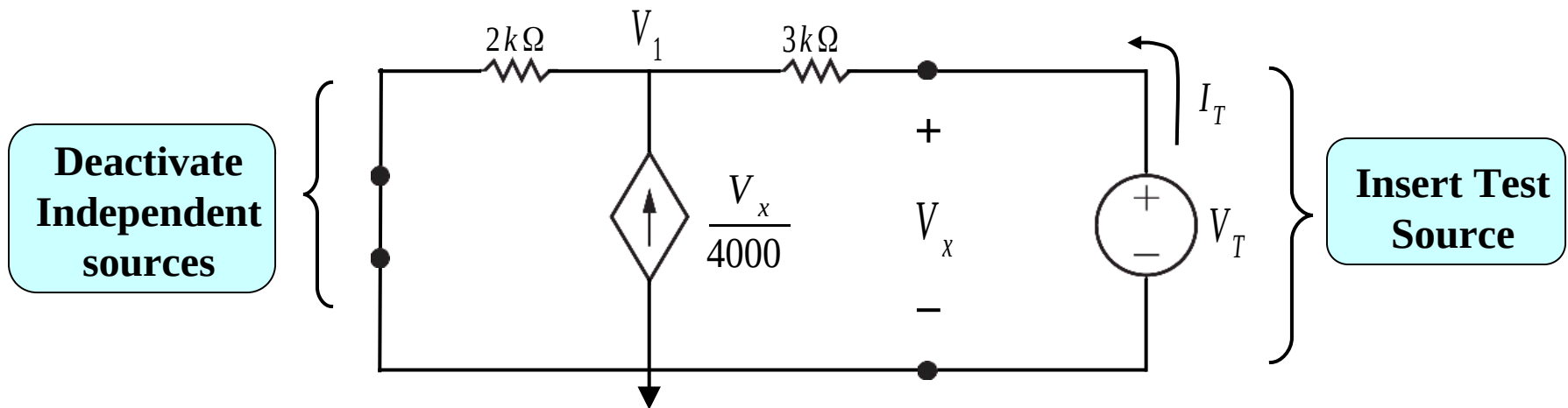
For Max Power

$$V_L = \frac{1}{2} \cdot 8$$

Voltage Division

$$\text{Maximum Power } p_{RL} = \frac{V_L^2}{R_L} = \frac{16}{10(k\Omega)} \Rightarrow p_{RL} = 1.6(mW)$$

Another method for Finding R_{Th} : Test Source



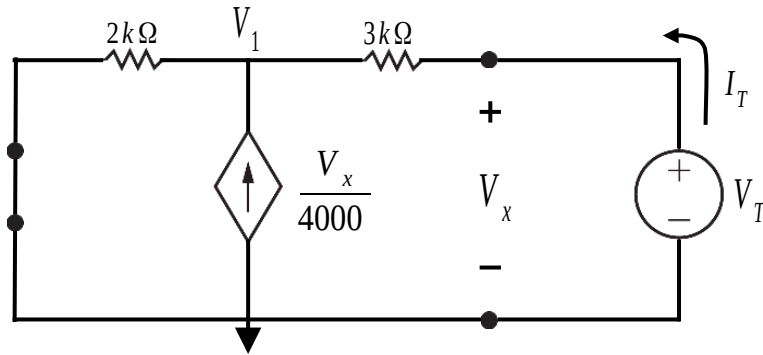
Nodal Equation

$$\textcircled{1} \quad \frac{V_1}{2K} - \frac{V_x}{4K} + \frac{V_1 - V_T}{3K} = 0 \quad \left. \vphantom{\frac{V_1}{2K}} \right\} \text{Node Equation}$$

$$\textcircled{2} \quad V_x = V_T \quad \left. \vphantom{V_x} \right\} \text{From Circuit}$$

$$\textcircled{3} \quad \frac{V_1}{2K} - \frac{V_T}{4K} + \frac{V_1 - V_T}{3K} = 0 \quad \left. \vphantom{\frac{V_1}{2K}} \right\} \text{Substitute } \textcircled{2} \text{ into } \textcircled{1}$$

Another method for Finding R_{Th} : Test Source (Contd.)



$$\textcircled{3} \quad 6V_1 - 3V_T + 4V_1 - 4V_T = 0 \quad \left. \vphantom{\textcircled{3}} \right\} \text{Multiply by 12K}$$

$$\textcircled{3} \quad 10V_1 - 7V_T = 0 \quad \left. \vphantom{\textcircled{3}} \right\} \text{Simplify}$$

Relate V_1 to I_T and V_T

$$\textcircled{4} \quad V_1 = -3KI_T + V_T \quad \left. \vphantom{\textcircled{4}} \right\} \text{KVL}$$

$$-30KI_T + 10V_T - 7V_T = 0 \quad \left. \vphantom{-30KI_T + 10V_T - 7V_T = 0} \right\} \text{Substitute } \textcircled{3} \text{ into } \textcircled{4}$$

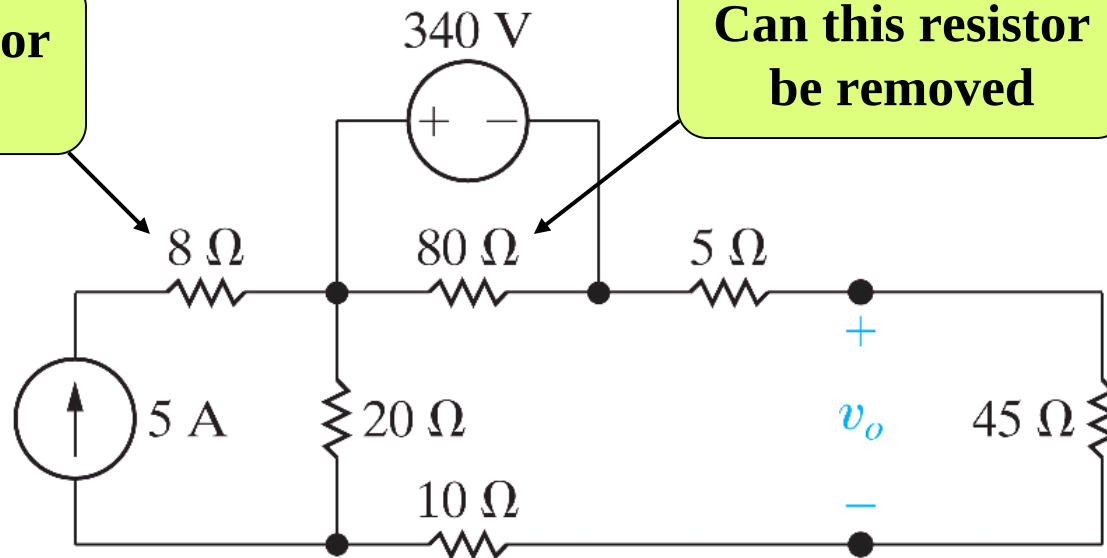
$$-30KI_T + 3V_T = 0 \quad \left. \vphantom{-30KI_T + 3V_T = 0} \right\} \text{Simplify}$$

$$V_T = \frac{30K}{3} I_T \quad \left. \vphantom{V_T = \frac{30K}{3} I_T} \right\} \text{Simplify}$$

$$\frac{V_T}{I_T} = \frac{30K}{3} = R_{Th} = 10(k\Omega)$$

Example: Source Transformation

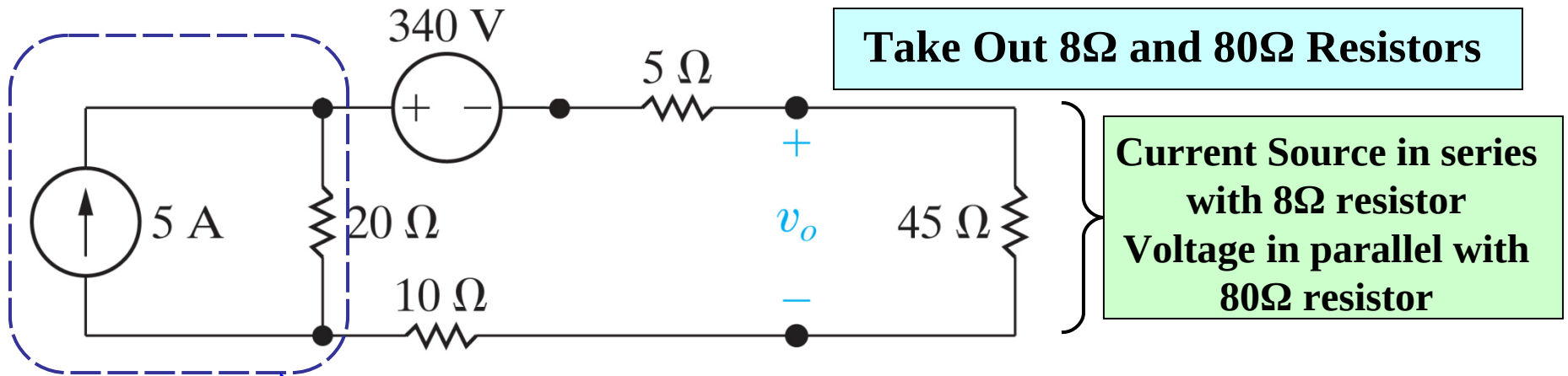
Can this resistor
be removed



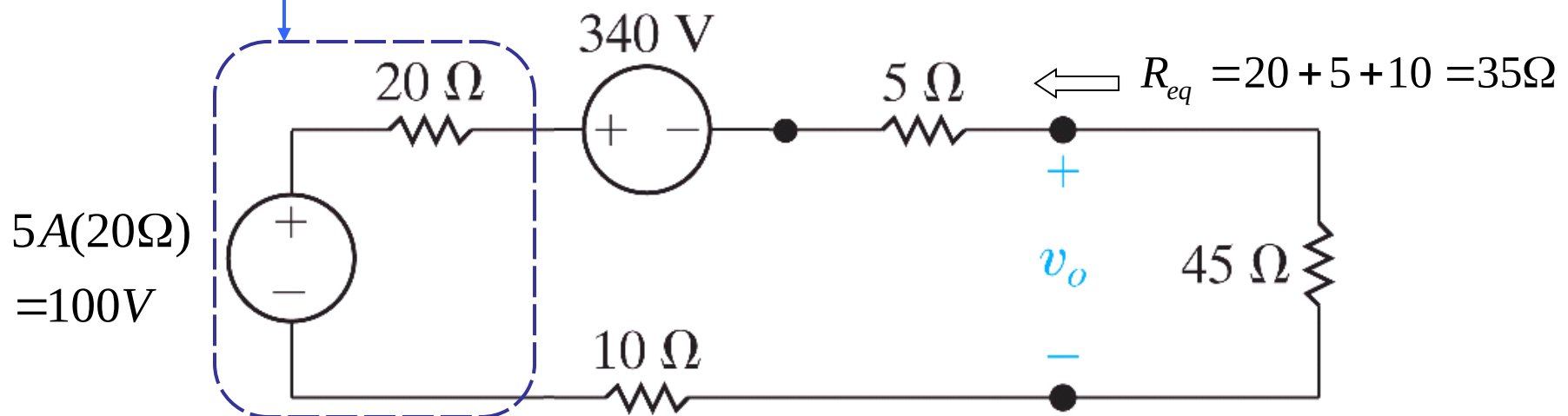
Can this resistor
be removed

Find V_o Using Source Transformations

Example: Source Transformation (Contd.)

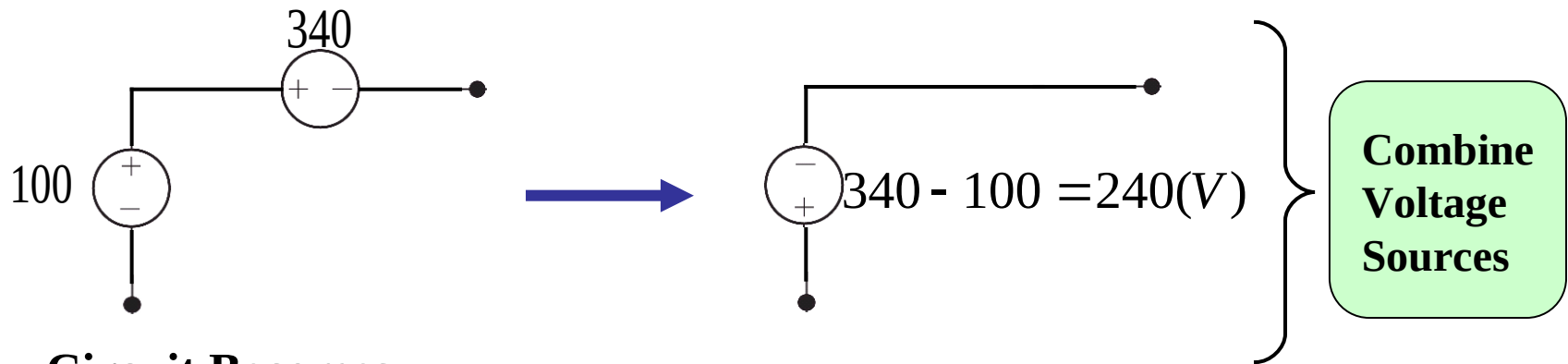


Source Transformation

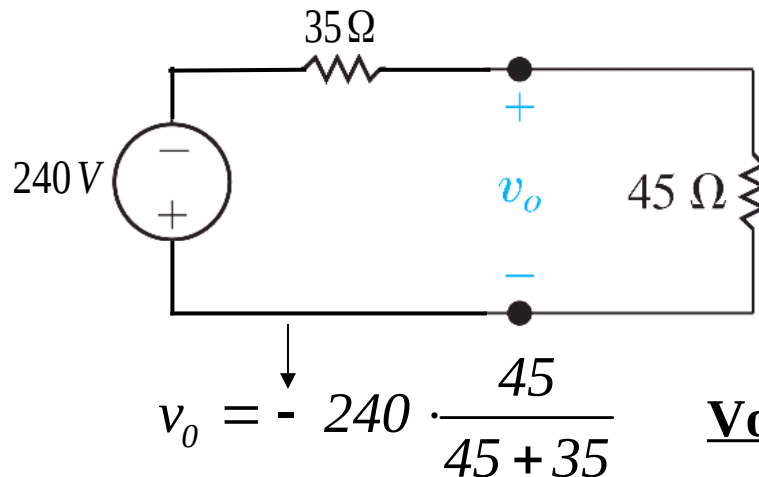


Example: Source Transformation (Contd.)

$$R_{eq} = 20\Omega + 5\Omega + 10\Omega = 35(\Omega)$$



Circuit Becomes



Voltage Division

$$v_o = - 135(V)$$